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Spatial Modelling of Satellite Data for the Development of a Wildfire Intensity Index in Italy

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Abstract

Wildfires represent a major environmental hazard affecting Mediterranean ecosystems, with important ecological, economic and social consequences. Their spatial occurrence is governed by the interaction between environmental conditions, climatic factors and human activities, making the identification and understanding of wildfire-prone areas a relevant objective for long-term territorial planning and prevention strategies.

This thesis develops a spatially explicit framework for modelling the long-term intensity of wildfire occurrence across Italy using a Bayesian Log-Gaussian Cox Process (LGCP).

The study was developed within the context of a research internship at the Italian Institute for Environmental Protection and Research (ISPRA), which provided access to national wildfire and environmental datasets and supported the development of a modelling framework applicable to the analysis of wildfire patterns at national scale.

Wildfire events are modelled as a spatial point process, allowing the contribution of environmental and anthropogenic covariates to be quantified while accounting for residual spatial dependence through a latent Gaussian spatial field. The spatial component is estimated using the Stochastic Partial Differential Equation (SPDE) approach within the Integrated Nested Laplace Approximation (INLA) framework. The analysis relies on harmonised geospatial datasets derived from satellite-based products and national databases, including the Burnt Areas Database – Italian Terrestrial Ecosystems (BAD-ITE), together with variables describing topography, vegetation characteristics, climate and human presence. The final model identifies temperature, elevation, tree cover density, distance from the coast and population as the main variables associated with the spatial distribution of wildfire occurrence. Temperature shows a positive linear association with wildfire intensity, whereas the remaining covariates exhibit non-linear relationships, highlighting the complexity of the interactions between landscape characteristics and human pressure. The latent spatial field further reveals residual spatial structures not fully explained by the observed covariates. The proposed framework provides a data-driven approach for analysing wildfire occurrence intensity, combining harmonised Earth Observation products with Bayesian spatial point-process modelling. Beyond the production of spatial intensity maps, the methodology offers an interpretable framework for investigating the environmental and anthropogenic factors shaping wildfire occurrence and represents a transferable approach for comparable analyses in other European regions.

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Chapter 1

Introduction and background

1.1 The phenomenon of wildfires in Europe

Wildfires are a global phenomenon that affects different regions of the world in a highly heterogeneous way. Beyond their natural occurrence, they represent a complex environmental, social and economic issue, as their impacts extend from immediate damages to long-term alterations of ecosystems and human systems [1].

In environmental terms, wildfires contribute to biodiversity loss, habitat fragmentation and soil degradation, and they can significantly alter hydrological processes [2]. At the same time, they are an important source of atmospheric emissions, releasing greenhouse gases and aerosols that interact with broader climate dynamics [3]. From a socio-economic perspective, wildfire events can have severe consequences on agriculture, infrastructure, tourism and human settlements, and in the most extreme cases they pose a direct threat to human safety [4].

Within Europe, wildfire occurrence exhibits a pronounced spatial gradient, with the highest concentration of fires occurring in the Mediterranean Basin. The combination of climatic conditions, such as prolonged summer droughts and high temperatures, together with land-use patterns and vegetation characteristics, creates an environment particularly prone to fire ignition and spread. As a result, countries such as Spain, Portugal, Italy, Greece and parts of the Balkans regularly experience the highest levels of wildfire activity [5].

The relevance of the phenomenon is also reflected in recent large-scale events. In 2024, for instance, approximately 419298 hectares were burned across Europe, a value slightly above the long-term average for the period 2006–2023. A substantial fraction of this burned area, about 35% (147017 hectares), affected Natura 2000 protected sites, highlighting how wildfires can impact areas of high ecological value.

Temporal variability in wildfire activity further complicates the interpretation of these phenomena at the European scale. While some regions show strong interannual fluctuations driven by climatic extremes, others are influenced by land management practices and socio-economic conditions. The 2024 fire season is a clear example of this heterogeneity, with a particularly high concentration of fire activity in Eastern Europe and the Balkans. In this context, Ukraine accounted for an unusually large share of both burned area and number of events. However, the interpretation of these statistics requires caution, as the country experienced an exceptional socio-political context during the study period, with wildfire dynamics potentially affected by factors not directly comparable with those characterising other European regions.

These patterns make it difficult to derive clear and consistent long-term trends in wildfire activity across Europe. Although several countries have been collecting fire data since the 1970s,

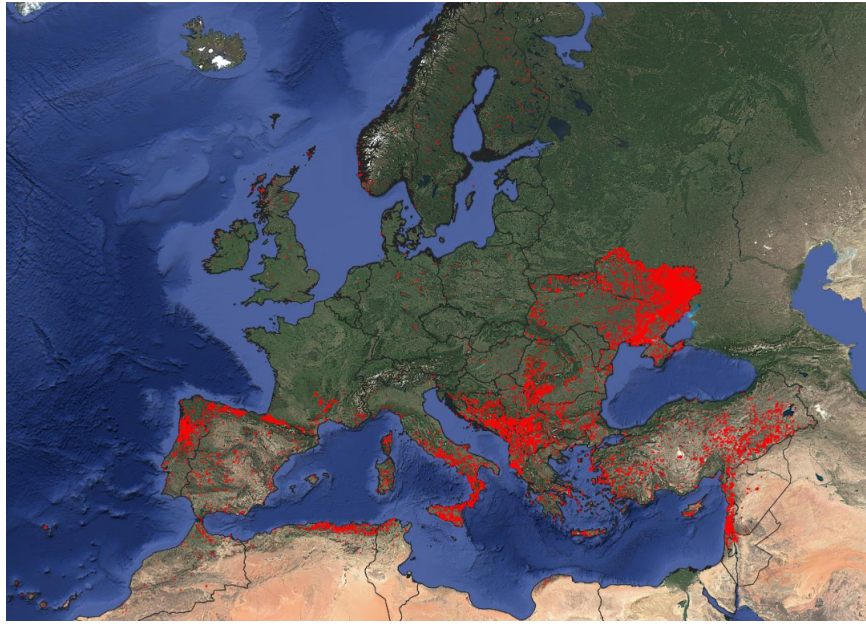


Figure 1.1. Burnt scars produced by wildland fires during the 2024 fire season. Image source: Advance report on forest fires in Europe, Middle East and North Africa, EFFIS [5]

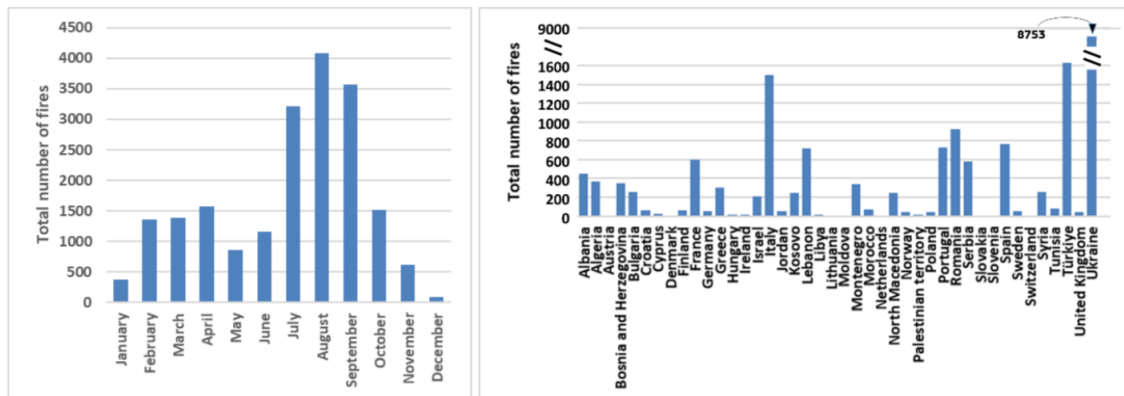


Figure 1.2. Total number of fires mapped by month and country in 2024. Image source: Advance report on forest fires in Europe, Middle East and North Africa, EFFIS [5]

the comparability of these datasets is often limited by differences in data collection methods, classification systems and temporal continuity [6]. Historical records were frequently produced using non-harmonised approaches, which further complicates cross-country analyses.

In this context, remote sensing has played a crucial role in improving the consistency and completeness of wildfire monitoring. Satellite-based observations have made it possible to obtain harmonised information over large spatial scales and with high temporal frequency, reducing some of the inconsistencies associated with ground-based reporting. In particular, the advent of higher-resolution satellite platforms, such as Landsat and Sentinel, has substantially improved the detection of small fire events that were previously underreported.

Despite these advances, important limitations still remain, and differences in national reporting systems and methodological choices continue to affect data comparability. For this reason, European-wide initiatives such as the European Forest Fire Information System (EFFIS) are

essential to provide harmonised datasets and a coherent framework for the analysis of wildfire regimes.

Overall, wildfire activity in Europe is characterised by strong spatial heterogeneity and temporal variability, but also by a set of recurrent patterns, including a pronounced concentration in Mediterranean regions and an increasing sensitivity to climatic and environmental variability. These aspects highlight the importance of reliable and harmonised monitoring systems for understanding and managing wildfire dynamics at the continental scale.

1.2 The European Forest Fire Information System (EFFIS)

The need for harmonised wildfire information across Europe led to the development of the European Forest Fire Information System (EFFIS), which today represents the main reference platform for wildfire monitoring and assessment at the European scale.

EFFIS was developed by the Joint Research Centre (JRC) of the European Commission in collaboration with the national forest and civil protection authorities of participating countries. Rather than being conceived simply as a harmonised wildfire database, EFFIS was developed with the broader objective of supporting the protection of European forests and natural ecosystems against wildfires. The system was established through a collaboration between the European Commission and the national forest fire administrations to assist the authorities responsible for wildfire prevention, preparedness and suppression across Europe. Consequently, EFFIS specifically focuses on wildland fires and their impacts on natural and semi-natural vegetation, providing information that supports not only emergency response but also ecological assessment and environmental management [5, 7].

Initially conceived as a research initiative in the late 1990s, EFFIS progressively evolved into an operational monitoring system and, since 2015, has been integrated within the Copernicus Emergency Management Service (CEMS). Copernicus is the Earth Observation Programme of the European Union, providing free and open satellite data and geospatial information through a constellation of dedicated Sentinel satellites together with complementary satellite and in situ observations.

Today, the EFFIS network includes 43 countries across Europe, North Africa and the Middle East, providing a common platform for wildfire monitoring independently of national administrative boundaries. Besides supporting national authorities during wildfire seasons, EFFIS supplies information to the European Commission, the Emergency Response Coordination Centre (ERCC) and other European institutions involved in disaster risk management.

One of the main strengths of EFFIS is the adoption of a standardised methodology for wildfire mapping. Burned areas are identified through satellite remote sensing, allowing the production of harmonised datasets that are directly comparable across countries. Since 2000, annual burned area maps have been generated using satellite imagery, while operational processing has progressively become automated to provide near-real-time information during the fire season.

A major improvement was introduced in 2018 with the integration of Sentinel-2 imagery into the operational workflow. Compared with the previously adopted MODIS imagery, Sentinel-2 provides substantially higher spatial resolution, allowing the identification of considerably smaller burned areas and a more accurate delineation of fire perimeters. As a result, EFFIS currently maps approximately 95% of the total annual burned area across Europe, providing one of the most comprehensive and consistent wildfire datasets currently available.

Beyond burned area mapping, EFFIS also provides daily fire danger forecasts based on the Canadian Fire Weather Index (FWI), near-real-time active fire detection, post-fire damage

assessment and information on the land-cover types affected by wildfires. These products make EFFIS not only an operational support tool for wildfire management but also a valuable resource for scientific analyses aimed at understanding wildfire regimes and their spatial and temporal evolution.

The harmonised burned-area products provided by EFFIS constitute the primary wildfire dataset used throughout this study. Their continental consistency, standardised processing chain and integration with Copernicus Earth Observation data make them particularly suitable for large-scale spatial analyses and for the development of reproducible wildfire indicators.

1.3 Wildfire occurrence in the context of climate change

Wildfire occurrence is the result of the interaction between multiple environmental and anthropogenic factors rather than a single controlling cause. Fire ignition and subsequent propagation require the simultaneous presence of three essential components, commonly referred to as the fire triangle: an ignition source, available fuel and favourable meteorological conditions. The absence of any of these elements prevents fire development, whereas their combination determines the likelihood of ignition and the potential intensity and spread of the event.

In Europe, ignition sources are overwhelmingly anthropogenic. Although natural causes such as lightning occur, particularly in mountainous regions, they represent only a minor fraction of wildfire events.

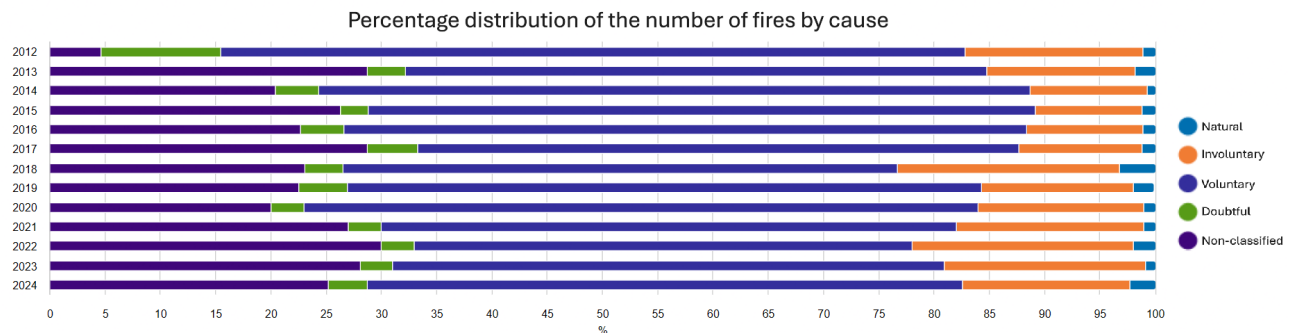


Figure 1.3. Percentage distribution of the number of fires by cause. Image source: ISPRA [8]

Figure 1.3 shows the distribution of wildfire causes in Italy between 2012 and 2024. During this period, natural ignitions accounted on average for only about 1.5% of the total number of recorded wildfires, while human activities were responsible for almost all events. Intentional fires alone represented approximately half of the recorded ignitions and exceeded 60% of all events in some years. These figures highlight that wildfire occurrence in Mediterranean countries is primarily determined by human presence and activities rather than by natural ignition mechanisms.

However, although human activities are responsible for most ignitions, the ignition source alone does not determine the magnitude of a wildfire. Once ignition has occurred, fire behaviour is governed predominantly by environmental conditions. Temperature, precipitation patterns, wind speed, fuel availability and vegetation moisture largely control fire propagation, determining whether an ignition remains a small event or develops into a large and severe wildfire [9].

Landscape characteristics further modulate wildfire behaviour. Vegetation type influences both the quantity and continuity of combustible material, while topography affects fire spread

through slope, aspect and elevation. Consequently, variables describing land cover, vegetation characteristics and terrain are routinely incorporated into wildfire susceptibility and risk models, together with climatic variables describing fire-conducive conditions [10].

Overall, wildfire occurrence reflects the complex interaction between human pressure, vegetation characteristics and environmental conditions. While human activities largely determine where and when ignitions occur, climate and landscape properties regulate the potential development and severity of the resulting fires.

Climate change is expected to modify several of these controlling factors. Although it cannot be considered a direct cause of increasing wildfire occurrence, it is widely recognised as capable of increasing the probability that favourable fire conditions occur simultaneously.

Rising air temperatures, changes in precipitation regimes and the increasing frequency of prolonged droughts reduce vegetation moisture content and increase fuel flammability, thereby creating environmental conditions that favour fire ignition and rapid fire spread.

Observed climatic trends across southern Europe indicate progressively warmer and drier summers, accompanied by more frequent and intense heat waves. These conditions enhance evapotranspiration, reduce soil moisture and prolong periods during which vegetation remains highly combustible. Strong wind events, another important driver of wildfire propagation, may further accelerate fire spread and reduce suppression effectiveness once a fire has started.

One of the clearest manifestations of these changing climatic conditions is the progressive extension of the wildfire season.

The fire season is commonly defined as the period of the year during which environmental conditions are sufficiently favourable for wildfire ignition and spread to require organised fire management activities [11]. Rather than representing a fixed calendar interval, its duration depends on annual meteorological conditions and fuel availability. Several studies have documented a significant increase in fire season length over recent decades. At the global scale, fire-conducive conditions have lengthened across most continents since the late 1970s [12]. In Mediterranean Europe, fire seasons have been estimated to have increased by approximately 12–19 days between 1979 and 2013, with interannual variations closely associated with annual burned area [13]. An extended fire season increases the temporal window during which large wildfires may occur while simultaneously reducing opportunities for fuel management practices such as prescribed burning.

The relationship between climate change and wildfires is also characterised by important feedback mechanisms. Wildfires release large quantities of greenhouse gases into the atmosphere and reduce the capacity of terrestrial ecosystems to store carbon, thereby contributing to further climate warming. Climate change, in turn, promotes environmental conditions that favour more frequent and potentially more severe wildfire events, creating a reinforcing interaction between climate and fire regimes [1].

The conceptual framework described above also provides the rationale for the selection of the environmental variables adopted in this study. The proposed analysis accounts for both the environmental factors regulating fuel availability and fire-conducive conditions and the anthropogenic component associated with wildfire occurrence, through variables that describe the main drivers identified in the literature.

1.4 Existing approaches to wildfire modelling

1.4.1 Definition of wildfire risk

Wildfire risk assessment has been addressed in the scientific literature through a wide range of methodologies developed to support both strategic planning and operational fire management. Although these approaches differ significantly in terms of objectives, spatial and temporal scales, and input data, they are generally framed within a broader conceptual understanding of wildfire risk as a multidimensional construct [14].

In particular, modern wildfire risk frameworks do not refer to a single measurable quantity, but rather to the interaction of several interconnected components, commonly including wildfire likelihood or hazard, exposure, vulnerability, and, in some formulations, coping capacity. Within this perspective, wildfire risk represents the integrated outcome of both environmental conditions that favour fire occurrence and spread, and the presence of exposed and vulnerable elements within the landscape. However, it is important to note that not all modelling approaches explicitly include all these components. Depending on the application context and data availability, many operational and scientific studies focus only on specific parts of this framework, such as ignition probability, fire behaviour, or exposure, without explicitly addressing potential impacts or vulnerability components.

Consequently, wildfire modelling should be understood as a broad field encompassing heterogeneous methodologies, each designed to answer different questions related to where fires are more likely to occur, under which conditions they develop and what elements are potentially exposed.

1.4.2 Long-term and short-term fire modelling approaches

Within this broad framework, wildfire modelling methodologies can be broadly distinguished into long-term and short-term approaches, which differ primarily in their temporal resolution and in the type of information used [15].

Long-term approaches aim to characterise the intrinsic susceptibility of a territory to wildfire occurrence by relying on variables that vary slowly over time. These typically include topography, land cover, vegetation structure and historical fire occurrence. Such approaches are widely used to identify spatial patterns of fire-prone areas and are particularly relevant for land-use planning, ecological management and long-term prevention strategies. Their main objective is therefore not to predict individual fire events, but rather to describe the spatial predisposition of a landscape to wildfire activity under average or climatological conditions.

In contrast, short-term approaches are designed to estimate fire danger at daily or near-real-time scales by incorporating dynamically changing meteorological information. These models explicitly account for weather conditions that directly affect fuel moisture and fire behaviour, such as air temperature, relative humidity, wind speed and recent precipitation. As a result, short-term indices are primarily used in operational contexts to support fire prevention services, resource allocation and emergency response during active fire seasons.

Among the most widely adopted short-term fire danger systems is the Canadian Forest Fire Danger Rating System (CFFDRS), and in particular its Fire Weather Index (FWI) component. The FWI estimates fire danger based exclusively on meteorological inputs, providing a set of indices that describe fuel moisture and potential fire behaviour. Owing to its robustness and transferability, the FWI system has become one of the reference models at international level and is also adopted by EFFIS to provide harmonised fire danger forecasts across Europe.

At national level, several countries have developed operational fire danger systems adapted to their specific environmental and vegetation conditions. In Italy, the RISICO (RISchio Incendi e

Coordinamento) model integrates meteorological observations and forecasts with information on vegetation and topography to produce daily fire danger assessments supporting civil protection activities. Similar frameworks are implemented in the United States through the National Fire Danger Rating System (NFDRS), while Australia adopts the Forest Fire Danger Index (FFDI), specifically designed for its climatic and ecological conditions.

1.4.3 Remote sensing and data-driven fire assessment

Alongside traditional meteorological and structural approaches, the increasing availability of Earth Observation data has substantially transformed wildfire assessment methodologies. Satellite remote sensing provides spatially continuous and temporally consistent information on vegetation conditions, land surface properties and post-fire effects, enabling the development of models that complement ground-based and meteorological observations.

Remote sensing products are widely used to derive indicators related to vegetation greenness, fuel moisture, burned area, burn severity and vegetation recovery. In particular spectral indices, which are numerical indicators derived from the combination of different spectral bands in multispectral satellite imagery [16]. They are widely used in remote sensing to enhance and quantify specific biophysical properties of the Earth's surface. By exploiting the different reflectance responses of materials across the electromagnetic spectrum, these indices allow the retrieval of information that is not directly observable in the visible range. An additional advantage of spectral indices is their ability to capture surface properties even under sub-optimal observational conditions, such as partial cloud cover or smoke, depending on the spectral configuration.

Among the most widely adopted spectral indices are the Normalized Difference Vegetation Index (NDVI), the Global Vegetation Moisture Index (GVMI) and the Normalized Burn Ratio (NBR). They have become standard proxies for assessing fuel conditions and vegetation stress over large spatial extents, or to compute severity after fire occurrences.

More recently, wildfire assessment methodologies have increasingly combined satellite-derived variables with meteorological data and ancillary geospatial information. These integrated approaches aim to exploit the complementary nature of Earth Observation and weather-based data, enabling more comprehensive representations of wildfire susceptibility and environmental conditions at different spatial and temporal scales. In parallel, the growing availability of computational resources has also encouraged the adoption of statistical and machine learning techniques for modelling wildfire occurrence and susceptibility patterns.

1.5 Objectives and positioning of the proposed methodology

The main objective of this thesis is to develop a spatially explicit model of wildfire occurrence intensity; aimed at describing the long-term patterns of wildfire occurrence across the study area. Unlike operational fire danger systems, which focus on short-term prediction of fire weather conditions, the present work is not intended for real-time forecasting or emergency response applications.

Instead, the study focuses on characterising the spatial heterogeneity of wildfire occurrence in relation to a set of environmental, climatic, vegetation-related and anthropogenic factors. The emphasis is placed on understanding how these variables influence the frequency and distribution of wildfire events across space, rather than on modelling fire behaviour, severity, or potential damages.

In this context, wildfire susceptibility is interpreted in terms of the propensity of a territory to experience wildfire events, expressed through spatial variation in observed fire occurrence patterns. The analysis does not aim to represent a complete wildfire risk framework, as it does not explicitly account for fire impacts or vulnerability components. Rather, it focuses on the environmental and socio-demographic conditions associated with the likelihood of wildfire occurrence.

The proposed framework is intended as a tool for exploratory spatial analysis, providing both a spatial representation of relative susceptibility to wildfire occurrence and a quantitative assessment of the association between wildfire occurrence and the considered environmental and anthropogenic covariates. In this way, the study aims not only to identify areas more prone to wildfire occurrence but also to improve the understanding of the factors that shape their spatial distribution.

This thesis was developed in the context of a research internship at the Italian Institute for Environmental Protection and Research (ISPRA). The collaboration provided access to national wildfire and environmental datasets and allowed the proposed modelling framework to be developed within an institutional research context focused on the analysis of wildfire patterns in Italy.

Chapter 2

Descriptive analysis of wildfire data

Before developing the spatial model of wildfire occurrence intensity, an exploratory analysis of the available data was carried out to better understand the characteristics of wildfire occurrence in Italy and to identify the main spatial and temporal patterns of the phenomenon.

Two complementary datasets were used throughout this study:

- the annual wildfire time series provided by the Italian Forest, Environmental and Agri-food Unit Command (CUFAA)¹, covering the period 1980–2023;
- the Burnt Areas Database for Italian Terrestrial Ecosystems (BAD-ITE)², derived from the European Forest Fire Information System (EFFIS) and reprocessed by ISPRA, covering the period 2018–2025.

The CUFAA historical archive spans 44 years and provides annual summaries of the total number of wildfires and the corresponding burned area at the national scale. Owing to its long temporal coverage, this dataset is particularly suitable for analysing long-term trends in wildfire occurrence.

In contrast, the BAD-ITE database provides a much finer spatial resolution. Rather than aggregated annual statistics, it contains the individual wildfire perimeters together with several attributes describing each event. During the ISPRA processing workflow, wildfire polygons were spatially intersected with the administrative boundaries provided by ISTAT, allowing the information to be analysed at the municipal level as well as over other territorial units, such as regions and protected areas.

Although BAD-ITE covers a considerably shorter time period than the CUFAA archive, its event-based structure makes it the most appropriate dataset for spatial modelling and detailed geographical analyses.

The two datasets should not be interpreted as directly comparable, since they are based on fundamentally different acquisition procedures and wildfire definitions.

The CUFAA archive is compiled from operational reports produced by the Italian Forest Corps and firefighting authorities. Consequently, wildfire events are recorded in the field whenever intervention is required, including relatively small fires that may affect only limited areas.

Conversely, the BAD-ITE database is derived from satellite observations acquired by the Sentinel-2 within the EFFIS framework. Only wildfires that produce detectable burned areas

¹Dataset can be downloaded at: <https://forest-fire.emergency.copernicus.eu/applications/data-and-services> in the section *Download the country totals (burnt areas & number of forest fires)*.

²Unprocessed dataset can be downloaded at: <https://forest-fire.emergency.copernicus.eu/applications/data-and-services> in the section *Download real-time updated Burnt Areas database*.

can therefore be included, implying that very small events may remain undetected. As a result, the total number of fires reported by CUFAA is expected to be substantially higher than that contained in BAD-ITE.

An additional source of discrepancy arises from the different definitions of wildfire adopted by the two systems. The National Wildfire Information Unit (NIAB) of CUFAA follows the definition established by Article 2 of the Italian Law No. 353/2000, according to which a wildfire is

“a fire with the potential to spread over forested, shrubland or wooded areas, including any anthropogenic structures and infrastructures located within these areas, or over cultivated, uncultivated and pasture lands adjacent to them” [17].

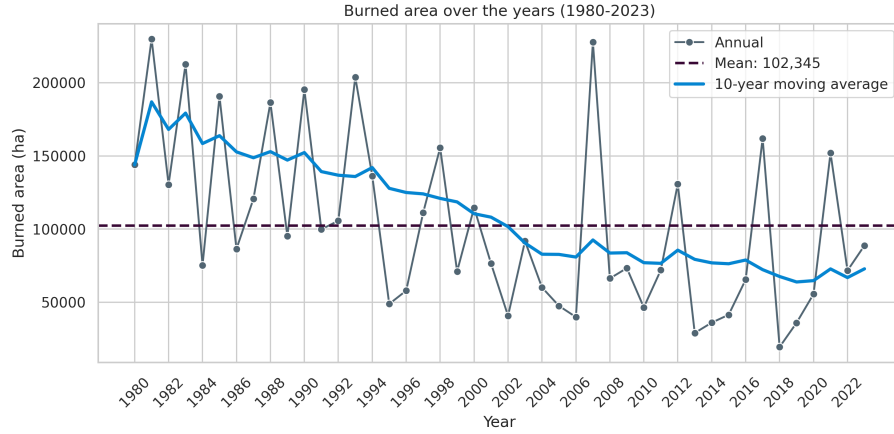
By contrast, within the European Forest Fire Information System (EFFIS), a wildfire is defined according to the land-cover classes included in the CORINE Land Cover (CLC) classification. Consequently, a wildfire is recorded only if the burned area effectively intersects one or more eligible wildland land-cover categories[17]. This distinction is substantial. Under the CUFAA definition, it is sufficient for a fire to threaten a forested area, whereas the EFFIS methodology requires the fire to actually burn at least part of the eligible land-cover classes. Together with the different acquisition methods, these conceptual differences explain why the two databases report different wildfire counts despite describing the same underlying phenomenon.

2.1 CUFAA time series from 1980 to 2023

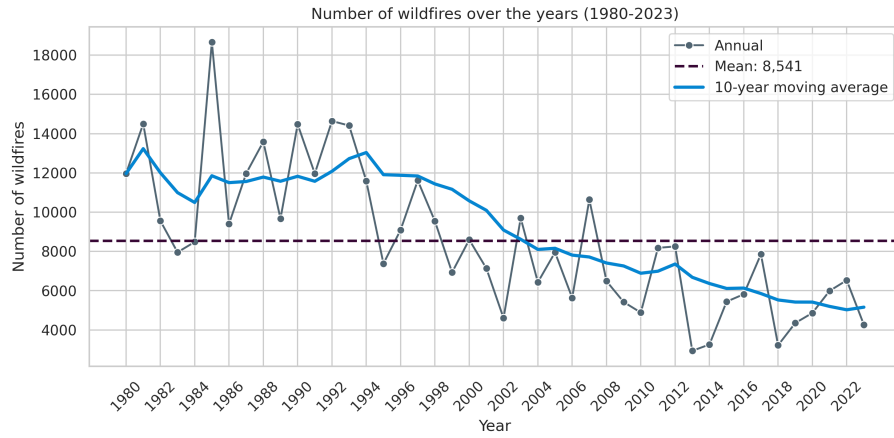
Figure 2.1 shows the annual burned area (panel a) and the annual number of wildfires (panel b) in Italy over the period 1980–2023.

Both time series exhibit pronounced interannual variability, with alternating years of relatively limited fire activity and exceptionally severe wildfire seasons. In particular, the burned area series is characterised by several marked peaks, corresponding to extreme fire years that substantially exceed the long-term average. Among the most severe wildfire seasons of the last two decades are 2007, 2017 and 2021, all associated with exceptionally large burned areas.

Over the entire study period, the mean annual burned area is approximately 102345 hectares, while the average annual number of wildfires is about 8541. These values provide a useful descriptive benchmark for wildfire activity in Italy, although they should not be interpreted as stationary characteristics because both variables display considerable year-to-year fluctuations. Despite this marked variability, a general decline in both the annual number of wildfires and the total burned area can be visually observed over the last two decades. Similar evidence has been reported by ISPRA, which attributes this tendency to the progressive strengthening of wildfire prevention policies, improvements in fire suppression capabilities and the implementation of coordinated firefighting strategies at both the national and European levels. In particular, the introduction of the Italian Wildfire Law (Law No. 353/2000) is considered to have played a major role in strengthening wildfire prevention and management [17].



(a) Time series of burned area over the years from 1980 to 2023 with long period mean and 10 years moving average.



(b) Time series of total number of wildfires over the years from 1980 to 2023 with long period mean and 10 years moving average.

Figure 2.1

2.1.1 Number of wildfires

To investigate the temporal evolution of the annual number of wildfires over the period 1980–2023, the percentage anomaly was computed for each year with respect to the long-term mean. Expressing the series as anomalies removes the influence of the measurement scale and allows each annual value to be interpreted as a relative deviation from the average conditions over the entire study period.

The percentage anomaly was calculated as

$$\text{Anomaly}(\%) = \left(\frac{\text{Observation} - \bar{X}}{\bar{X}} \right) \times 100,$$

where $\bar{X}$ denotes the long-term mean annual number of wildfires.

Temporal trends were analysed using quantile regression by estimating the conditional 50th percentile (median) and the 90th percentile as functions of time. Unlike ordinary least squares

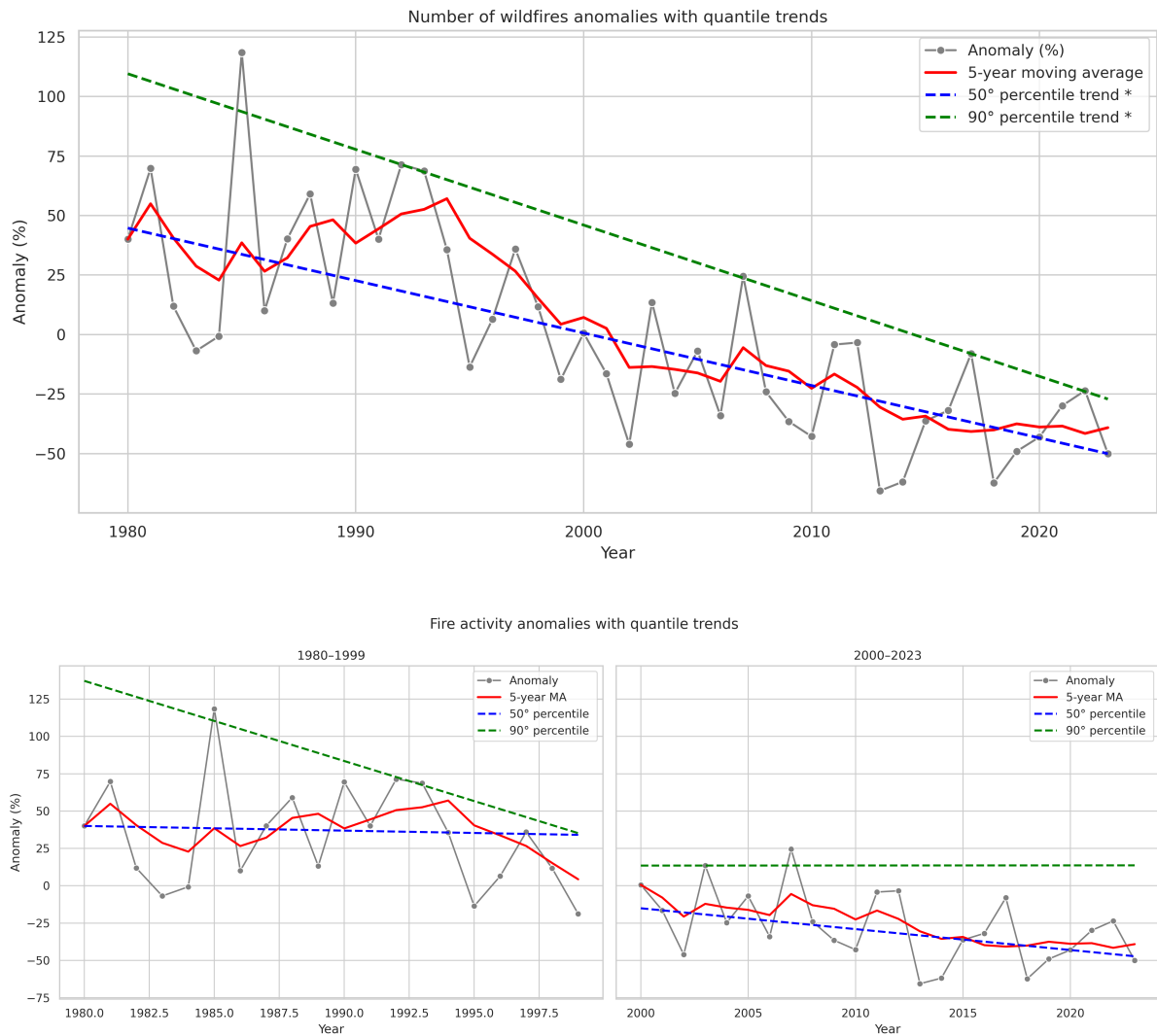


Figure 2.2. Percentage anomalies of the annual number of wildfires with fitted quantile regression lines. Top: full study period (1980–2023). Bottom: separate analyses for 1980–1999 and 2000–2023. Anomalies are computed with respect to the long-term mean over the entire period.

regression, which models the conditional mean, quantile regression estimates different parts of the response distribution. This approach provides a more comprehensive description of temporal changes and is less sensitive to extreme observations, making it particularly suitable for wildfire data, which are often characterised by highly variable and occasionally extreme years.

The fitted median regression line represents the typical annual wildfire activity expected for a given year, whereas the 90th percentile describes the evolution of the upper tail of the distribution, corresponding to years with unusually high wildfire occurrence.

For the complete 1980–2023 period, both the median and the 90th percentile exhibit negative and statistically significant slopes. This indicates that the temporal decline is not limited to the central tendency of the distribution but also affects the most severe wildfire seasons, suggesting an overall downward shift in the distribution of annual wildfire counts.

To investigate whether this trend remained stable over time, the analysis was repeated separately for the periods 1980–1999 and 2000–2023, while retaining the same long-term mean as the

anomaly reference. In the first sub-period, the median exhibits an approximately flat trend, whereas the 90th percentile shows a decreasing but non-significant tendency. During the second sub-period, the median displays a slight negative slope and the 90th percentile remains essentially stable; however, neither trend is statistically significant. These results suggest that the significant long-term decline observed over the full study period is primarily driven by changes occurring across the entire 44-year record rather than by a persistent trend within either sub-period considered separately.

2.1.2 Total burned area

The same methodology was applied to the annual burned area time series over the period 1980–2023. Percentage anomalies were computed with respect to the long-term mean, and temporal trends were investigated by means of quantile regression, estimating the conditional 50th and 90th percentiles.

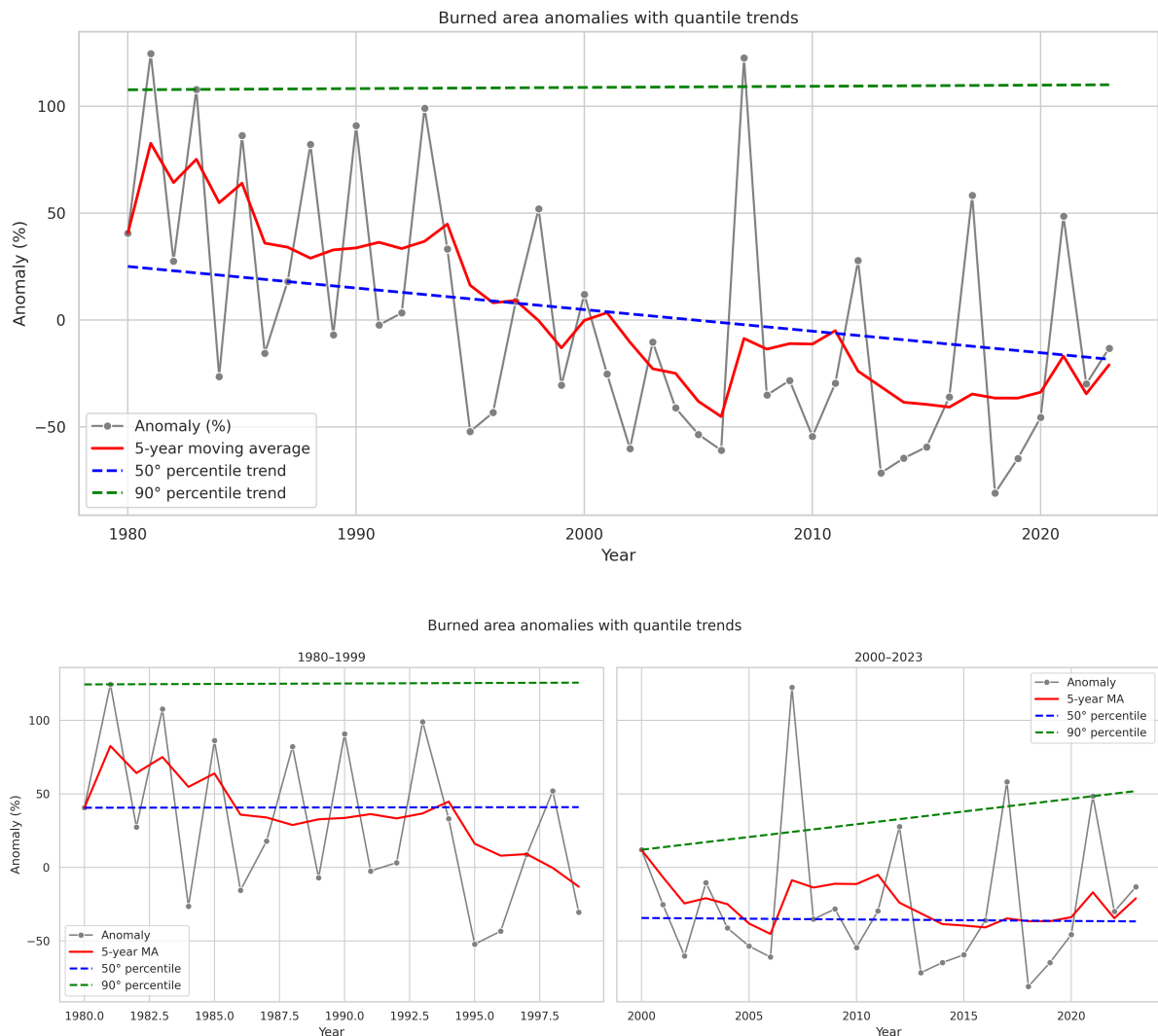


Figure 2.3. Percentage anomalies of the annual burned area with fitted quantile regression lines. Top: full study period (1980–2023). Bottom: separate analyses for 1980–1999 and 2000–2023. Anomalies are computed with respect to the long-term mean over the entire period.

Unlike the annual number of wildfires, the burned area does not exhibit statistically significant temporal trends for either the median or the 90th percentile over the complete study period. Although the fitted median regression line shows a slight negative slope, the trend is not statistically significant, indicating that the typical annual burned area has remained broadly stable once the considerable year-to-year variability is taken into account.

Similarly, the estimated 90th percentile remains approximately constant over time, suggesting that the most severe wildfire seasons have not experienced a systematic increase or decrease during the study period. Instead, the upper tail of the distribution continues to be dominated by a limited number of exceptionally large fire seasons.

The analysis was then repeated separately for the periods 1980–1999 and 2000–2023, while retaining the same long-term mean as the anomaly reference. In the first sub-period, both the median and the 90th percentile are essentially flat, with no statistically significant temporal trend. During the second sub-period, the median remains approximately constant, whereas the 90th percentile exhibits a slight positive slope; however, this increase is also not statistically significant.

Overall, these results suggest that, unlike wildfire occurrence, the annual burned area has not undergone a clear long-term temporal change. Instead, its evolution appears to be driven primarily by strong interannual fluctuations and the occurrence of occasional extreme wildfire seasons rather than by a persistent monotonic trend.

2.1.3 Forest / non-forest

While remaining in the context of total burned area, it is available the repartition of burned area between forested and non forested area, as shown in Figure 2.4.

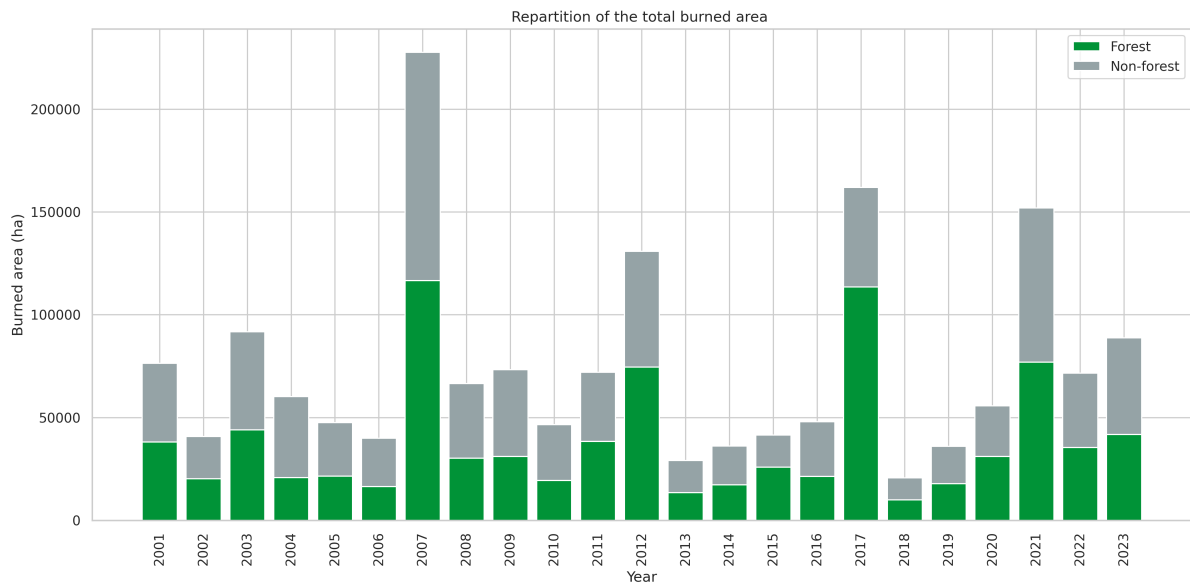


Figure 2.4. Time series of the repartition of total burned area between forest and non forest

Between 2001 and 2012, forests accounted for 46.6% of the total burned area, whereas non-forest areas represented 53.4%. During the period 2013–2023, the proportion of burned forest increased to 52.0%, while non-forest areas accounted for 48.0%. To assess whether the proportion of

burned forest differed between the two periods, a Mann–Whitney U test was performed³. Although this corresponds to an increase of approximately 5.4 percentage points in the average proportion of burned forest, the Mann–Whitney U test did not detect a statistically significant difference between the two periods ($p = 0.23$). Therefore, the available data do not provide sufficient evidence to conclude that the relative contribution of forested areas to the total burned area has changed over time. These results should be interpreted with some caution, as annual observations form a time series and may exhibit temporal dependence, whereas the Mann–Whitney U test assumes independent observations.

³The Mann–Whitney U test is a non-parametric test that evaluates the null hypothesis that the two independent samples originate from the same underlying distribution.

2.2 BAD-ITE from 2018 to 2025

The second wildfire occurrence data source used in this study were obtained from the *Burnt Areas Database – Italian Terrestrial Ecosystems* (BAD-ITE), a geodatabase developed by the Italian Institute for Environmental Protection and Research (ISPRA). The database provides spatially explicit information on wildfire-affected areas across Italy and supports both spatial and temporal analyses of fire impacts on terrestrial ecosystems, with particular emphasis on areas of conservation interest.

BAD-ITE is derived from the real-time updated *Burnt Areas Database* distributed by the European Forest Fire Information System (EFFIS). The original EFFIS product consists of vector polygons delineating burned areas detected from multispectral optical satellite imagery acquired by the MODIS and Sentinel-2 MSI sensors, with spatial resolutions of 250 m and 20 m, respectively. The database is updated on a daily basis and contains wildfire perimeters across Europe from the year 2000 onwards.

To ensure consistency with the improved spatial detail provided by Sentinel-2 imagery, ISPRA adopts the post-2018 EFFIS products for the production of its official wildfire statistics. Consequently, the present study considers wildfire events occurring between 2018 and 2025.

2.2.1 Pre-processing

The original EFFIS wildfire database is designed for pan-European monitoring and therefore provides limited detail at national scale. To enable a more detailed spatial characterization of wildfire events, ISPRA applies a preprocessing workflow based on the spatial integration of fire perimeters with national geospatial datasets.

All wildfire perimeters are projected into the ETRS89 / LAEA Europe coordinate reference system (EPSG:3035) and spatially intersected with ancillary spatial layers. These include administrative boundaries provided by ISTAT, which allow attribution of wildfire events to municipalities and higher-level territorial units, as well as national and international protected areas, such as Natura 2000 sites, Ramsar, the Official List of Protected Areas (EUAP) and other designated conservation areas.

In addition, wildfire perimeters are intersected with ecosystem classification layers based on the ISPRA Ecosystems Classification Model for Forests (ECM-F4) [18]. This classification distinguishes major forest types, including broadleaved deciduous forests (T1), broadleaved evergreen forests (T2), evergreen coniferous forests (T3), subalpine deciduous conifer forests (T34), and unclassified forest areas (TNC). This integration enables the characterization of wildfire occurrence across different ecological contexts and supports subsequent spatial analyses. The pre-processing workflow applied by ISPRA is shown in a graphical representation in Figure 2.5.

The preprocessing procedures described above resulted in a geospatial dataset containing 13,433 records and 29 attributes, that can be inspected in detail in Table 2.1.

A key aspect of the dataset concerns the interpretation of the variable *EFFIS_ID*. This identifier uniquely represents a wildfire event within the EFFIS database. However, the BAD-ITE dataset is enriched through overlays with administrative and environmental layers and is therefore stored at a sub-fire level. Consequently, when a burned-area polygon intersects multiple municipalities, the same wildfire is represented by multiple records, each corresponding to a different spatial portion of the event and containing the attributes associated with the intersected municipality. As a result, the number of rows in the database does not correspond to the number of individual wildfires. For the purposes of the present analysis, where the unit of observation is the wildfire

Table 2.1. Main attributes available in the BAD-ITE database.

Variable	Description
EFFIS_ID	Unique identifier assigned to the wildfire event by EFFIS.
DATA	Date of wildfire occurrence.
AGGIORNAMENTO	Date of the latest update of the record.
AREA_HA	Total burned area (hectares) of the unique wildfire event (EFFIS_ID) .
CLASS	Fire classification provided by EFFIS.
EPOCH	Temporal reference ID associated with the event.
YEAR	Year of occurrence.
DOY	Day Of Year.
WEEK	Calendar week of occurrence.
REGIONE	Italian administrative region.
PROVINCIA	Italian province.
COMUNE	Italian municipality.
lon	Longitude of the wildfire location.
lat	Latitude of the wildfire location.
elevation	Elevation associated with the burned area (m).
AREA_AU_HA	Total area of the interested municipalities (hectares).
BIOGEOREG	Biogeographical region.
RAMSAR	Indicator of overlap with a Ramsar site.
RN2K	Indicator of overlap with the Natura 2000 network.
FV	Indicator of overlap with protected areas classified as FV.
EUAP	Indicator of overlap with areas included in the Official List of Protected Areas (EUAP).
AREA	Area of the wildfire polygon (ha) recomputed by ISPRA in the single interested municipality (the sum of all AREA of records with the same EFFIS_ID gives back the same value reported on AREA_HA).
ECM_TNC	Burned area within unclassified forest ecosystems (ECM-F4).
ECM_T1	Burned area within deciduous broadleaved forests (ECM-F4 class T1).
ECM_T2	Burned area within evergreen broadleaved forests (ECM-F4 class T2).
ECM_T3	Burned area within evergreen conifer forests (ECM-F4 class T3).
ECM_T34	Burned area within subalpine deciduous conifer forests (ECM-F4 class T34).
ECM_For	Total burned forest area according to the ECM-F4 classification.
geometry	Polygon geometry representing the burned area.

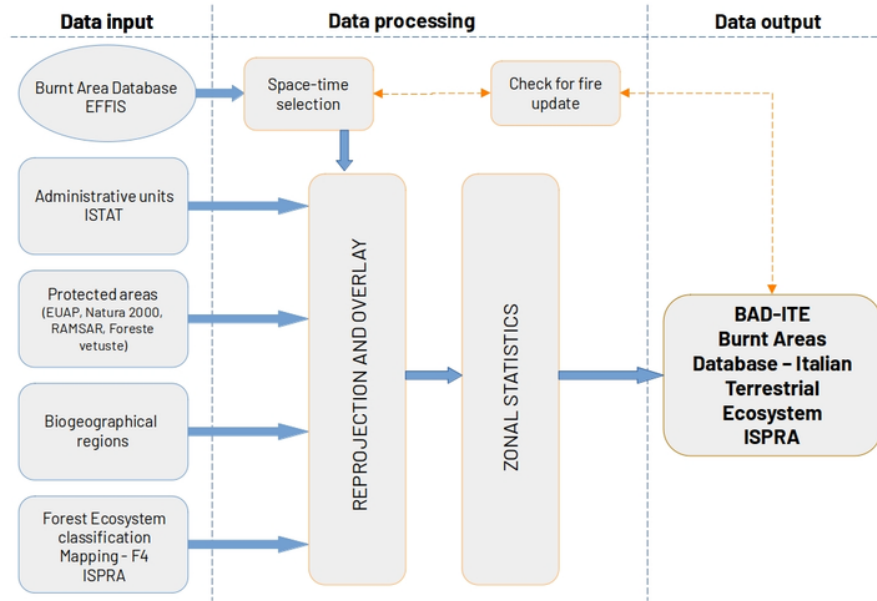


Figure 2.5. ISPRA's pre-processing workflow

event, it was therefore necessary to reconstruct a one-to-one correspondence between records and fires. This was achieved by aggregating all records sharing the same *EFFIS_ID* using the *dissolve* operation implemented in the **GeoPandas** library.

During this procedure, geometric features belonging to the same wildfire were merged into a single polygon, while attribute values were aggregated according to their nature. Continuous variables such as elevation and coordinates were averaged, area-related variables were summed, logical indicators describing protected-area membership were combined using logical operators, and categorical administrative information was retained by concatenating the unique values associated with each event.

The aggregation process reduced the number of records from 13433 to 9561, corresponding to the total number of distinct wildfire events considered in the study. No missing values were present in either the original or the aggregated dataset.

Before proceeding with the descriptive analysis, the influence of individual years was assessed. Particular attention was given to the year 2018, which corresponds to a transition period associated with the introduction of the Sentinel-2 satellite products. As a consequence, the available data for that year are incomplete and do not provide a reliable representation of wildfire activity.

For this reason, 2018 was excluded from the subsequent descriptive analyses in order to avoid introducing an artificial bias. The influence of each annual observation was further evaluated using Cook's distance. The value obtained for 2018 ($D = 0.556$) exceeded the commonly adopted threshold of $4/n = 0.500$, indicating that this observation had a disproportionate influence on the fitted regression trend. This result provides additional quantitative support for its exclusion, which was primarily motivated by the known incompleteness of the underlying data.

By contrast, the year 2021 also emerged as an influential observation because of the exceptionally large number of wildfires and burned area recorded. Unlike 2018, however, this observation reflects a genuine extreme wildfire season rather than a data collection issue. It was therefore

retained in the dataset, as such extreme years constitute an integral component of the natural variability of wildfire occurrence. The final dataset covers the period 2019–2025. In its aggregated form, based on the EFFIS_ID spatial unit, the dataset consists of 9414 rows and 29 variables. In addition to this aggregated version, the dataset is also retained in its original (non-aggregated) form, which consists of 13199 rows, and it will be used depending on the specific modelling or diagnostic requirement. This dual representation allows for flexibility in the analysis, enabling both pixel-level and aggregated-scale investigations of wildfire occurrence patterns.

Table 2.2. Dataset record count before and after removing the 2018 outlier.

Dataset Version	Original (with 2018)	Filtered (without 2018)
Non-aggregated wildfires	13433	13199
Aggregated dataset	9561	9414

2.3 Exploratory analysis of recent years wildfire data

The study of wildfire occurrence in the context of climate variability and climate change is inherently complex and requires long-term observational records to support robust climatological conclusions. In general, climate-scale assessments typically rely on time series of at least 30 years in order to distinguish long-term signals from interannual variability.

In this study, the available dataset spans a considerably shorter period, which does not allow for formal attribution or inference in a climatological sense. For this reason, no claims regarding climate change impacts are made in a strict statistical or climatological framework.

Nevertheless, the dataset provides valuable information for exploring spatial and ecological patterns of wildfire occurrence. In particular, it allows the investigation of how fire activity varies across ecosystems, ecoregions and provides useful insights for the interpretation of the subsequent spatial modelling framework.

The following sections therefore focus on an exploratory analysis of wildfire counts and burned areas across different ecological and geographical strata, without aiming at long-term climatic attribution.

2.3.1 Total number of wildfires

Figure 2.6 presents the choropleth map of the cumulative number of wildfires recorded by EFFIS over the period 2019–2025 across the Italian territory.

The highest concentrations of wildfire events are observed in Sicily, which clearly emerges as the most affected region, followed by Calabria. High wildfire occurrence is also evident in Campania and Lazio, while intermediate values are found in Apulia, Basilicata and Sardinia.

By contrast, Northern Italy is characterised by substantially lower wildfire frequencies.

While the previous map highlights the spatial distribution of wildfire occurrence, it is also informative to examine how wildfire activity is distributed over time. Figure 2.7 summarises the annual number of wildfires recorded in the EFFIS dataset during the study period.

Panel (a) reports the absolute number of wildfire events for each year, while panel (b) shows the corresponding percentage contribution of each year to the total number of recorded wildfires.

The annual number of wildfire events exhibits noticeable variability over the study period, while also showing an overall increase towards the most recent years. In particular, 2025 accounts for approximately 20.4% of all wildfire events recorded in the dataset, representing the largest

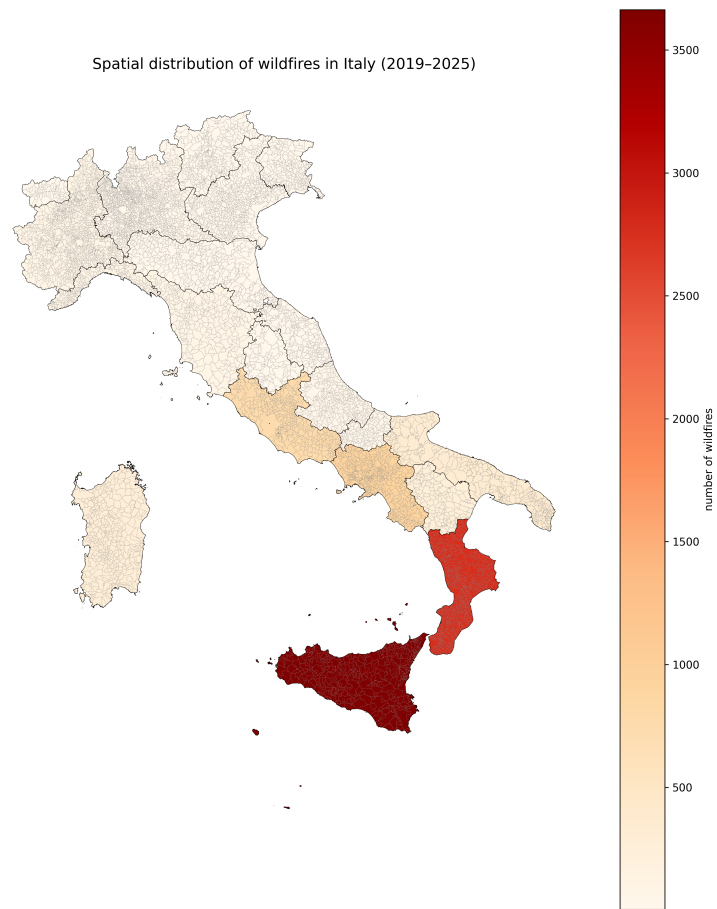


Figure 2.6. Distribution of wildfires in Italy from 2019 to 2025

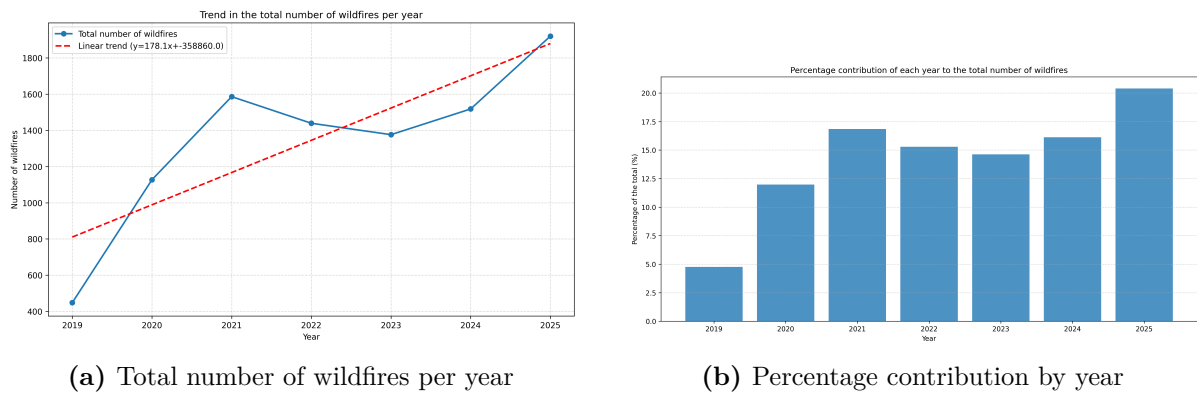


Figure 2.7. Distribution of wildfires in Italy from 2019 to 2025

annual contribution. Although lower than 2025, the year 2021 stands out as the second most active wildfire season in terms of the number of recorded events. Conversely, the first years of the series contribute a substantially smaller proportion of the total. Given the limited temporal coverage of the dataset, these results should be interpreted as a descriptive summary of the observed period rather than as evidence of a long-term temporal trend.

2.3.2 Total burned area

The temporal pattern of the total burned area differs from that observed for the number of wildfire events.

Although the fitted linear trend shows a slight positive slope, the series is primarily characterised by pronounced interannual variability rather than by a consistent increase over time.

A particularly striking feature is the exceptional wildfire season of 2021, which alone accounts for approximately 28.5% of the total burned area recorded over the entire period. This highlights the exceptional severity of that fire season relative to the rest of the study period.

Overall, the burned area appears to be driven more by the occurrence of exceptional fire seasons than by a gradual year-to-year change, emphasising the highly episodic nature of wildfire activity.

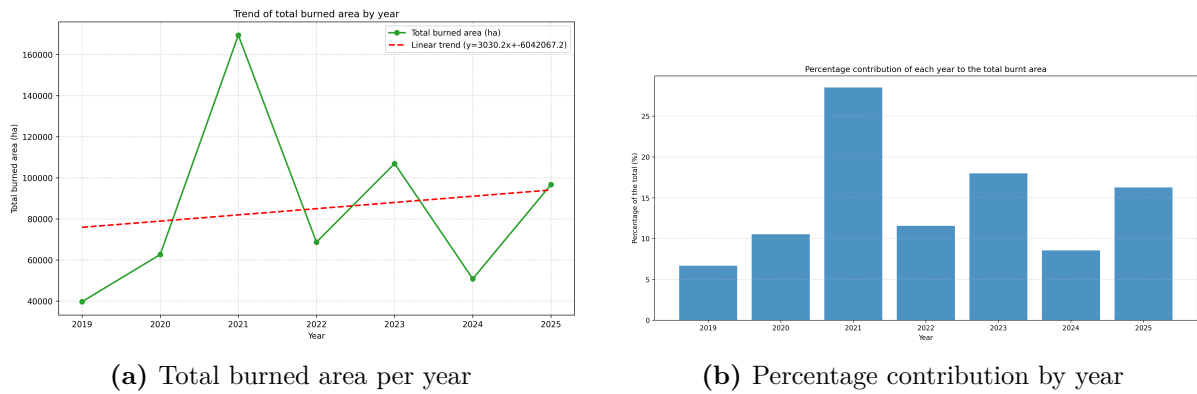


Figure 2.8. Distribution of total burned area in Italy from 2019 to 2025

2.3.3 Average wildfire extent

50th percentile (median) of wildfire extent

To characterise the typical size of wildfire events, the median burned area per fire is considered rather than the mean.

The median provides a more robust measure of central tendency, as it is less sensitive to extreme values, which are frequent in wildfire size distributions and can substantially bias the average.

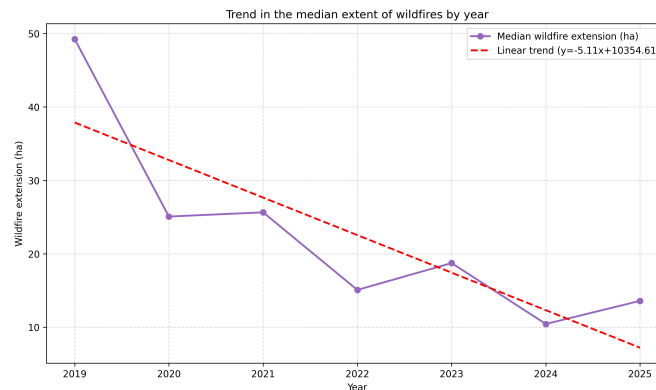


Figure 2.9. Median extension of wildfires

The results indicate a decreasing pattern in the median wildfire extent over the study period, suggesting a reduction in the typical size of individual fire events in the BAD-ITE dataset.

90th percentile of wildfire extent

Beyond average conditions, it is also informative to examine the behaviour of the upper tail of the wildfire size distribution. Instead of focusing on the single largest event per year, which may be overly sensitive to outliers and unrepresentative of annual conditions, the 90th percentile of burned area is used as a more robust indicator of large-fire behaviour.

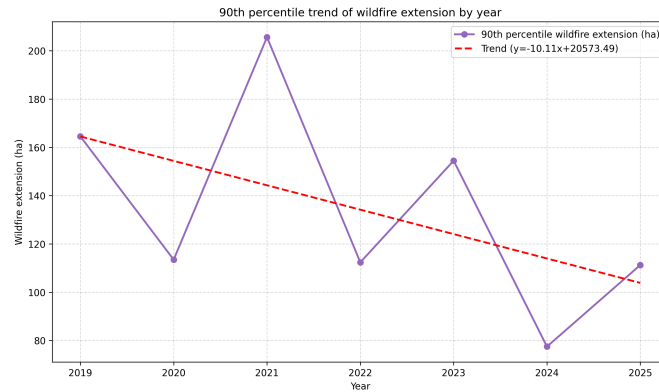


Figure 2.10. 90° percentile extension of wildfires

The temporal evolution of the 90th percentile shows a decreasing trend over time, indicating a reduction in the magnitude of larger wildfire events over the considered period.

An exception is observed in 2021, where the 90th percentile is substantially higher than in all other years. This reinforces the evidence that 2021 was characterised not only by a high number of wildfire events and a large total burned area, but also by unusually large individual fires compared to the rest of the period.

2.3.4 Spatial expansion of the phenomenon

As a final exploratory analysis on the aggregated dataset, the spatial expansion of wildfire activity was investigated by assessing whether fires tend to recur in the same administrative areas or progressively affect new locations over time.

For this purpose, the number of affected administrative units was computed for each year at three spatial levels: municipalities, provinces and regions.

The results show an overall increase across all three spatial scales over the study period, indicating a broader geographic distribution of wildfire occurrence in the later years of the analysis.

At the municipal level, the temporal pattern closely mirrors that of the total number of wildfire events, reflecting the finer spatial granularity of this unit of analysis. In this context, 2021 emerges as the year with the largest spatial expansion, affecting the highest number of distinct municipalities.

At broader administrative levels, different peak years are observed. The year 2022 shows the maximum spatial extent in terms of both provinces and regions affected by wildfires. Notably, in 2022 all Italian regions experienced at least one wildfire event, indicating a nationwide spatial coverage for that year.

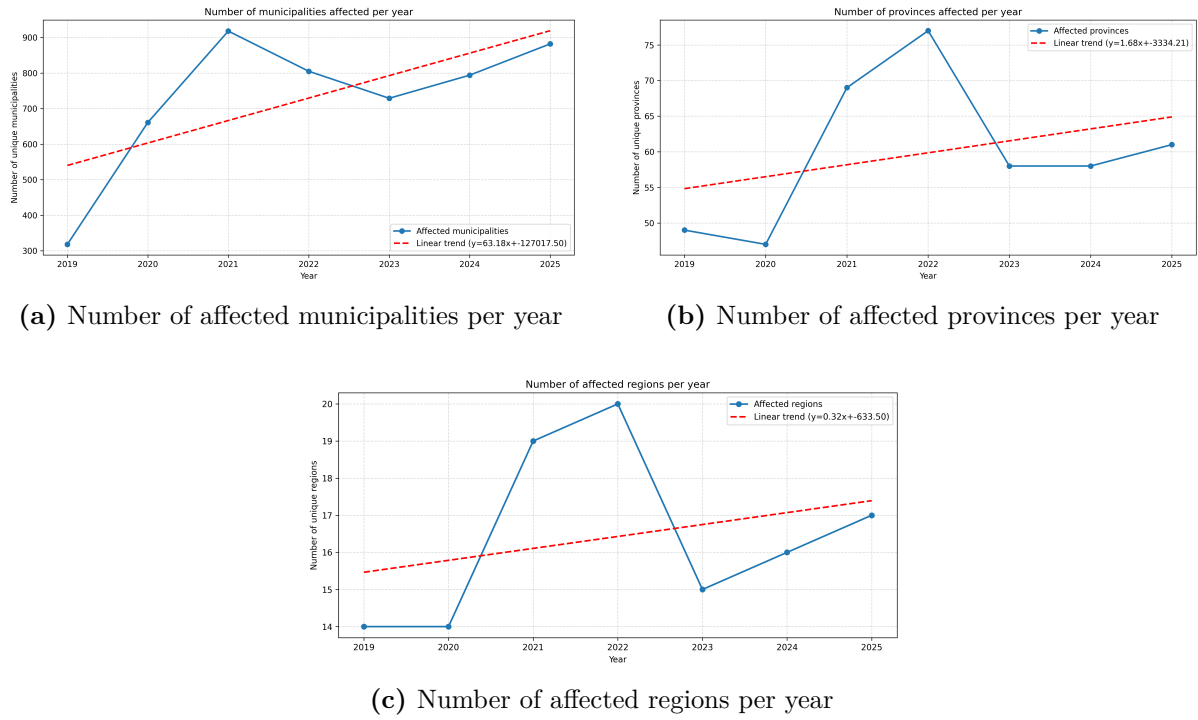


Figure 2.11. Spatial extent of wildfires in Italy over time

2.4 Ecoregions

After describing wildfire activity at the national scale, the analysis is further refined by investigating its spatial distribution across the main Italian ecoregions. This allows the assessment of whether wildfire occurrence differs among ecologically homogeneous areas characterised by distinct environmental and biogeographical conditions.

Ecoregions are relatively homogeneous territorial units within which species and natural communities interact consistently with the physical characteristics of the environment. They therefore represent areas with similar ecological potential and provide an appropriate spatial framework for interpreting ecological processes, disturbance regimes, vegetation patterns and landscape structure [19].

The Italian ecoregion classification adopted in this study was provided by ISTAT and is available at: https://download.mase.gov.it/Natura2000/Lista%20Rossa%20degli%20Ecosistemi%20Italiani/_prodotto_D/.

Five ecoregions were considered, together covering the entire Italian territory:

- **Alpine Ecoregion** (18.09%)
- **Po Valley Ecoregion** (16.64%)
- **Apennine Ecoregion** (28.09%)
- **Tyrrhenian Ecoregion** (28.36%)
- **Adriatic Ecoregion** (8.82%)

Figure 2.12 illustrates the spatial distribution of the five ecoregions across Italy, which constitute the geographical framework adopted for the following analyses.



Figure 2.12. Ecoregions map

2.4.1 Total number of wildfires

The analyses presented in this section were performed using the non-aggregated version of the BAD-ITE dataset. Burned-area polygons were spatially intersected with the ecoregion layer, after which the resulting records were aggregated by both *EFFIS_ID* and ecoregion. Consequently, within each ecoregion, each record corresponds to a distinct wildfire event.

It is important to note that an individual wildfire may intersect more than one ecoregion. Since they are represented as georeferenced burned-area polygons, a single *EFFIS_ID* can therefore contribute to multiple ecoregions following the spatial intersection. This behaviour is intentional, as it allows each wildfire to be associated with every ecoregion affected by the event, regardless of the proportion of burned area falling within each category.

Figure 2.13 reports the cumulative number of wildfire events recorded in each ecoregion.

The same colour scheme adopted in Figure 2.12 is retained throughout the following figures to facilitate interpretation.

The Tyrrhenian ecoregion clearly stands out as the most affected, with 7951 wildfire events, whereas all remaining ecoregions record fewer than 850 events, highlighting a marked imbalance in the spatial distribution of wildfire occurrence across the Italian ecoregions.

Figure 2.14 further illustrates the annual distribution of wildfire events. Panel (a) reports the absolute number of wildfires per ecoregion, while panel (b) shows the corresponding annual

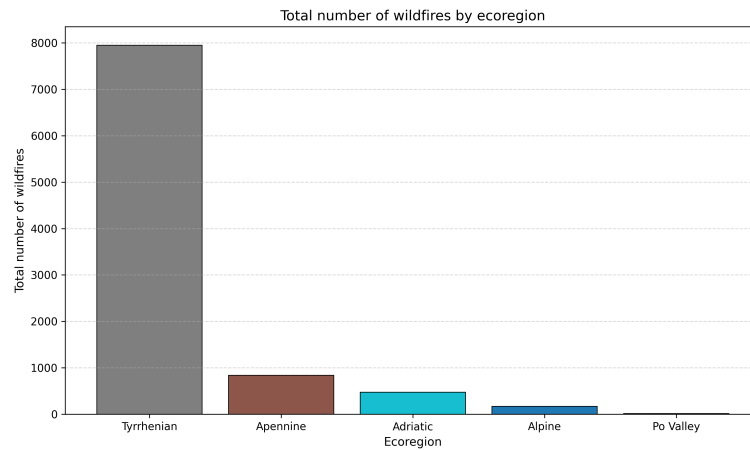


Figure 2.13. Number of wildfires in each ecoregion

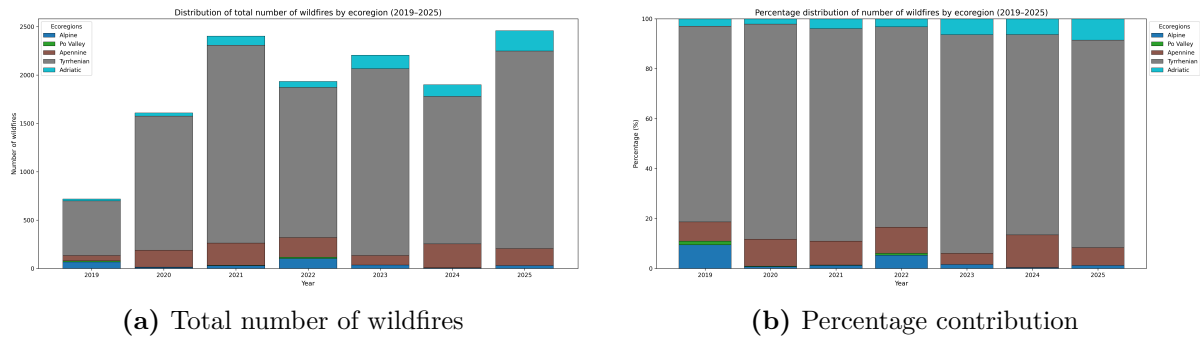


Figure 2.14. Total number of wildfires per year in each ecoregion

percentage contribution.

Overall, the relative contribution of each ecoregion remains fairly stable throughout the study period. The main exception is the Alpine ecoregion, which exhibits relatively higher wildfire occurrence in 2019 and 2022. A slight temporal variation is also observed for the Adriatic ecoregion, which shows a slight increase over the study period.

To further investigate these temporal patterns, annual wildfire counts were modelled separately for each ecoregion using Generalized Additive Models (GAMs). Unlike ordinary linear regression, GAMs do not impose a predefined functional form and allow the relationship between time and wildfire occurrence to be estimated flexibly through smooth functions. Given the limited number of annual observations, the smooth curves are intended exclusively as descriptive graphical summaries and are not interpreted inferentially. This approach is particularly useful for detecting gradual changes or non-linear temporal patterns that may not be adequately captured by a single linear trend. In addition, the fitted smooths are accompanied by 95% confidence intervals, providing a graphical assessment of the associated uncertainty.

The fitted GAMs suggest an increasing temporal pattern for both the Adriatic and Tyrrhenian ecoregions. The upward trend is more evident for the Adriatic ecoregion, whose fitted smooth is accompanied by a relatively narrow 95% confidence band, indicating lower uncertainty in the estimated temporal evolution. A similar increasing pattern is also observed for the Tyrrhenian ecoregion; however, it appears visually less pronounced because of the much larger scale of the annual wildfire counts. In addition, its confidence intervals are wider than those of the Adriatic

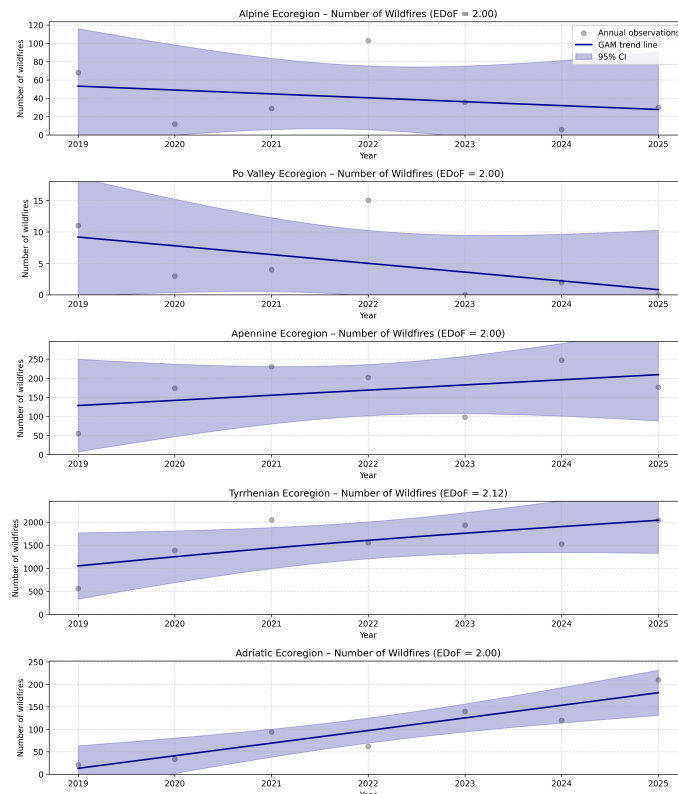


Figure 2.15. GAM function

ecoregion, reflecting greater uncertainty around the estimated trend. By contrast, the remaining ecoregions exhibit either relatively flat temporal patterns or larger year-to-year fluctuations, with no evident long-term increase over the study period.

2.4.2 Total burned area

The left panel in Figure 2.16 shows the cumulative total burned area (in hectares) for each ecoregion over the study period. The pattern is broadly consistent with that observed for the number of wildfire events, although a notable reordering between the Adriatic and Apennine ecoregions can be observed.

This suggests potential differences in the average size of wildfire events across ecoregions, with the Apennine area exhibiting a higher number of events but a comparatively lower total burned area. This aspect is further explored in subsequent sections.

The right panel in Figure 2.16 provides a normalised measure of burned area, which accounts for the different spatial extent of the ecoregions. Given the substantial differences in size among ecoregions, absolute burned area values are not directly comparable, as larger regions have a higher baseline probability of being affected. For this reason, the proportion of burned area is computed as: $\frac{\text{burned area in ecoregion}}{\text{total area of ecoregion}}$

This normalisation allows for a more meaningful comparison across regions.

Under this perspective, the Tyrrhenian ecoregion remains the most affected area, both in absolute and relative terms, with approximately 6.04% of its surface burned over the study

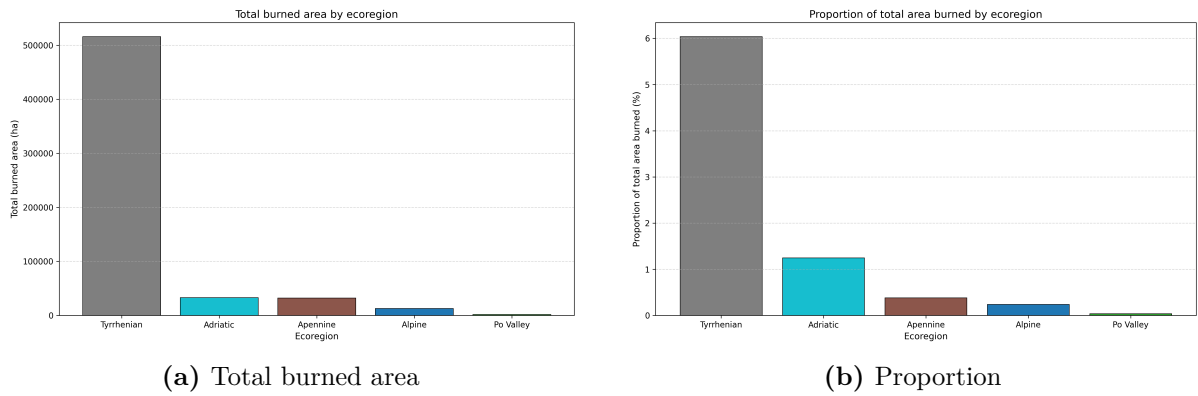


Figure 2.16. Total burned area per each ecoregion

period⁴. The Adriatic ecoregion follows with 1.25%, despite being the smallest in spatial extent. In relative terms, the Tyrrhenian ecoregion exhibits approximately 4.8 times higher burned-area proportion than the Adriatic ecoregion and about 15.8 times higher than the Apennine ecoregion.

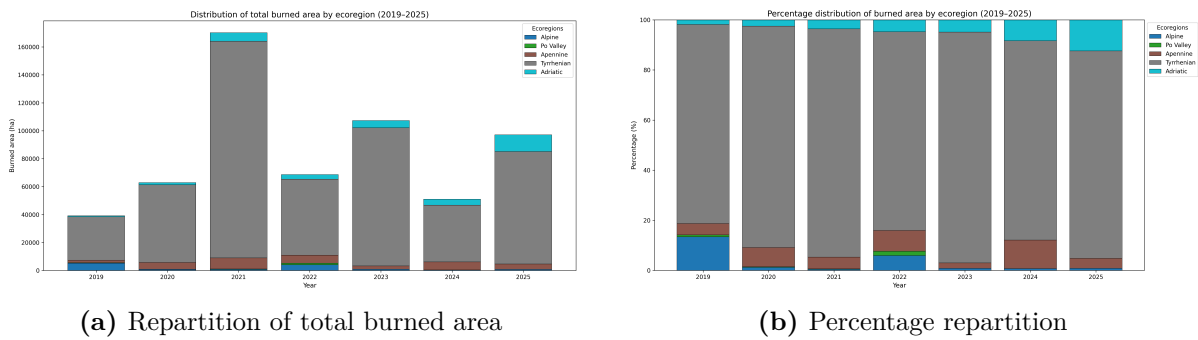


Figure 2.17. Repartition of total burned area per year in each ecoregion

Figure 2.17 illustrates the annual distribution of burned area across the five ecoregions, both in absolute terms and as a relative percentage contribution.

The observed pattern is broadly consistent with that previously described for the number of wildfire events, indicating a strong correspondence between the spatial distribution of fire occurrence and total burned area across ecoregions.

Annual trends in total burned area were further investigated for each ecoregion using Generalized Additive Models (GAMs), with 95% confidence intervals used to quantify the associated uncertainty.

The results indicate an increasing trend for the Adriatic ecoregion, similar to that observed for the total number of fires, accompanied by relatively narrow confidence intervals, suggesting a more stable and well-defined temporal pattern. In contrast, the remaining ecoregions exhibit either approximately flat trends or wider confidence bands, indicating higher variability and less pronounced temporal structure in burned area over time.

⁴It should be noted that the burned area is defined as the cumulative sum of wildfire polygons over the period 2019–2025. Therefore, it should not be interpreted as permanent land loss. In addition, the dataset does not account for post-fire recovery processes or fire severity.

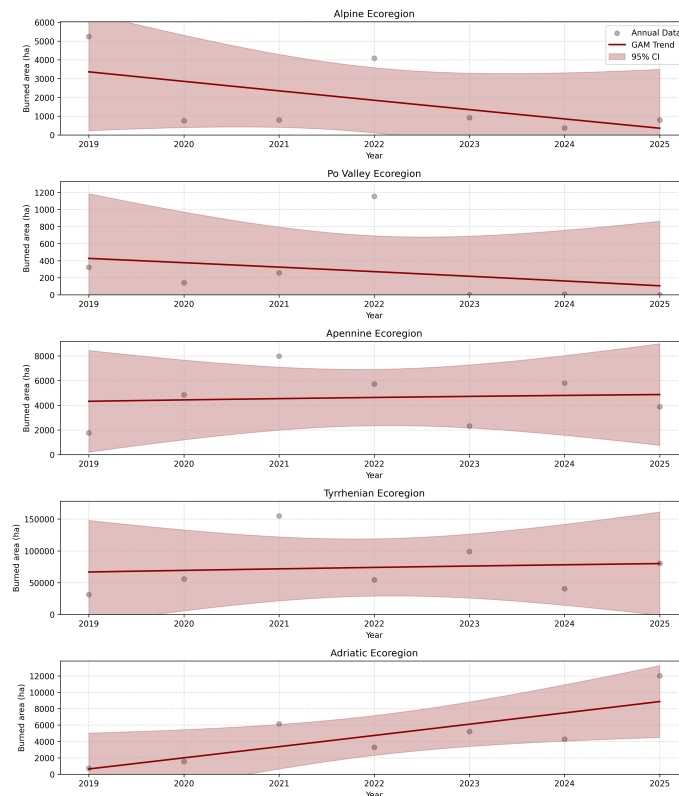


Figure 2.18. GAM function

2.4.3 Average wildfire extension

The size of individual wildfires was subsequently analysed across the five Italian ecoregions. Since wildfire areas exhibit highly skewed distributions with numerous extreme values, the median was considered the most informative measure of central tendency, although both the mean and the median are reported in Figure 2.19.

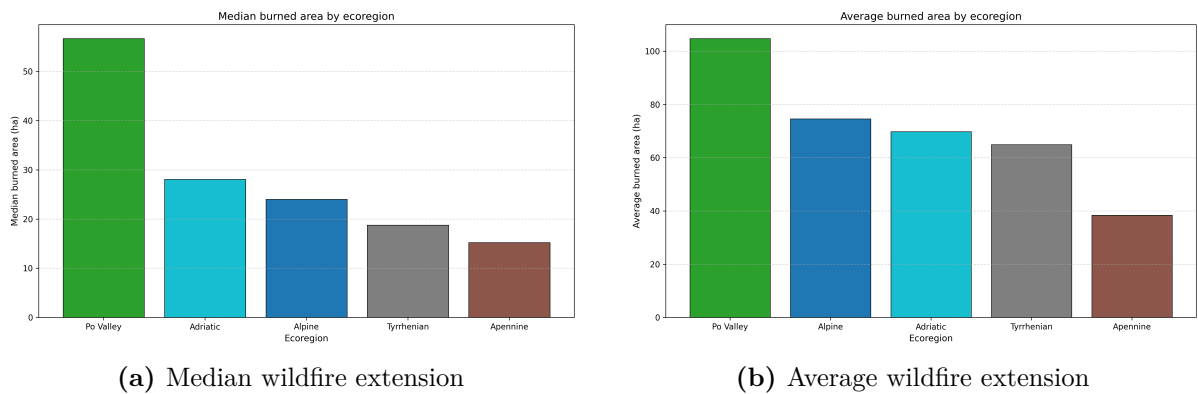


Figure 2.19. Median and average wildfire extension in each ecoregion

To assess whether wildfire extension differs among ecoregions, a non-parametric statistical approach was adopted. Overall differences were evaluated using the Kruskal–Wallis test⁵. This

⁵The Kruskal–Wallis test is the non-parametric analogue of one-way ANOVA and tests the null hypothesis that

choice is appropriate given the marked departure from normality observed in wildfire sizes.

A methodological consideration concerns the spatial overlay procedure used to assign wildfire events to ecoregions. Only 0.78% of wildfire events intersect two ecoregions, while no wildfire spans more than two ecological units. Consequently, although a small degree of dependence between observations may be introduced, the potential pseudo-replication is considered negligible and is not expected to materially affect the results of the statistical analyses.

Wildfire size differs substantially among ecoregions. The Apennine ecoregion exhibits the smallest wildfire extension, with a median burned area of approximately 15 ha. This confirms the pattern suggested by the previous analyses, namely that this ecoregion experiences relatively frequent wildfires that are, on average, smaller than those occurring elsewhere. The Kruskal–Wallis test, followed by Dunn’s post-hoc comparisons, indicates that wildfire extension in the Apennine ecoregion is significantly lower than in each of the remaining ecoregions.

In contrast, the Padanian ecoregion records the lowest number of wildfire events and the smallest cumulative burned area, both in absolute terms and relative to its surface. Nevertheless, when wildfires occur, they tend to be considerably larger than elsewhere, with a median burned area of approximately 56.6 ha, more than twice that observed in the Adriatic ecoregion (28 ha).

The Tyrrhenian ecoregion remains the most affected by wildfires in terms of both the cumulative burned area and the total number of events. However, despite its high wildfire frequency, the median wildfire size is among the lowest after the Apennine ecoregion, suggesting that its large cumulative burned area primarily reflects the occurrence of numerous individual events rather than exceptionally large fires.

The Adriatic ecoregion ranks second in terms of median wildfire size, and the pairwise comparisons indicate a statistically significant difference between the two coastal ecoregions (Tyrrhenian and Adriatic). Finally, the Alpine ecoregion shows intermediate wildfire sizes, with a median burned area of approximately 24 ha.

Across all ecoregions, mean wildfire sizes are consistently larger than the corresponding medians, reflecting the strong right-skewness of the distributions and the presence of a limited number of exceptionally large wildfire events.

all groups originate from the same underlying distribution. Because it does not identify which groups differ, significant results were followed by pairwise Dunn post-hoc comparisons with Holm correction for multiple testing.

2.5 Ecosystems

The same analytical framework was subsequently applied using the ecosystem classification instead of the ecoregional subdivision.

Unlike ecoregions, which partition the Italian territory into geographically contiguous ecological units, the ecosystem classification groups areas according to their dominant land cover, vegetation characteristics and ecological features. Consequently, the same ecosystem type may occur in geographically distant areas, providing a complementary perspective on wildfire occurrence across different environmental settings.

The ecosystem map was obtained from the official classification released by ISTAT and is available at https://download.mase.gov.it/Natura2000/Lista%20Rossa%20degli%20Ecosistemi%20Italiani/_prodotto_D/. The original dataset distinguishes 98 ecosystem classes, whose complete description is available in the official documentation⁶.

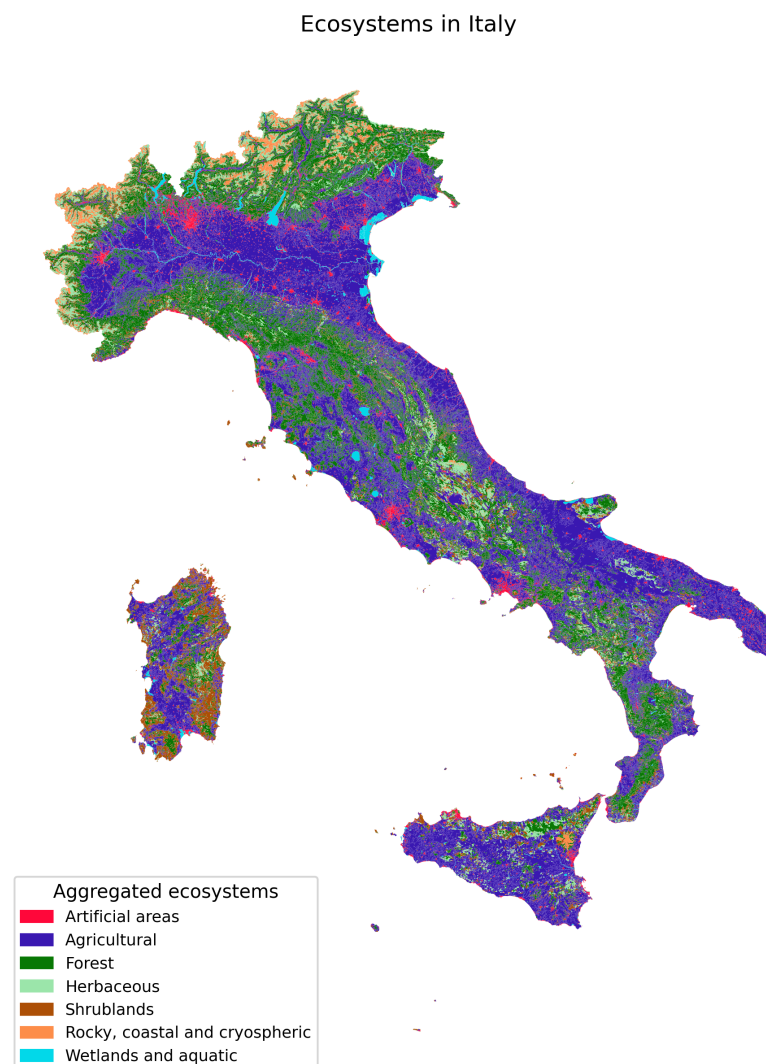


Figure 2.20. Ecosystems map

⁶https://download.mase.gov.it/Natura2000/Lista%20Rossa%20degli%20Ecosistemi%20Italiani/_prodotto_D/Legenda_Ecosistemi_V2.pdf.

For the purposes of this study, the original classes were aggregated into seven broad ecosystem categories:

- Artificial areas (5.50%)
- Agricultural (52.11%)
- Forest (26.20%)
- Herbaceous (5.77%)
- Shrublands (7.12%)
- Rocky, coastal and cryospheric (1.69%)
- Wetlands and aquatic (1.62%)

The analyses presented in the following sections were conducted using these aggregated ecosystem classes in order to facilitate interpretation while preserving the main ecological characteristics of the original classification.

As a first exploratory visualization, Figure 2.21 shows the wildfire polygons overlaid on the corresponding ecosystem map. This representation allows a qualitative assessment of the spatial correspondence between fire occurrences and ecosystem types, providing an initial overview of how the phenomenon is distributed across different environmental contexts.

Wildfires overlaid on Italian ecosystems (2019–2025)

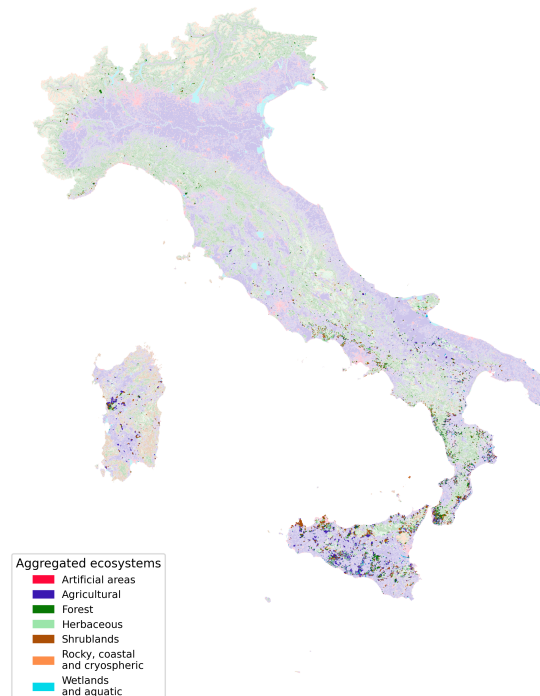


Figure 2.21. Wildfire polygons over ecosystems map

2.5.1 Number of wildfires

The analysis begins by considering the number of wildfires across ecosystem types. As in the case of ecoregions, it is important to note that the dataset is derived from a spatial overlay between wildfire polygons and ecosystem boundaries. Consequently, individual wildfire events may be counted more than once when intersecting multiple ecosystem classes. Given the inherently finer and more fragmented spatial structure of ecosystem units compared to ecoregions, this effect is expected to be more pronounced in the present case.

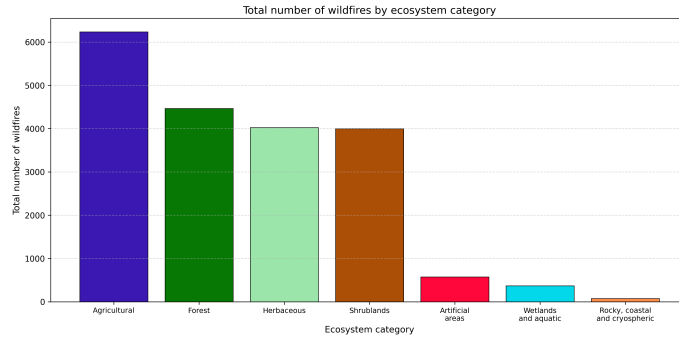


Figure 2.22. Total number of wildfires for each ecosystem

The agricultural ecosystem exhibits the highest number of wildfire occurrences (6233 events), followed by forest ecosystems (4463 events). Herbaceous and shrubland ecosystems show comparable numbers of events, while artificial areas, wetlands and aquatic systems, and rocky, coastal and cryospheric ecosystems are characterised by substantially lower wildfire frequencies. Figure 2.23 illustrates the annual distribution of wildfire events across ecosystems.

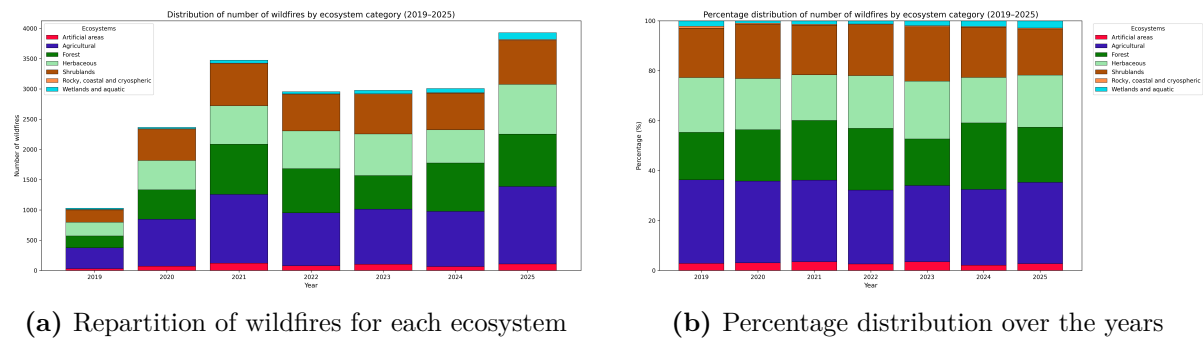


Figure 2.23. Total number of wildfires

Overall, the proportional contribution of each ecosystem remains relatively stable over time. This behaviour appears even more homogeneous than that observed for ecoregions, where certain classes exhibited more pronounced interannual variability. Given the absence of clear diverging patterns in the relative contributions of the different ecosystem classes over time, no Generalized Additive Models were fitted at this level of analysis. The proportional distributions remain broadly stable across the study period, suggesting that any apparent increase in absolute counts is largely driven by the overall temporal evolution of wildfire activity rather than by ecosystem-specific trends. For this reason, the analysis is kept at a descriptive level, focusing on cross-ecosystem comparisons rather than on formal modelling of temporal dynamics.

2.5.2 Total burned area

Panel (a) in Figure 2.24 reports the total burned area (in hectares) for each ecosystem type over the study period.

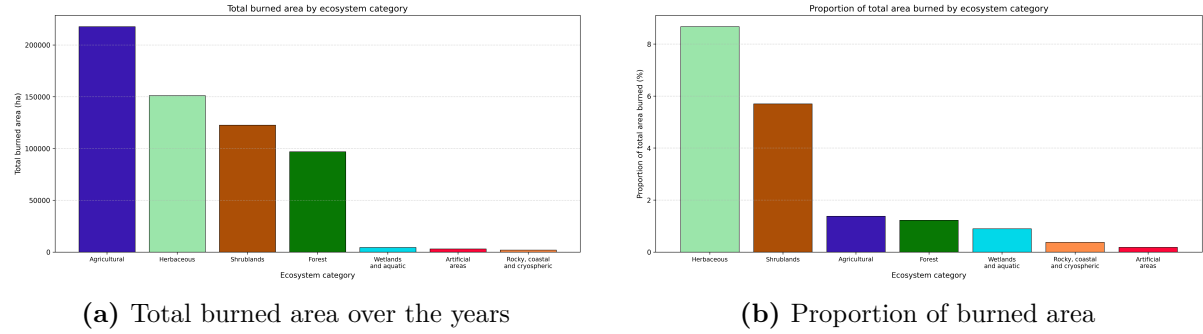


Figure 2.24. Total burned area

The agricultural ecosystem exhibits the largest total burned area (217670 hectares), followed by herbaceous ecosystems (151068 hectares).

To account for differences in ecosystem extent, a proportion of ecosystem area burned is computed as the ratio between burned area and total ecosystem area: $\frac{\text{burned area per ecosystem}}{\text{total ecosystem area}}$.

When this measure is considered, herbaceous ecosystems emerge as the most affected class, with 8.67% of their total area burned, followed by shrublands at 5.70%. Although agricultural ecosystems account for the largest absolute burned area, their relative vulnerability is comparable to that of forest ecosystems, which occupy an intermediate position.

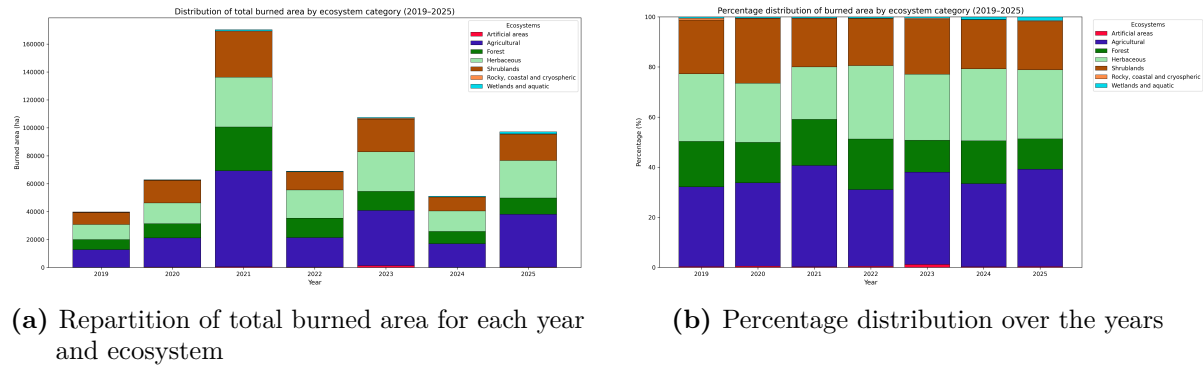


Figure 2.25. Repartition of total burned area

Figure 2.25 illustrates the annual distribution of burned area across ecosystems, both in absolute and relative terms. Overall, the proportional contribution of each ecosystem remains relatively stable over time, with only minor interannual fluctuations across categories.

2.5.3 Wildfire size within ecosystems

Although the spatial aggregation by ecosystems allows for a detailed characterization of burned areas, the analysis of wildfire size within ecosystem classes is not reported in this study.

This choice is motivated by the structure of the underlying dataset, which is derived from the spatial intersection between wildfire polygons and ecosystem boundaries. As a consequence,

approximately 78% of wildfire events intersect more than one ecosystem class, with an average of 2.10 intersected ecosystems and individual fires spanning up to six different categories.

In this context, the variable representing the burned area in each ecosystem does not represent the full size of a wildfire, but rather the portion of each fire falling within a given ecosystem. Therefore, the resulting values are strongly influenced by the spatial fragmentation of ecosystem boundaries and by the geometry of the intersection process, rather than by the intrinsic size distribution of wildfire events within each ecosystem.

For this reason, any statistical comparison of wildfire size across ecosystems would not provide a direct ecological interpretation of fire magnitude, but rather reflect the partitioning of fire polygons across heterogeneous spatial units.

Consequently, the analysis focuses on more robust and interpretable indicators, such as total burned area, relative burned area and wildfire frequency per ecosystem, which better capture differences in ecosystem exposure and vulnerability without introducing ambiguities related to spatial disaggregation.

Having described the main characteristics of the BAD-ITE dataset and the spatial and temporal patterns of wildfire occurrence, the next chapter focuses on the methodological framework adopted in this study. It introduces the statistical foundations of the modelling approach that will subsequently be used to develop the wildfire intensity index based on the BAD-ITE data analysed in this chapter.

Chapter 3

Methods: Point Processes and SPDE–INLA

3.1 Introduction to spatial data

The increasing availability of georeferenced information has led to a growing interest in spatial data analysis across a wide range of scientific disciplines. Advances in data acquisition technologies, computational resources and statistical methodologies have made it possible to explicitly incorporate the spatial location of observations into statistical analyses, providing a more comprehensive understanding of complex natural and human-related processes. Today, spatial data play a fundamental role in numerous fields, including environmental sciences, climatology, ecology, public health, agriculture, urban planning and economics [20].

Spatial data consist of observations associated with a specific geographical location within a study region. Unlike classical statistical data, spatial observations are generally not independent, since nearby locations often exhibit similar characteristics as a consequence of underlying environmental or social processes. This phenomenon, commonly referred to as spatial dependence or spatial autocorrelation, represents one of the defining features of spatial data and motivates the use of dedicated statistical models capable of accounting for spatial correlation.

Spatial information can be obtained from a variety of sources. Satellite imagery and other remote sensing technologies provide continuous measurements of environmental variables such as land cover, vegetation indices, temperature and precipitation over large spatial domains. Ground-based monitoring stations collect observations of climatic and environmental variables, including air temperature, rainfall and air pollution, at specific locations. Additional sources include administrative records, field surveys, census data, mobile phone positioning data and social media platforms, all of which provide georeferenced information describing different environmental, ecological and socio-economic phenomena [21].

Depending on the nature of the spatial variable and the observation process, spatial data are commonly classified into three main categories: areal data, point-referenced (or geostatistical) data and point pattern data.

Areal data

They arise when observations are associated with predefined spatial units, such as municipalities, administrative regions or cells of a regular grid. In this case, the study domain is partitioned into a finite set of non-overlapping areas, and each observation refers to an entire spatial unit rather

than to an exact location. Areal data frequently originate from the aggregation of individual observations or from raster-based products derived from remote sensing.

Point-referenced (or geostatistical) data

They consist of measurements collected at a finite number of known spatial locations within a continuous study domain. Here, the variable of interest is observed at selected sampling sites, while its value at unsampled locations is unknown and can be predicted by exploiting the spatial correlation between observations. Typical examples include measurements of air pollution, precipitation or temperature collected by environmental monitoring stations.

Point pattern data

They differ fundamentally from the previous two categories because the observations themselves correspond to the spatial locations of randomly occurring events. Rather than measuring a variable at predetermined locations, the primary information consists of the geographical positions where events occur. The objective is therefore to characterise the underlying spatial process generating the observed pattern and to determine whether the events exhibit complete spatial randomness, clustering or inhibition. In some applications, additional information may be associated with each event, giving rise to marked point patterns.

The present work focuses on this third class of spatial data. Specifically, wildfire occurrences are represented as a spatial point pattern, where each event is identified by its geographical location within the study region. This representation provides the basis for the point process modelling application that will be introduced in the following chapters.

3.2 Spatial point pattern

Spatial point processes are statistical models designed to describe the randomness associated with the spatial locations of observed events. Given a bounded region ($D \subset \mathbb{R}^2$), a realisation of a spatial point pattern can be represented as a set of locations $S = \{s_1, s_2, \dots, s_n\}$, where both the number of points (n) and their spatial coordinates ($s_i \in D$) are treated as random variables generated by an underlying stochastic mechanism [20].

In practical applications, the study region (D) is typically a fixed and well-defined geographical domain, such as a forest, a country or an administrative area. This setting is particularly relevant in applied spatial analysis, as it naturally allows the incorporation of covariate information defined over the entire domain.

A central idea in spatial point process theory is that the observed configuration of points should be interpreted in relation to a reference model of spatial randomness. The most basic benchmark is given by *Complete Spatial Randomness (CSR)*, which assumes that events occur independently and uniformly over the study domain. Under CSR, there is no underlying spatial structure governing the locations of events and deviations from this assumption indicate the presence of spatial dependence. In most real applications, CSR is too restrictive and the main objective of modelling is precisely to characterise and explain such deviations from randomness. Departures from CSR may arise due to the influence of environmental conditions, geographical constraints or interactions between events.

From a probabilistic perspective, a spatial point pattern can be viewed as a complex random object defined over the space of all possible configurations of points within (D). Direct specification of its full probability distribution is generally intractable. For this reason, it is often more

convenient to adopt a generative formulation. In this framework, a spatial point pattern can be described through two components: a probability distribution for the number of events in the region $N(D)$ and a mechanism describing the joint spatial distribution of event locations conditional on $N(D) = n$.

Because points are unordered, the joint distribution of their locations is symmetric when describing the joint configuration of points within the bounded domain.

Although a full formalisation of this distribution is complex, this decomposition highlights the key structure underlying spatial point processes: randomness in both the number of events and their spatial arrangement. An alternative and particularly useful representation of a spatial point pattern is given in terms of counting measures. For any region B within the study domain D , the random variable

$$N(B) = \sum_{i=1}^n \mathbf{1}\{s_i \in B\}$$

denotes the number of events occurring inside B , where $B \subset D$, $n = N(D)$ and $\mathbf{1}(\cdot)$ is the indicator function. This formulation characterises the point pattern in terms of the number of events observed over different regions of the study domain, rather than by their individual locations. The two representations are equivalent, since the collection of counts over all regions uniquely determines the underlying point pattern. This counting measure provides the mathematical foundation for defining spatial point process models, as discussed in the following section.

3.2.1 Intensity function

The first-order intensity function $\lambda(s)$ of a spatial point process is defined such that, for a small region ds around location s , $\lambda(s)ds$ approximates the expected number of events occurring within that region. More generally, for any bounded subset $B \subset D$,

$$\mathbb{E}\{N(B)\} = \int_B \lambda(s) ds.$$

The intensity therefore represents the expected number of events per unit area and constitutes the fundamental quantity of interest in spatial point process modelling. Different point process models mainly differ in the assumptions they make about the form of the intensity function and the dependence structure between events.

3.3 Poisson processes

The Poisson process represents the fundamental probabilistic model for spatial point patterns and provides the natural starting point for modelling the occurrence of events in space [20]. Its defining assumption is that the number of events observed within any region follows a Poisson distribution. The spatial distribution of events is governed by the intensity function, which determines the expected number of events per unit area.

Depending on whether the intensity function is assumed to be constant or allowed to vary over space, two main classes of models can be distinguished: the Homogeneous Poisson Process (HPP) and the Non-Homogeneous Poisson Process (NHPP).

Figure 3.1 illustrates the difference between these two processes using simulated point patterns.

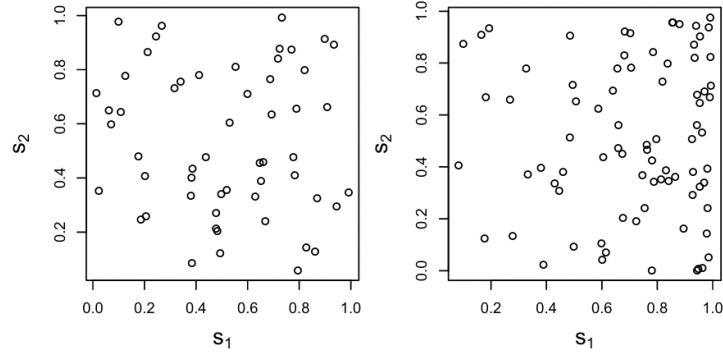


Figure 3.1. Simulated point patterns from (left) a homogeneous Poisson process (HPP), (right) a non-homogeneous Poisson process (NHPP), source:[20]

3.3.1 Homogeneous Poisson Processes (HPP)

The simplest spatial point process is the Homogeneous Poisson Process (HPP), which assumes complete spatial randomness (CSR). Under this model, events occur independently of one another and are equally likely to occur anywhere within the study region. Consequently, the expected number of events is proportional only to the area considered and does not depend on location.

Formally, for any bounded subset $B \subset D$,

$$N(B) \sim \text{Poisson}(\lambda|B|),$$

where $|B|$ denotes the area of region B and $\lambda > 0$ is constant over the entire domain.

Under this assumption, the intensity function is indeed spatially constant,

$$\lambda(s) = \lambda,$$

so every location has the same expected event rate. Because event locations are also conditionally independent, the HPP provides the classical definition of CSR.

3.3.2 Non-Homogeneous Poisson Processes (NHPP)

Unlike the HPP, the NHPP preserves the assumption of conditional independence between event locations, while allowing the expected event density to vary across the study region. The Homogeneous Poisson Process can therefore be generalized by allowing the intensity function to vary with location, leading to the Non-Homogeneous Poisson Process (NHPP).

$$N(B) \sim \text{Poisson} \left(\int_B \lambda(s) ds \right).$$

The intensity function $\lambda(s)$ is no longer constant but varies continuously over space, allowing different regions to exhibit different expected event rates. It is indeed evident also from the simulation in Figure 3.1 that areas where the intensity is high tend to produce more points, and areas where the intensity is low tend to produce fewer points. A common specification assumes that the logarithm of the intensity is a linear function of spatially referenced covariates,

$$\log \lambda(s) = \beta_0 + \mathbf{x}(s)^T \boldsymbol{\beta},$$

where $\mathbf{x}(s)$ denotes the vector of covariates observed at location s and β the corresponding regression coefficients. Under this formulation, changes in the covariates explain systematic spatial variation in the expected occurrence of events.

Likelihood

Inference for Poisson point processes is based on the likelihood of the observed event locations. Let $S = \{s_1, \dots, s_n\}$ denote the observed point pattern within the study region D . The likelihood is given by

$$L(\lambda; S) = \exp \left\{ - \int_D \lambda(u) du \right\} \prod_{i=1}^n \lambda(s_i).$$

The first term represents the probability of observing no additional events over the study region, whereas the second term evaluates the intensity at each observed event location. Consequently, the likelihood depends on the entire intensity surface over the domain rather than only on the observed points. As a consequence, inference requires evaluating the integral of the intensity over the whole observation domain, a quantity that generally has no closed-form expression and is therefore approximated numerically.

Although the NHPP allows the intensity to vary across space through observed covariates, it still assumes that, conditional on the intensity function, event locations are independently distributed. In many environmental applications, however, important drivers of the spatial process are either unavailable or only partially observed. Consequently, residual spatial dependence may remain even after accounting for the available covariates.

3.4 Cox processes

Cox processes overcome the limitations of the Poisson framework by allowing the intensity function itself to be random. This represents a key generalisation compared to the Non-Homogeneous Poisson Process, where spatial variation is fully explained through a deterministic function of location or covariates. In a Cox process, the event intensity is therefore modelled as a stochastic quantity varying across space. More precisely, the intensity is represented as a random field defined over the spatial domain D , which introduces an additional layer of randomness beyond the Poisson assumption.

A random field is a collection of random variables indexed by space. Formally, given a probability space (Ω, F, P) , a random field $T(s)$, with $s \in R^d$, is a function such that for every fixed location s , $T(s)$ is a random variable on that probability space [22]. In other words, each spatial location is associated with a random quantity, and the entire field describes how these random quantities are jointly distributed over space.

Among random fields, a particularly important class is that of Gaussian random fields.

A random field $T(s)$ is Gaussian if any finite collection of values $(T(s_1), \dots, T(s_m))$ follows a multivariate normal distribution. This property implies that the field is fully characterised by its mean function and covariance structure, making Gaussian fields particularly convenient for statistical modelling.

In spatial statistics, Gaussian random fields are widely used to describe latent spatial structure, as they provide a flexible way to model smooth spatial variation and correlation between nearby locations.

Conditional on a given realisation of the intensity surface, the point process is a Poisson process. This construction is often referred to as a doubly stochastic process, since randomness operates both at the level of the intensity and at the level of point occurrence.

A particularly important subclass is obtained when the logarithm of the intensity is modelled as a Gaussian random field. This leads to the Log-Gaussian Cox Process (LGCP).

3.4.1 Log-Gaussian Cox Processes (LGCP)

Log-Gaussian Cox processes (LGCPs) constitute a flexible class of stochastic models for spatial point patterns characterised by complex spatial heterogeneity and dependence structures. They can be interpreted as a specific subclass of Cox processes, obtained when the underlying random intensity is driven by a Gaussian random field.

In this framework, the intensity function $\lambda(s)$ is no longer a deterministic function of spatial covariates, but a stochastic process defined over the spatial domain. In particular, let $Z(s)$ denote a Gaussian random field indexed by location $s \in D$, representing the latent log-intensity of the process. The observed intensity is then obtained through an exponential transformation, ensuring positivity:

$$\lambda(s) = \exp\{Z(s)\}.$$

When considering covariates, the structure becomes:

$$Z(s) = \eta(s) = \beta_0 + x(s)^\top \beta + w(s),$$

where $\eta(s)$ is the latent linear predictor.

This specification implies that the spatial variability of the point pattern is partly explained by $w(s)$, which is structured latent effects captured by the Gaussian field, allowing for smooth spatial dependence and unobserved heterogeneity.

Let S denote the resulting point process on D , conditional on a realisation of the latent field $Z(s)$, the process is an inhomogeneous Poisson process with intensity function $\lambda(s)$. Marginally, however, the model inherits additional variability compared to a standard NHPP, allowing for clustering and over-dispersion induced by the latent spatial structure.

This makes the LGCP a particularly suitable modelling framework for the present application on wildfire occurrences, where part of the spatial variation is expected to be explained by available environmental covariates, while additional unobserved or unmeasured factors may induce residual spatial heterogeneity in the observed event pattern.

3.5 Bayesian inference of LGCP and INLA

In Bayesian inference for complex spatial models, such as point processes and Cox processes, posterior distributions are typically analytically intractable, as they require the evaluation of high-dimensional integrals.

The most widely used general-purpose approach to address this problem is the Monte Carlo Markov Chain (MCMC) framework. MCMC methods approximate posterior distributions through stochastic sampling schemes and are extremely flexible, allowing inference for a very broad class of models. However, their main limitation is computational cost, which can become prohibitive for large spatial datasets or models involving latent fields with high dimensionality. To overcome these limitations, a computationally efficient alternative has been developed: Integrated Nested Laplace Approximation (INLA).

INLA is specifically designed for the class of latent Gaussian models and provides a deterministic approximation to posterior marginals. Rather than targeting the full joint posterior distribution, INLA focuses directly on marginal posterior distributions of parameters and latent effects, making it particularly suitable for spatial hierarchical models.

A key feature of INLA is that it is built upon the concept of latent Gaussian models, which naturally include spatial and temporal dependence structures through an unobserved stochastic field. Both Cox processes and Log-Gaussian Cox processes fall within this modelling framework [23].

3.5.1 Latent Gaussian Models

Latent Gaussian Models (LGMs) constitute the class of Bayesian hierarchical models for which INLA is applicable. They are composed of three main components: a likelihood function, a latent Gaussian field and a set of hyperparameters.

Let $y = \{y_1, \dots, y_n\}$ denote the observed data, a fundamental assumption is that observations are conditionally independent given a latent Gaussian structure. The dependence among observations is therefore entirely captured by the latent field.

The model is defined through a linear predictor of the form:

$$\eta_i = \mu + \sum_j \beta_j z_{ij} + \sum_k w_k f^k(u_{ik}), \quad (3.1)$$

where:

- μ is the intercept,
- z_{ij} are observed covariates with linear effects β_j ,
- $f^k(\cdot)$ are flexible latent components used to represent nonlinear effects, spatial effects or other structured random effects,
- w_k are known weights.

The latent Gaussian field is then defined as the collection of all unknown components in the linear predictor, typically written as: $x = \{\eta, \mu, \beta, f^1, f^2, \dots\}$, which is assigned a multivariate Gaussian prior structure.

Conditionally on the latent field and hyperparameters θ , the model can be written as:

$$\begin{aligned} y|x, \theta &\sim \prod_i \pi(y_i|\eta_i, \theta) \\ x|\theta &\sim \mathcal{N}(0, Q^{-1}(\theta)) \\ \theta &\sim \pi(\theta) \end{aligned}$$

where $Q(\theta)$ is the precision matrix¹ of the latent Gaussian field. The structure of $Q(\theta)$ is crucial: in spatial applications it is typically sparse², which allows efficient computation within INLA. The posterior distribution of the latent field and hyperparameters is then given by:

¹The precision matrix is the inverse of the covariance matrix.

²A sparse matrix is a matrix where the vast majority of the elements are zero.

$$\pi(x, \theta|y) \propto \exp \left(-\frac{1}{2} x^T Q(\theta) x + \sum_i \log \pi(y_i|\eta_i, \theta) + \log \pi(\theta) \right). \quad (3.2)$$

Although this joint posterior $\pi(x, \theta|y)$ is high-dimensional and not analytically tractable, INLA does not attempt to compute it directly. Instead, it focuses on approximating marginal posterior distributions of interest, such as the posterior of the latent field $\pi(x_i|y)$ and the hyperparameters $\pi(\theta_j|y)$, which are typically sufficient for inference in spatial models [23].

3.5.2 INLA computational approach

The objective of the INLA framework is to approximate the posterior marginal distributions of the latent Gaussian model in a computationally efficient manner. Rather than estimating the full joint posterior distribution, INLA focuses directly on the posterior marginals of the hyperparameters and of the latent field, which are generally the quantities of primary inferential interest [23].

Specifically, inference aims at approximating the posterior marginals of the hyperparameters,

$$\pi(\theta_j|y) = \int \pi(\theta|y) d\theta_j \quad (3.3)$$

and of the latent field,

$$\pi(x_i|y) = \int \pi(x_i|\theta, y) \pi(\theta|y) d\theta \quad (3.4)$$

The main computational challenge is that the latent field x is typically very high-dimensional, making direct integration computationally infeasible. In contrast, the vector of hyperparameters θ is usually of much lower dimension. The key idea underlying INLA is therefore to avoid integration over the latent field by constructing accurate approximations to both the posterior distribution of the hyperparameters $\pi(\theta|y)$ and the full conditional density $\pi(x_i|\theta, y)$.

The first step consists of approximating the posterior distribution of the hyperparameters. Starting from Bayes' theorem,

$$\pi(\theta|y) = \frac{\pi(x, \theta|y)}{\pi(x|\theta, y)} \propto \frac{\pi(y|x, \theta) \pi(x|\theta) \pi(\theta)}{\pi(x|\theta, y)}, \quad (3.5)$$

the numerator can be evaluated directly, whereas the denominator is generally unavailable in closed form. INLA approximates (3.5) at a specific value θ^k of the hyperparameters vector and replaces the denominator with a Gaussian approximation centred at the mode of the conditional posterior distribution of the latent field, obtaining:

$$\tilde{\pi}(\theta^k|y) \propto \frac{\pi(y|x, \theta^k) \pi(x|\theta^k) \pi(\theta^k)}{\tilde{\pi}_G(x|\theta^k, y)}, \quad (3.6)$$

where $\tilde{\pi}_G(x|\theta^k, y)$ denotes the Gaussian approximation of the conditional posterior distribution. This approximation corresponds to a Laplace approximation of the marginal posterior distribution of the hyperparameters.

Once it has been approximated, the next step consists of approximating the conditional posterior distribution of each component of the latent field. To this end, INLA starts from the identity

$$\pi(x_i | \theta, y) = \frac{\pi(x | \theta, y)}{\pi(x_{-i} | x_i, \theta, y)} \propto \frac{\pi(y | x, \theta) \pi(x | \theta)}{\pi(x_{-i} | x_i, \theta, y)}, \quad (3.7)$$

where x_{-i} denotes the latent field excluding its i -th component. As in the previous step, the denominator is approximated using Laplace techniques. INLA provides three alternative approximations of $\pi(x_i|\theta, y)$: a Gaussian approximation, a full Laplace approximation and a simplified Laplace approximation. The latter improves the Gaussian approximation by correcting for location and skewness by a Taylor's series expansion around the mode of the Laplace approximation, while remaining considerably less computationally demanding than the full Laplace approximation. For this reason, it is adopted as the default option in the R-INLA implementation, but all the three are available.

Finally, the posterior marginals of the latent field are obtained by numerically integrating over the hyperparameter space,

$$\tilde{\pi}(x_i|y) = \sum_{k=1}^K \tilde{\pi}(x_i|\theta^k, y) \tilde{\pi}(\theta^k|y) \Delta_k, \quad (3.8)$$

where $\{\theta^1, \dots, \theta^K\}$ are support points selected in regions of high posterior density and Δ_k are the corresponding numerical integration weights.

The name *Integrated Nested Laplace Approximation* reflects the computational strategy adopted by the algorithm. The term *nested* refers to the successive Laplace approximations used to approximate first the posterior distribution of the hyperparameters and subsequently the conditional posterior distributions of the latent field. The term *integrated* refers to the final numerical integration over the hyperparameter space through which the marginal posterior distributions are obtained.

The overall INLA computational scheme can therefore be summarized as follows:

1. Approximate the posterior distribution of the hyperparameters $\pi(\theta|y)$, find the mode and identify a set of support points $\{\theta^1, \dots, \theta^K\}$ in regions of high density of $\tilde{\pi}(\theta|y)$;
2. Evaluate the approximated posterior density $\tilde{\pi}(\theta^k|y)$ at each support point using (3.6);
3. For each selected value θ^k , approximate the conditional posterior distribution $\pi(x_i|\theta^k, y)$ using either the Gaussian, Laplace or simplified Laplace approximation;
4. Obtain the posterior marginals of the latent field by numerically integrating over the hyperparameter space using the equation (3.8).

3.5.3 SPDE approximation

A key component of the Log-Gaussian Cox Process is the latent Gaussian field, which models the logarithm of the intensity surface continuously over the spatial domain. Although this formulation provides considerable modelling flexibility, direct Bayesian inference for continuous Gaussian random fields is computationally demanding. In particular, inference based on the covariance matrix of the field becomes impractical for large datasets, since covariance matrices are typically dense³ and their manipulation requires substantial computational resources.

To overcome this limitation, Lindgren et al. [24] introduced the Stochastic Partial Differential Equation (SPDE) approach, establishing an explicit connection between Gaussian random fields with Matérn covariance and Gaussian Markov Random Fields (GMRFs). Rather than modelling the covariance function directly, the Gaussian field is represented as the solution of a stochastic partial differential equation. This representation makes it possible to approximate

³A dense matrix is a matrix in which most of the elements are non-zero.

the continuous field with a GMRF whose precision matrix is sparse, allowing efficient Bayesian inference through INLA.

From the Matérn covariance to the SPDE representation

The Matérn covariance family is one of the most widely adopted covariance models in spatial statistics because it provides a flexible description of spatial dependence while allowing direct interpretation of its parameters. An important theoretical result due to Whittle, later exploited by Lindgren et al. [24], shows that a Gaussian random field with Matérn covariance can be equivalently represented as the solution of the stochastic partial differential equation

$$(\kappa^2 - \Delta)^{\alpha/2} (\tau\omega(s)) = W(s), \quad (3.9)$$

where Δ denotes the Laplacian operator, $W(s)$ is Gaussian spatial white noise, $\kappa > 0$ is a scale parameter controlling the spatial correlation, $\tau > 0$ regulates the marginal variability of the field and $\alpha = \nu + d/2$ determines its smoothness.

The importance of this result is primarily computational rather than analytical. Instead of specifying and manipulating the covariance matrix of the Gaussian field, inference can be performed through the SPDE representation, which naturally leads to a sparse Gaussian Markov Random Field approximation.

Mesh representation of the latent field

The SPDE does not admit a closed-form solution over an arbitrary spatial domain and therefore requires a numerical approximation. This is achieved by discretising the study region through an irregular triangular mesh, which provides the computational support for the latent Gaussian field.

The transition from regular grids to irregular meshes represents one of the main advantages of the SPDE approach [25]. Traditional lattice-based approximations require the study region to be partitioned into cells of equal size, forcing the same spatial resolution throughout the entire domain. This may result in an unnecessarily large number of cells in areas where little spatial detail is needed, while failing to adequately capture local variability in regions where observations are highly concentrated. In contrast, an irregular mesh partitions the spatial domain into non-overlapping triangles whose vertices define the locations at which the latent field is represented.

The flexibility of the triangular mesh offers several practical advantages. First, the mesh resolution can be adapted to the spatial distribution of the data: smaller triangles can be used in regions where observations are dense or where rapid spatial variation is expected, whereas larger triangles can be employed in sparsely sampled areas, substantially reducing the computational burden. Second, irregular meshes naturally accommodate complex geographical boundaries, such as coastlines or administrative borders, avoiding the staircase effect typically produced by regular grids. Finally, the mesh is usually extended beyond the study region through an outer boundary composed of relatively large triangles. This extension reduces the influence of artificial boundary conditions imposed by the SPDE and limits edge effects on the area where inference is performed [24].

Once the mesh has been constructed, the continuous latent Gaussian field ($\omega(s)$) is approximated through a finite-dimensional basis expansion,

$$\omega(s) \approx \sum_{k=1}^{n_v} \psi_k(s) \omega_k, \quad (3.10)$$

where n_v is the number of mesh vertices, $\psi_k(s)$ are piecewise linear basis functions defined on the vertices of the triangulation and ω_k are Gaussian-distributed weights associated with each vertex.

Each basis function is equal to one at its corresponding mesh vertex, decreases linearly over the neighbouring triangles and is equal to zero outside its local neighbourhood. Consequently, the latent field is represented as a smooth interpolation of the values assigned to the mesh vertices, so that inference is reduced to estimating the finite vector of weights ω rather than the field at every location in the spatial domain.

A fundamental consequence of the SPDE representation is that the vector of weights forms a Gaussian Markov Random Field (GMRF). Owing to the local support of the basis functions, each weight depends directly only on neighbouring vertices, inducing conditional independence relationships among distant nodes. As a result, the corresponding precision matrix is sparse, dramatically reducing both memory requirements and computational cost. This sparse representation is the key feature that allows INLA to perform efficient Bayesian inference for spatial models defined on continuous domains.

Spatial hyperparameters

Although the SPDE formulation is expressed in terms of the parameters κ and τ , spatial dependence is generally interpreted through two more intuitive hyperparameters: the spatial range ρ and the marginal standard deviation σ .

The spatial range represents the distance beyond which spatial correlation becomes negligible. Locations separated by distances smaller than the range tend to exhibit similar values of the latent field, whereas locations farther apart become approximately independent. Assuming the commonly adopted Matérn model with $\alpha = 2$, the range is related to the SPDE parameter κ through

$$\rho = \frac{\sqrt{8\nu}}{\kappa}. \quad (3.11)$$

The marginal standard deviation controls the overall variability of the latent Gaussian field and therefore determines the magnitude of spatial fluctuations in the log-intensity surface. Under the same parameterisation,

$$\sigma = \frac{1}{\sqrt{4\pi\kappa\tau}}. \quad (3.12)$$

These two quantities provide a direct interpretation of the latent spatial effect and are generally the hyperparameters reported when fitting spatial models using the SPDE approach [24].

Prior specification

Within the Bayesian framework, prior distributions must be assigned to the spatial hyperparameters. In modern implementations of the SPDE approach, including R-INLA, this is typically achieved through Penalized Complexity (PC) priors [26, 27].

PC priors provide a principled framework for prior specification in hierarchical Bayesian models. The central idea is to define each model component as an extension of a simpler reference model,

referred to as the *base model*. The prior is then constructed by measuring the deviation from this base model using the Kullback–Leibler divergence, which quantifies the loss of information incurred when approximating a flexible model with a simpler one. As a result, the prior assigns the highest probability to the simplest model configuration, corresponding to weak or absent spatial structure, while increasingly flexible spatial fields are progressively penalised through lower prior probability [27].

For Matérn Gaussian random fields represented through the SPDE formulation, this general construction can be expressed in terms of the spatial hyperparameters governing the latent field [26]. In particular, the base model for the marginal standard deviation corresponds to the absence of spatial variation ($\sigma = 0$), while the base model for the spatial range corresponds to an infinitely large range ($\rho \rightarrow \infty$), which implies a spatially constant latent field over the domain. This formulation induces shrinkage towards simpler spatial structures unless the data provide sufficient evidence for more complex dependence.

An important practical advantage of this approach is that the priors are expressed directly in terms of the interpretable spatial hyperparameters, namely the spatial range ρ and the marginal standard deviation σ , rather than the internal SPDE parameters κ and τ . They are specified through probability statements of the form

$$P(\rho < \rho_0) = p_\rho, \quad P(\sigma > \sigma_0) = p_\sigma,$$

where ρ_0 and σ_0 are user-defined thresholds reflecting prior knowledge about the expected spatial scale and variability of the process. These probability statements control the degree of regularisation while maintaining invariance to reparameterisation [26, 28].

Approximation of the likelihood

The SPDE representation not only provides an efficient approximation of the latent Gaussian field, but also makes it possible to approximate the likelihood of a Log-Gaussian Cox Process without discretising the observations onto a regular lattice.

It is convenient to work with the log-likelihood, which for a LGCP takes the form

$$\log L(\lambda; S) = \sum_{i=1}^n \log \lambda(s_i) - \int_D \lambda(s) ds, \quad (3.13)$$

where the first term accounts for the contribution of the observed events and the second term represents the integral of the intensity over the study domain, i.e. the expected number of events.

A key advantage of the SPDE framework is that the latent field is defined continuously over space. Consequently, the contribution of the observed points can be evaluated directly at their observed coordinates, avoiding the need to aggregate events into artificial grid cells. The only intractable component is therefore the integral of the intensity function, which generally has no analytical solution.

To approximate this integral, Simpson et al. [25] proposed introducing a finite set of integration (quadrature) points associated with the mesh. Each integration point is assigned a weight representing the area of the corresponding dual mesh cell, allowing the integral to be approximated by a weighted sum,

$$\int_D \lambda(s) ds \approx \sum_{k=1}^m w_k \lambda(\tilde{s}_k), \quad (3.14)$$

where $\tilde{s}_k$ denotes the quadrature nodes and w_k their associated integration weights.

The resulting approximation transforms the continuous likelihood into a finite-dimensional expression involving only evaluations of the latent field at the observed locations and at the quadrature points. This formulation is fully compatible with the latent Gaussian modelling framework adopted by INLA, allowing efficient Bayesian inference while preserving the continuous interpretation of the underlying intensity surface.

Finally, Simpson et al. [25] demonstrated that both the numerical approximation of the integral and the SPDE approximation of the latent field converge as the mesh is refined, providing theoretical justification for the proposed approach.

In the following chapter, the methodology developed in the previous sections is applied to the BAD-ITE wildfire dataset. In particular, the data preparation steps required to obtain an INLA-ready dataset are presented, together with a detailed description of the environmental covariates. The chapter then introduces the modelling framework adopted for this application, with particular emphasis on model specification, hyperparameter choices and computational settings within the INLA–SPDE framework.

Chapter 4

Materials and modelling framework

4.1 Data source: BAD-ITE

The wildfire occurrence dataset adopted in this study is the *Burnt Areas Database – Italian Terrestrial Ecosystems* (BAD-ITE), described in detail in Chapter 2. The database was generated through the integration of multiple remote sensing products and national cartographic sources, followed by a harmonisation and preprocessing procedure aimed at producing a consistent wildfire inventory for the Italian territory.

The same dataset is employed in the present analysis, with one important difference. Unlike the application presented in Chapter 2, where the year 2018 was excluded because it corresponds to the transition period following the operational introduction of Sentinel-2 imagery into the EFFIS processing chain, here it was retained. Although 2018 does not represent a complete annual record, its inclusion is expected to have a more limited impact on the spatial analysis than on the temporal analysis, although the incomplete observation period remains a limitation. After including the 2018 observations, the final dataset consists of 9561 individual wildfire events. Before the statistical modelling, however, additional preprocessing steps were required to transform the wildfire inventory into a format suitable for spatial point process analysis within the INLA framework.

4.1.1 Construction of the INLA-ready dataset

Each wildfire event in BAD-ITE, as described earlier, is represented as a burned-area polygon. To adapt the dataset to a spatial point process formulation, each polygon was transformed into a single spatial point by extracting its centroid. This approximation allows wildfire occurrence to be modelled in terms of event locations while necessarily discarding information on burned area extent, internal spatial heterogeneity and fire geometry. This modelling choice implies indeed that the response of interest is wildfire occurrence, not burned area or fire severity.

The resulting dataset was further refined to ensure consistency in the definition of the spatial support of the process. In particular, all geometries were projected into the UTM reference system (EPSG:32632, with coordinates converted from metres to kilometres for model implementation) to ensure numerical stability in the subsequent SPDE formulation. This projected coordinate system was selected because it provides metric coordinates, allowing Euclidean distances, areas and spatial relationships required by the SPDE approach to be computed directly. Although a continental or national equal-area projection such as ETRS89 / LAEA Europe (EPSG:3035) would represent a more appropriate choice for analyses covering the entire Italian territory, the use of UTM 32N was maintained consistently throughout all preprocessing steps, mesh

construction and model estimation. Since the same coordinate system was applied to all spatial operations, the internal coherence of the modelling framework was preserved¹.

Domain

A crucial step in the modelling framework concerns the definition of the spatial domain over which the wildfire point process is assumed to be defined. Following the operational definition adopted by the European Forest Fire Information System (EFFIS) and reported by ISPRA [17], a wildfire event is considered relevant when it affects natural or semi-natural land-cover classes. Consequently, the statistical population of interest is not the entire geographic surface, but specifically the subset of natural areas where wildfire processes are ecologically meaningful. To implement this definition, land-cover information was used from the CLCplus Backbone Raster Product, downloaded from the Copernicus website. This dataset provides a detailed land cover classification with the following categories:

- 1: Sealed;
- 2: Woody – needle leaved trees;
- 3: Woody – broadleaved deciduous trees;
- 4: Woody – broadleaved evergreen trees;
- 5: Low-growing woody plants (bushes, shrubs);
- 6: Permanent herbaceous;
- 7: Periodically herbaceous;
- 8: Lichens and mosses;
- 9: Non- and sparsely-vegetated;
- 10: Water;
- 11: Snow and ice;
- 253: Coastal seawater buffer;
- 254: Outside area.

Following the EFFIS definition of natural land, only the following classes were retained as part of the analysis domain: 2, 3, 4, 5, 6, 7 and 9 (the ones in green), while all remaining classes were considered non-natural and excluded from the modelling domain. The resulting land-cover map was converted into a binary raster distinguishing natural from non-natural areas. It was subsequently aggregated to a spatial resolution of 1 km using a majority-rule criterion (mode), ensuring consistency with the spatial resolution adopted for the environmental covariates. Finally, the raster was converted into a polygon representation defining the support of the point process. To improve the numerical stability of the spatial model, minor islands were removed from the domain. Although they belong to the Italian territory, these smaller islands are geographically disconnected from the mainland and the major islands and therefore

¹The influence of the selected projection on spatial scale estimates and intensity surfaces could be further assessed through sensitivity analyses.

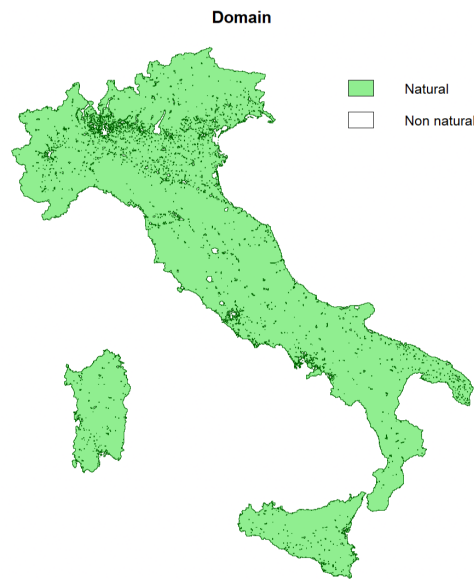


Figure 4.1. Domain of the point process

are not expected to share the same spatial dependence structure with the coast. In addition, the number of observed wildfire events in these areas is very limited (26 wildfires), making their contribution to the estimation of the spatial intensity negligible. Their removal avoids the introduction of artificial connections in the latent spatial field and improves the construction of the integration scheme.

The resulting dataset considers only the points that fall inside the defined polygonal domain, shown in figure 4.1, and it constitutes the final input for the spatial modelling stage, where wildfire occurrences are treated as a realisation of an inhomogeneous spatial point process defined over the natural land surface of Italy.

4.1.2 Covariates

The aim of the model is to investigate the relationship between wildfire occurrence and a set of environmental and anthropogenic predictors. To this end, a collection of topographic, climatic, vegetation-related and human-pressure covariates was assembled to characterise the spatial heterogeneity of wildfire occurrence across the study area.

All covariates were harmonised to a common spatial resolution of 1km and projected at the coordinate reference system (EPSG:32632, converted from meters to kilometres). To improve comparability and numerical stability they were standardised.

Most of the rasters were obtained from the ISPRA geospatial database and from the Copernicus website. Since the analysis is purely spatial and aggregates wildfire events occurring during the 2018–2025 period, the selected covariates should be interpreted as long-term environmental descriptors rather than as meteorological conditions observed at the time of individual fire events.

The covariates considered in the analysis are elevation, annual cumulative precipitation, average monthly precipitation, temperature, land surface temperature, latitude, tree cover density, solar radiation, logarithmic distance from the coastline, population and a composite human pressure index. The rasters of the covariates considered are shown and described more in detail below.

Elevation

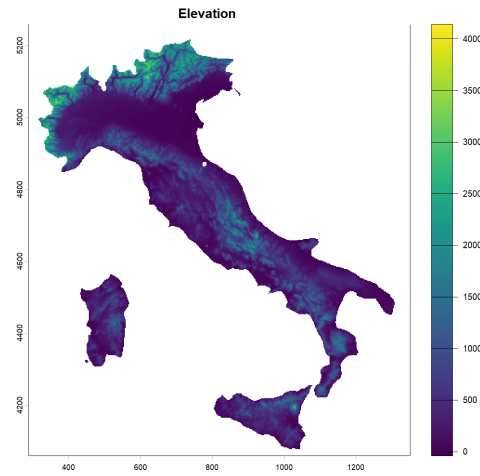


Figure 4.2. Elevation, unit of measure: m (a.s.l)

The elevation covariate was derived from the ISPRA Digital Elevation Model (DEM), which provides a continuous representation of terrain elevation across the Italian territory. The original dataset has a spatial resolution of 20 m and is expressed in metres above sea level (m a.s.l.). As described above, the dataset was harmonised to the common analysis grid and standardised prior to model fitting.

Elevation was included in the model as a topographic descriptor of the study area, representing broad-scale environmental gradients potentially associated with wildfire occurrence.

Precipitation

The precipitation variable considered in this study is represented by two alternative raster datasets, both used to characterize rainfall conditions at different temporal scales.

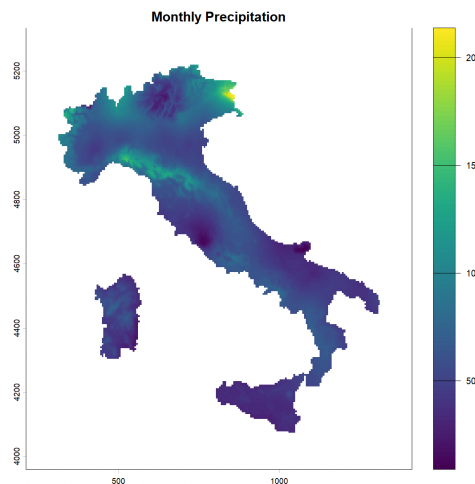


Figure 4.3. Average monthly precipitation, unit of measure: mm/month

The first variable, shown in Figure 4.3, consists of average monthly precipitation, obtained from the WorldClim historical monthly climate database. The dataset is based on the CRU-TS

4.09 observations produced by the Climatic Research Unit (University of East Anglia) and distributed as downscaled and bias-corrected gridded climate surfaces. The original product has a spatial resolution of 2.5 arc-minutes (approximately 4.5 km at the equator) and provides total monthly precipitation expressed in millimetres (mm). For the purposes of this study, monthly precipitation values were averaged over the 2018–2024 period, before being harmonised to the common analysis grid. The second precipitation covariate, shown in Figure 4.4, is the

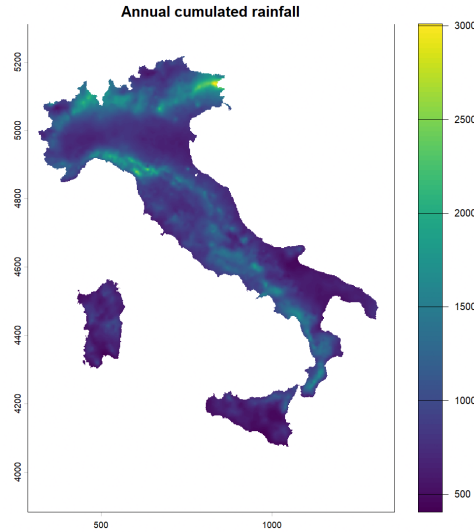


Figure 4.4. Annual Cumulated Rainfall, unit of measure: mm/year

Annual Cumulated Rainfall (CRF), obtained from the ISPRA database (SCIA [29]). This product was developed within the GIS-based procedure for the national groundwater balance assessment and represents the climatological annual cumulative precipitation averaged over the period 1996–2015. The original raster has a spatial resolution of 20 m and is expressed in millimetres per year (mm/year) [30]. As for the other predictors, the dataset was aggregated to the common modelling resolution.

Since the proposed LGCP is a purely spatial model and does not explicitly account for temporal dynamics, both precipitation variables should be interpreted as descriptors of the long-term spatial distribution of rainfall rather than as meteorological conditions associated with individual wildfire events. Monthly precipitation captures the average rainfall regime during the study period, whereas annual cumulated rainfall represents the long-term climatological water availability across the study area. Both variables were included to account for spatial differences in moisture conditions potentially associated with wildfire occurrence.

Temperature

The temperature covariate was provided by ISPRA and consists of the Normalized Annual Average Air Temperature on Elevation raster, originally developed by Fioravanti et al. [31]. The dataset provides a gridded representation of the spatial distribution of mean annual air temperature across Italy, derived from long-term observations collected by national and regional meteorological networks over the period 1961–2009. The temperature surfaces were generated through a regression-kriging interpolation procedure, which combines ground observations with topographic predictors to produce continuous temperature estimates. The original raster has a spatial resolution of 20 m and represents a normalized temperature index. Prior to the analysis,

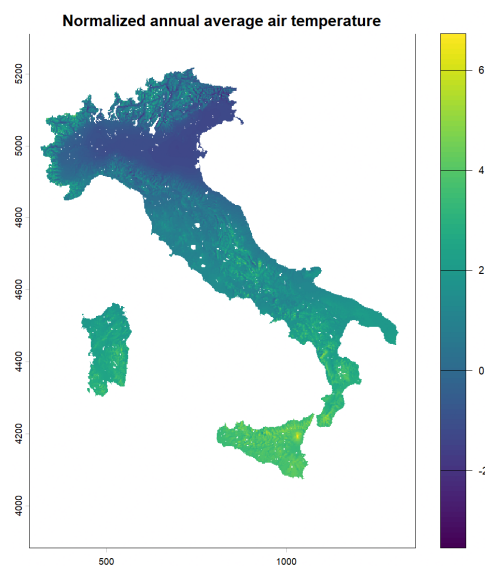


Figure 4.5. Normalized Annual Average Air Temperature on Elevation, unit of measure: dimensionless (normalized index)

the raster was aggregated to the common modelling resolution of 1 km and standardised before model fitting. This covariate is interpreted as a climatological descriptor of the thermal conditions characterising each location rather than as the temperature observed at the time of individual wildfire events.

Land surface temperature

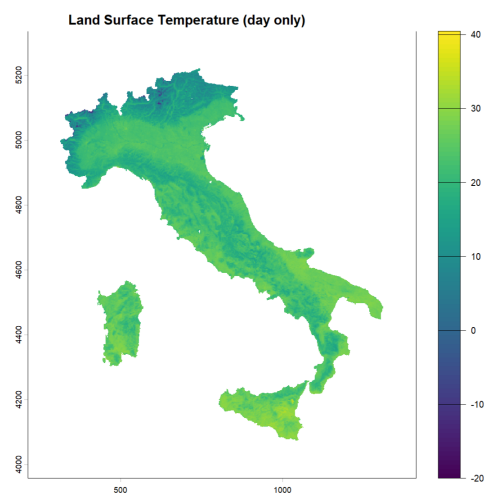


Figure 4.6. Land Surface Temperature, unit of measure: Celsius degrees (C°)

The land surface temperature covariate was obtained from the Copernicus Climate Data Store as part of the ESA Land Surface Temperature Climate Change Initiative (LST CCI). The dataset provides monthly clear-sky land surface temperature estimates derived from multiple infrared satellite sensors, including ATSR-2, AATSR, Terra MODIS and Sentinel-3 SLSTR. Observations are available separately for daytime and nighttime acquisitions, corresponding

approximately to 10:00 and 22:00 local solar time.

The original product is distributed on a regular latitude–longitude grid with a spatial resolution of approximately 0.01° (≈ 1 km), and temperature values are expressed in Kelvin. Data availability is affected by cloud cover, which introduces spatial gaps in the retrievals. In particular, nighttime observations exhibit a substantially higher proportion of missing values due to persistent cloud contamination, resulting in incomplete spatial coverage. For this reason, only daytime LST observations were retained for the analysis.

For the present study, monthly daytime values were averaged over the period 2018–2024 to obtain a single spatial covariate representing mean land surface thermal conditions. Temperatures were converted to degrees Celsius prior to aggregation and then standardised before model fitting. This covariate is interpreted as a descriptor of typical land surface thermal conditions over the considered period.

Tree cover density

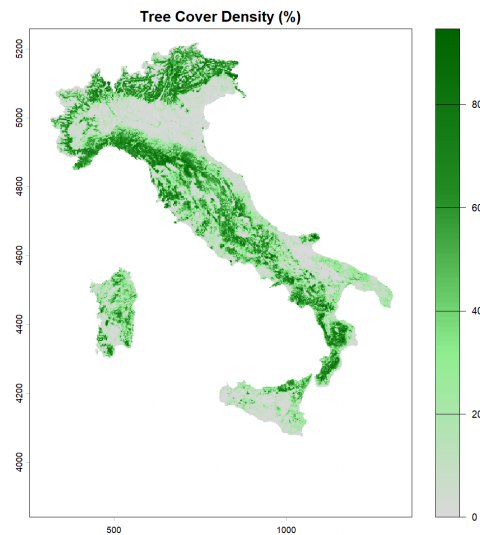


Figure 4.7. Tree cover density, unit of measure: %

The Tree Cover Density covariate used in this study represents the percentage of tree canopy cover and was obtained from the ISPRA database, originating from the Copernicus Land Monitoring Service. The dataset describes the spatial distribution of tree canopy density derived from satellite-based land cover products and high-resolution remote sensing observations.

The original product is provided at pan-European scale with spatial resolutions of 10 m, and reports tree cover density values ranging from 0% to 100% for the 2018 reference year. As for all covariates, the dataset was harmonised to the common 1 km modelling grid. This covariate was included as a proxy for spatial variability in vegetation structure and forest cover.

Solar radiation

The solar radiation covariate used in this study represents the incoming shortwave solar radiation reaching the Earth's surface, expressed in Wh m^{-2} . The dataset was obtained from the ISPRA database and corrected using a long-term cloud climatology.

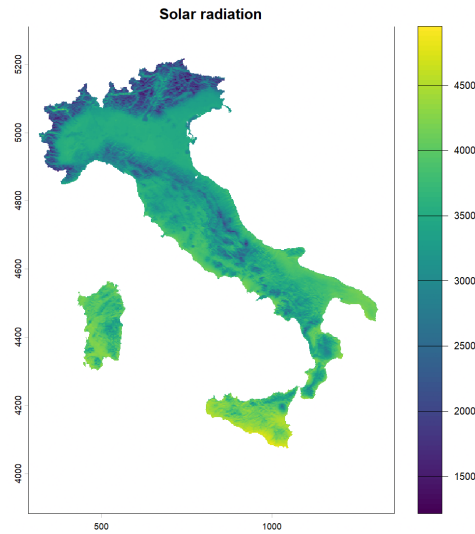


Figure 4.8. Solar radiation, unit of measure: Wh m^{-2}

The original dataset was generated at 20 m spatial resolution using the `r.sun` module implemented in GRASS GIS, which estimates direct, diffuse and reflected solar radiation from a digital elevation model (DEM), while explicitly accounting for terrain geometry and topographic shading [32]. To obtain realistic long-term estimates, the theoretical solar radiation was subsequently corrected using a climatology of daytime cloud cover derived from the ESA Climate Change Initiative cloud dataset (2004–2014), thereby accounting for the attenuation of direct radiation and the contribution of diffuse radiation under cloudy conditions.

For the purposes of this study, the raster was subsequently harmonised to the common 1 km modelling grid adopted for all covariates and standardised. Solar radiation was included to represent the spatial variability of surface insolation, which influences local energy balance, vegetation conditions and fuel drying processes.

Distance from coast

The coastline distance raster represents the logarithm (base 10) of the Euclidean distance from each grid cell to the nearest shoreline and was obtained from the ISPRA geospatial database. The variable was provided in log-transformed form, already accounting for the strong spatial skewness of raw distance values. The original dataset has a spatial resolution of 20 m. As for all covariates, the raster was harmonised to the common 1 km modelling grid. This variable was included to account for spatial gradients associated with coastal proximity.

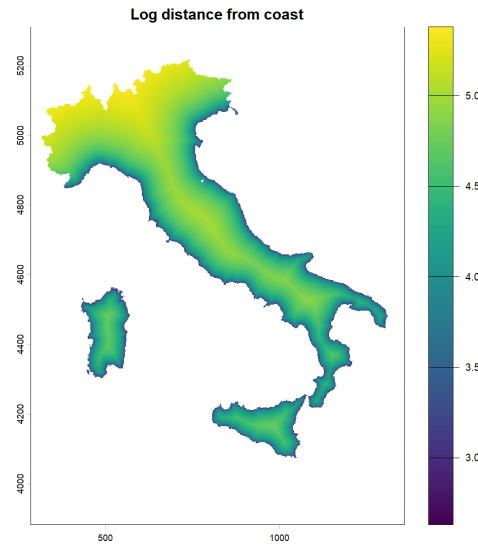


Figure 4.9. Logarithmic distance from coast, unit of measure: $\log_{10}(\text{m})$

Latitude

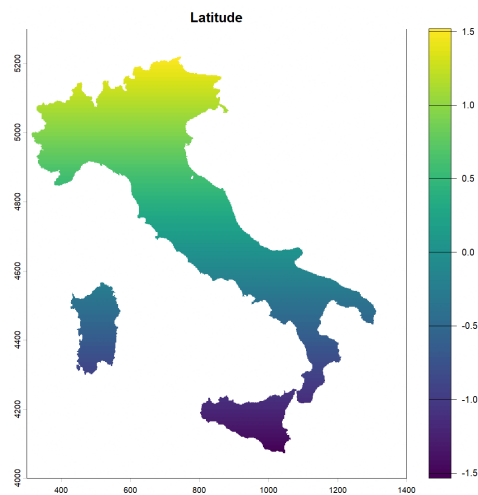


Figure 4.10. Latitude, unit of measure: dimensionless

The latitude raster was generated by extracting the north–south coordinate (Y-axis) from the projected study area grid. This coordinate, defined in the UTM reference system, was then standardised using a z-score transformation, resulting in a dimensionless variable representing relative position along the north–south axis within the study domain. The resulting raster does not represent geographic latitude in degrees, but rather a normalised spatial coordinate derived from the projected reference system. It should be interpreted as a proxy of the north–south gradient rather than true geographic latitude. This variable was included as a spatial indicator to account for broad-scale north–south gradients not captured by the other environmental covariates, reflecting large-scale spatial structure across the study domain.

Population

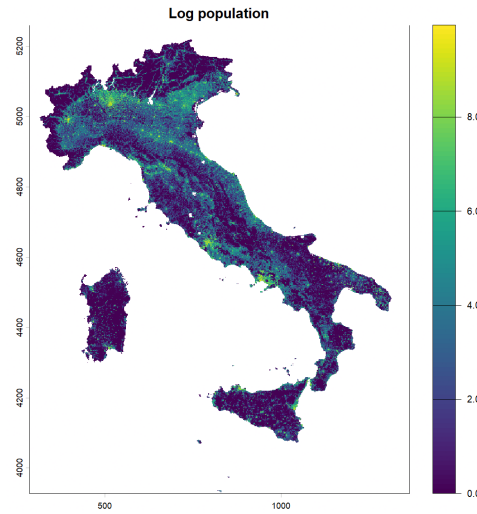


Figure 4.11. Logarithmic population, unit of measure: log-transformed population

The population covariate was obtained from the WorldPop Global High Resolution Population Denominators dataset, which provides estimates of the total number of people per grid cell. The dataset is produced using a Random Forest-based dasymetric redistribution approach and is constrained to match national population totals from the United Nations World Population Prospects (2020 revision).

The original product is provided in GeoTIFF format at a spatial resolution of approximately 3 arc-seconds (≈ 100 m at the equator) in a Geographic Coordinate System (WGS84), with units expressed as number of people per pixel. For the present analysis, the raster was reprojected to the study coordinate reference system and harmonised to the common 1 km modelling grid. Aggregation was performed using a summation approach, so that the resulting values represent the total number of people within each grid cell. A $\log(1 + x)$ transformation was then applied to reduce the strong right-skewness of the distribution and to account for the presence of zero values in the raster. This covariate was included as a proxy for the spatial distribution of human and infrastructure presence within the study area.

Human pressure

The human pressure raster, unlike the other covariates, was constructed as a composite index combining the population and tree cover density rasters. It is intended as a proxy for anthropogenic pressure on natural and semi-natural environments.

Specifically, it was computed as:

$$HP(s) = \log(1 + \text{Population}(s)) \cdot \left(\frac{\text{Tree Cover Density}(s)}{100} \right)$$

The population component was first log-transformed using a $\log(1 + x)$ transformation, while the tree cover component was retained in its original percentage form. The two components were then combined multiplicatively to obtain the human pressure index. The resulting raster was subsequently standardised before being included in the statistical model.

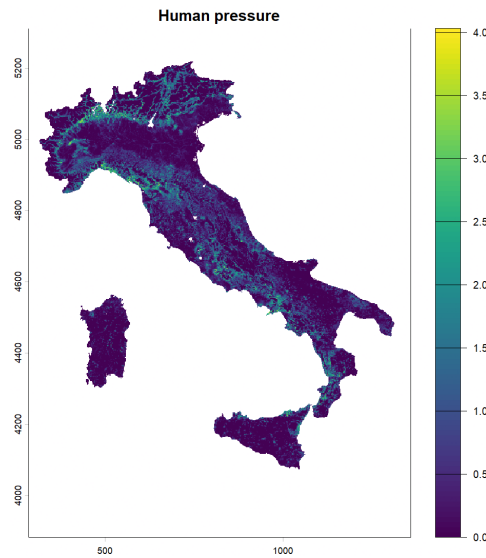


Figure 4.12. Human pressure, unit of measure: dimensionless

This multiplicative formulation was adopted to represent the spatial co-occurrence of human presence and vegetated land cover, such that high values arise only in areas where both components are simultaneously present. In this way, the index captures an interaction effect between anthropogenic presence and landscape structure, avoiding the compensatory behaviour that would arise from an additive formulation. The resulting variable is dimensionless and it is intended as a synthetic indicator of combined human presence and vegetation structure within the study domain.

An overview of all the covariates considered in the analysis, including their sources, units of measurement, temporal reference and spatial resolution, can be found in Table 4.1.

4.1.3 Covariate transformation for non-linear effects

Continuous covariates included in the modelling framework could be specified either as linear effects or as non-linear effects through second-order random walk (RW2) models. Since RW2 effects are defined over an ordered sequence of covariate values, continuous raster predictors require a preliminary discretisation step.

For covariates considered for non-linear modelling, the original raster layers were first smoothed using a 3×3 focal mean operation in order to reduce local pixel-scale variability while preserving the main spatial gradients. The smoothing was applied before standardisation. The resulting continuous distributions were then discretised into ordered classes, using either equal-width intervals or quantile-based classification depending on the covariate distribution and the number of observations available in each class.

The resulting ordered covariate classes were subsequently used as input for the RW2 formulation, allowing flexible estimation of potentially non-linear relationships between environmental and anthropogenic drivers and wildfire occurrence intensity.

Table 4.1. Summary of the covariates

Covariate	Source	Unit	Temporal reference	Original resolution
Elevation	ISPRA	m a.s.l.	Static	20m
Annual cumulative precipitation	ISPRA	mm/year	Long-term average (1996–2015)	20m
Average monthly precipitation	WorldClim	mm/month	2018–2024 average	2.5 arc-minutes ($\approx 4.5\text{km}$)
Temperature	ISPRA	dimensionless (normalized index)	Climatological average (1961–2009)	20m
Land Surface Temperature (LST)	Copernicus	$^{\circ}\text{C}$	2018–2024 average	1km
Latitude	Coordinates derived	dimensionless	Static	1km
Tree Cover Density	Copernicus	%	2018	10m
Solar radiation	ISPRA	WH/m^{-2}	Long-term average (2004–2014)	20m
Log distance from the coast	ISPRA	$\log_{10}(\text{m})$	Static	20m
Population count	WorldPop	dimensionless	2020	3 arc-seconds ($\approx 100\text{m}$)
Human pressure index	Derived	dimensionless	2020	1km

4.1.4 Correlation analysis

Correlation analysis is conducted to examine the relationships between the variables considered. This approach helps to identify and exclude variables that are highly correlated with each other, thereby minimizing issues related to multicollinearity. In this study, Spearman’s rank correlation coefficient was used to assess the strength and direction of monotonic relationships between variables.

Spearman’s rank correlation

Spearman’s rank correlation coefficient is a non-parametric measure used to assess the strength and direction of a monotonic relationship between two variables. Unlike Pearson’s correlation coefficient, it does not assume a linear relationship between variables but evaluates whether, as one variable increases, the other tends to increase or decrease in a consistent manner. The method is based on the ranked values of the data rather than their original magnitudes. Ranks correspond to the ordered positions of observations when the variables are sorted. This formulation makes Spearman’s correlation less sensitive to outliers and appropriate for variables that do not follow a normal distribution. It is therefore suitable for identifying monotonic, including non-linear, relationships between environmental variables. The mathematical formulation of Spearman’s rank correlation coefficient is defined as follows:

$$r_S = \frac{\sum_{i=1}^n (r_i - \bar{r})(s_i - \bar{s})}{\sqrt{\sum_{i=1}^n (r_i - \bar{r})^2 \cdot \sum_{i=1}^n (s_i - \bar{s})^2}}$$

where:

- r_i = i-th rank of X

- $\bar{r}$ = average rank of X
- s_i = i-th rank of Y
- $\bar{s}$ = average rank of Y

4.2 Modelling approach

4.2.1 LGCP applied to wildfire occurrences

Wildfire occurrences can be represented as a spatial point pattern, where each event is identified by its geographical location within the study area. The availability of satellite-derived spatial information makes it possible to preserve the exact geographical position of each wildfire event, rather than aggregating observations into administrative units or regular grids. In this context, the objective is not only to describe the observed distribution of wildfire occurrences, but also to estimate the underlying spatial intensity governing their occurrence. For this purpose, a Log-Gaussian Cox Process (LGCP) was adopted. LGCPs are particularly suitable for wildfire data because they directly model point events while allowing the occurrence intensity to vary continuously across space as a function of environmental covariates and latent spatial processes. Wildfire events are not expected to occur independently in space, since for their nature they are linked to environmental conditions, vegetation characteristics and human-related factors, that often exhibit spatial continuity, leading neighbouring locations to share similar levels of wildfire susceptibility. The LGCP framework accounts for this behaviour through the inclusion of a latent Gaussian spatial field, which captures residual spatial dependence not explained by the observed covariates. Consequently, the model separates the contribution of measured environmental factors from unobserved spatially structured effects, providing a flexible representation of wildfire occurrence across the study area.

Furthermore, LGCPs have been widely applied in spatial ecology, epidemiology and environmental sciences for the analysis of point events distributed irregularly in space, making them a well-established framework for modelling wildfire occurrence.

Building on the motivation outlined above, wildfires are modelled as a realization of a spatial point process defined over the study domain. It is important to note that the resulting point pattern represents the spatial distribution of burned-area centroids rather than wildfire ignition locations. The primary quantity of interest is the spatial intensity function $\lambda(s)$, which is additive on the log-scale. Let $\lambda(s)$ denote the intensity of the point process at location s within the spatial domain, it represents the expected number of events per unit area over the entire observed period (2018-2025). In a Log-Gaussian Cox Process framework, the logarithm of the intensity is expressed as a latent additive model combining fixed and random components:

$$\log(\lambda(s)) = \eta(s), \quad s \in D$$

where $\lambda(s)$ is the intensity of the point process at point s , inside the domain D , and $\eta(s)$ is the latent linear predictor. A general form of the model can be written as:

$$\eta(s) = \beta_0 + \sum_{k=1}^p \beta_k x_k(s) + \sum_{j=1}^q f_j(z_j(s)) + w(s)$$

and therefore:

$$\lambda(s) = \exp \left(\beta_0 + \sum_{k=1}^p \beta_k x_k(s) + \sum_{j=1}^q f_j(z_j(s)) + w(s) \right)$$

where:

- β_0 is the intercept;
- $x_k(s)$ are covariates included with linear effects;
- β_k are the corresponding coefficients;
- $f_j(\cdot)$ are any non-linear or smooth effects;
- $z_j(s)$ are covariates modelled non-linearly;
- $w(s)$ is a zero-mean Gaussian random field capturing residual spatial dependence.

4.2.2 Computational settings for the SPDE-INLA approach

The spatial analysis was implemented in R using the INLA and `inlabru` packages. The modelling framework described in Chapter 3 was adopted throughout the analysis, while the following paragraphs summarise the computational settings used in this specific application.

The latent spatial field was represented through the SPDE approach using a triangulated mesh constructed over mainland Italy together with Sicily and Sardinia. A finer mesh resolution was adopted over the study area, where the spatial process is estimated, while a coarser resolution was used over the surrounding sea to reduce computational cost without affecting inference. The resulting mesh is shown in Figure 4.13.

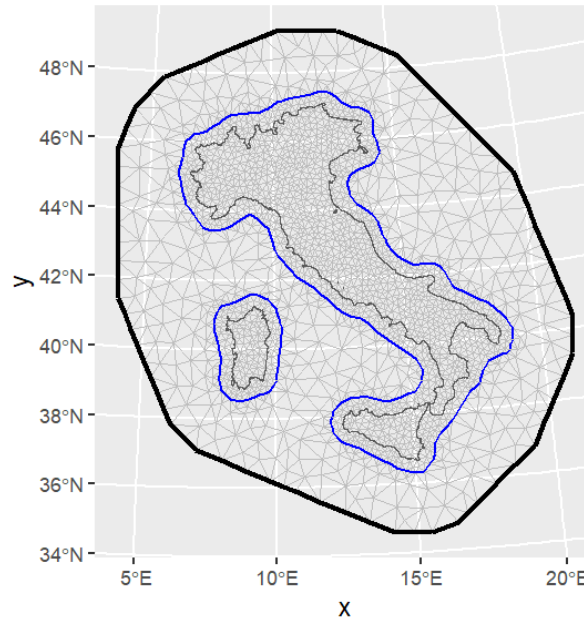


Figure 4.13. Triangulated mesh adopted for the SPDE approximation.

Penalised Complexity (PC) priors were assigned to the SPDE hyperparameters following previous applications of LGCP models to wildfire occurrence in Italy. The adopted prior specifications were

$$\mathbb{P}(\rho < 100 \text{ km}) = 0.5, \quad \mathbb{P}(\sigma > 1) = 0.5$$

In particular, the prior on the spatial range (ρ) was defined such that there is a 50% probability that spatial correlation becomes negligible at approximately 100 km, while the prior on the marginal standard deviation (σ) was defined such that there is a 50% probability that it exceeds 1.

These specifications reflect expected spatial dependence structures at regional scales and allow for residual spatial variability not explained by the available covariates.

The polygonal domain, constructed as described in section 4.1.1 and shown in Figure 4.1, was then used within the SPDE-INLA framework to define the integration points required for likelihood approximation. Locations outside the selected domain were assigned zero integration weight and therefore did not contribute to the likelihood. This ensures that the intensity function is estimated only over the population of interest, consistently with the wildfire definition adopted in the study.

4.2.3 Model selection criteria

Competing LGCP models were compared using Bayesian model selection criteria in order to identify the specification providing the best balance between model fit, predictive performance and model complexity. Model comparison was primarily based on the Watanabe–Akaike Information Criterion (WAIC), while the Deviance Information Criterion (DIC) was used as a complementary diagnostic measure. Posterior credible intervals were also examined to assess the stability and interpretability of the estimated covariate effects.

To facilitate a systematic and reproducible comparison, an automated fitting procedure was developed to estimate a predefined set of candidate models and extract the corresponding model diagnostics. This ensured that all candidate models were evaluated under a consistent fitting framework. To reduce potential multicollinearity and avoid including redundant information, covariates representing substantially overlapping processes were not included simultaneously within the same candidate model.

Watanabe–Akaike Information Criterion (WAIC)

The Watanabe–Akaike Information Criterion (WAIC) is a fully Bayesian measure of predictive performance that estimates the expected out-of-sample predictive accuracy while accounting for model complexity. Unlike DIC, WAIC averages over the full posterior distribution of the model parameters rather than relying on a point estimate [33].

$$WAIC = -2(lppd - p_{WAIC})$$

Models with lower WAIC values are preferred, as they provide a better trade-off between goodness of fit and predictive performance. In this study, WAIC was adopted as the primary criterion for comparing candidate models.

Deviance Information Criterion (DIC)

The DIC is based on the model deviance [34], defined as:

$$D(\theta) = -2\log(p(y|\theta))$$

where y denotes the observed data, θ the model parameters, and $p(y|\theta)$ the likelihood. Smaller values of the deviance indicate a better agreement between the model and the observed data.

Model fit is summarised through the posterior mean deviance, that is given by:

$$\bar{D} = \mathbb{E}_{\theta|y}[D(\theta)]$$

which represents the average lack of fit over the posterior distribution of the parameters. To account for model complexity, the effective number of parameters is computed as:

$$p_D = \mathbb{E}_{\theta|y}[D(\theta)] - D(\bar{\theta}) = \bar{D} - D(\bar{\theta})$$

where $\bar{\theta} = \mathbb{E}_{\theta|y}[\theta]$ denotes the posterior mean of the parameters, while The quantity p_D measures the effective complexity of the model and acts as a penalty term. The DIC is then defined as:

$$DIC = \bar{D} + p_D$$

Models with lower DIC values are generally preferred, as they provide a better trade-off between model fit and complexity. Although WAIC was used as the primary model selection criterion, DIC was also computed as a complementary measure of model fit and complexity. Agreement between the two criteria was considered an additional indication of model robustness.

Posterior credible intervals

Posterior credible intervals were inspected to evaluate the direction, magnitude and uncertainty associated with the estimated covariate effects. They were not used as an automatic criterion for variable inclusion or exclusion. Instead, they were interpreted together with the overall model performance, allowing the contribution of each covariate to be assessed within the complete Bayesian model.

These diagnostics were used to support model interpretation rather than to drive the model selection procedure, which is instead based on predictive performance criteria and presented in the following chapter.

Chapter 5

Results

This chapter presents the results of the spatial point process modelling analysis. First, the final point pattern and the modelling domain are summarised. Then, the correlation structure among covariates is examined to define a reduced and nonredundant set of candidate predictors. Finally, candidate LGCP models are compared, the selected model is interpreted and the estimated spatial intensity surface is presented.

5.1 Final modelling dataset

The data preprocessing workflow described in Chapter 2 resulted in the final point pattern used for the spatial analysis. Following the filtering, cleaning and spatial processing steps previously presented, the modelling dataset consists of 9510 wildfire events, represented by their centroids. Figure 5.1 shows the spatial distribution of the final event locations across the Italian territory.

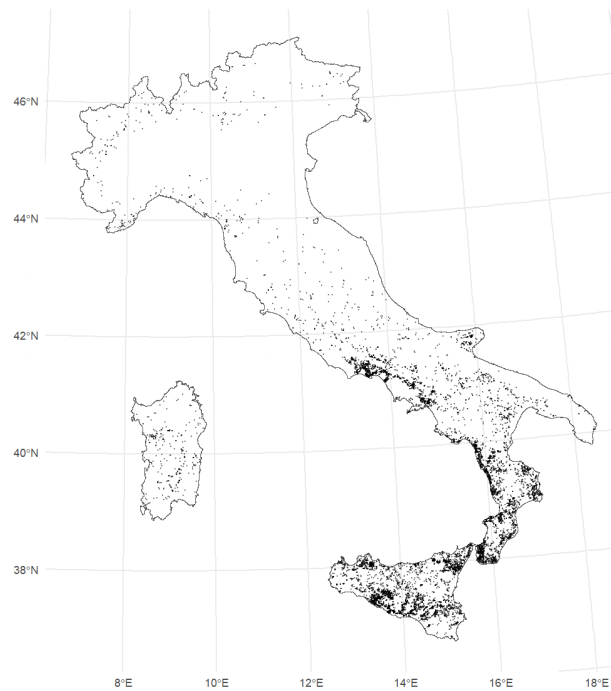


Figure 5.1. Final dataset of wildfire centroids

The point pattern reveals a marked spatial heterogeneity, with areas of high event concentration alternating with regions characterised by a much lower occurrence of wildfires.

The study domain covers the natural areas of the Italian territory, details on how the domain is built are reported in Chapter 4. Only centroids falling within the natural and semi-natural modelling domain were retained. Therefore, the fitted intensity must be interpreted as the intensity of wildfire occurrence over the wildfire-relevant natural domain, not over the whole Italian territory. The following table shows a summary of some main features regarding the final modelling dataset.

Quantity	Value
Original BAD-ITE records	13433
Distinct wildfire events after aggregation by EFFIS_ID	9561
Distinct wildfire events after removing minor islands	9535
Final wildfire centroids inside the modelling domain	9510
Excluded centroids after removing minor islands	26
Excluded centroids after defining the modelling domain on natural areas	25
Total number of excluded centroids	51
Spatial resolution of covariates	1 km
Coordinate reference system	EPSG:32632

Table 5.1. Summary of the final modelling dataset.

5.2 Preliminary covariate screening

Before fitting the LGCP models, a preliminary screening of the candidate covariates was performed to identify highly correlated predictors and reduce potential multicollinearity. As described in the Methods chapter, pairwise associations were assessed using Spearman's rank correlation coefficient. The resulting correlation matrix is reported in Figure 5.2, where each cell indicates the correlation coefficient, ranging from -1 (perfect negative correlation) to 1 (perfect positive correlation).

The correlation matrix reveals the presence of distinct groups of interrelated variables. A highly correlated cluster of variables is observed among climatic predictors, particularly between solar radiation, temperature, land surface temperature and latitude, indicating that these variables share a common large-scale environmental gradient. Due to the very high correlation between land surface temperature and solar radiation (0.91), land surface temperature was excluded from subsequent analyses to reduce multicollinearity. The decision to retain solar radiation rather than LST is also supported by the fact that LST data suffer from missing values due to cloud cover, as previously discussed in the chapter on materials. Precipitation variables, annual and monthly respectively, also exhibit strong intercorrelation, reflecting their shared representation of the same underlying hydrological process. In this context, monthly precipitation was retained for the modelling framework. Another pair of variables showing a high correlation is the one formed by population and human pressure. This was expected due to the construction of the latter from population data, however, it does not appear to be highly correlated with tree cover density, despite also being constructed from it. For this reason, these variables were both considered in the analysis, but with the intention of not including them simultaneously in the same models to avoid redundancy. On the other hand elevation acts as a linking variable across multiple environmental domains, showing moderate correlations with

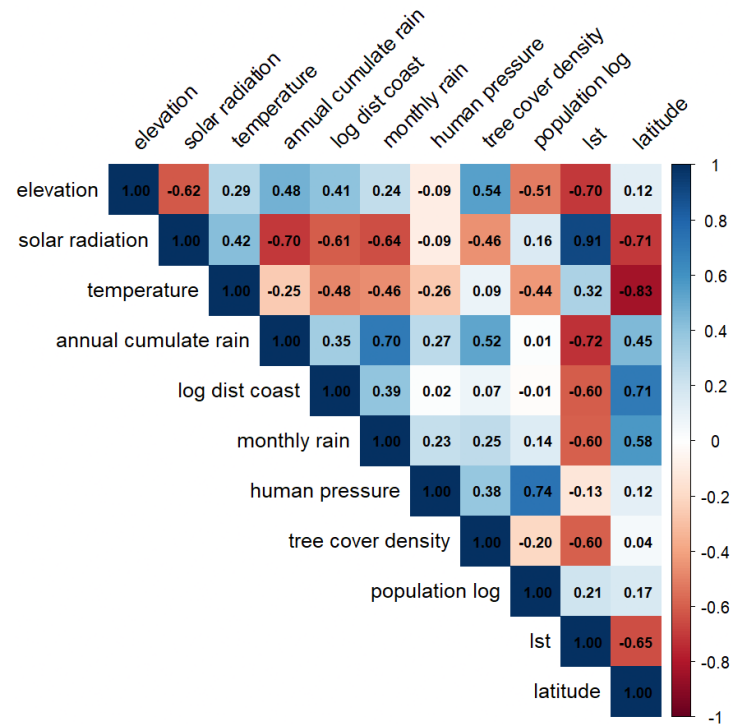


Figure 5.2. Correlation matrix

both climatic and vegetation-related predictors. Overall, the correlation structure highlights the presence of coherent environmental gradients and supports the selection of a reduced set of non-redundant predictors for modelling. Table 5.2 summarizes the decisions taken after the preliminary screening. Variables representing essentially the same environmental information were reduced to a single representative predictor, whereas covariates showing moderate or even relatively high correlations, but describing distinct environmental processes were retained as candidate predictors. In these cases, potential multicollinearity was addressed during the model selection stage, by avoiding the inclusion of highly correlated variables within the same model formulation. Consequently, the correlation analysis was used as a preliminary screening tool rather than as a strict variable selection procedure.

Table 5.2. Summary of the covariates selection

Covariate	Decision	Reason
Elevation	Retained	Moderate correlations; ecologically relevant predictor
Annual cumulative precipitation	Excluded	Highly correlated with Monthly Precipitation
Average monthly precipitation	Retained	Preferred representation of precipitation variability
Temperature	Retained	Candidate climatic predictor; evaluated during model selection
Land Surface Temperature	Excluded	Highly correlated with solar radiation ($r_S = 0.91$); affected by missing values
Latitude	Retained	Proxy for large-scale climatic gradient; evaluated during model selection,
Tree Cover Density	Retained	Ecologically relevant fuel availability indicator
Solar radiation	Retained	Candidate climatic predictor; evaluated during model selection
Log distance from the coast	Retained	Represents coastal influence
Population count	Retained*	Not included together with Human Pressure
Human pressure index	Retained*	Not included together with Population

(* indicates mutually exclusive in candidate models.)

5.3 Candidate model set

Following the preliminary variable screening described in Chapter 4, an extensive model comparison procedure was performed to identify the most appropriate specification of the Log-Gaussian Cox Process.

The set of candidate covariates was defined after an initial correlation analysis, which was used to remove highly collinear predictors and retain a subset of non-redundant environmental and anthropogenic variables. The final pool of candidate predictors included the predictors selected during the correlation analysis.

Because population and human pressure describe closely related aspects of anthropogenic influence and showed substantial redundancy in the preliminary analysis, these two variables were never included simultaneously in the same model. Apart from the Population–Human Pressure pair, no combinations were explicitly excluded a priori. Other correlated predictors were allowed to appear together, although preference was ultimately given to more parsimonious specifications whenever correlated variables provided only marginal improvements in model fit. Before model comparison, an initial functional form was assigned to each continuous predictor based on previous literature, ecological considerations and exploratory analyses. In the initial specification, temperature, latitude, precipitation and solar radiation were modelled as linear effects, whereas elevation, distance from the coast and tree cover density were represented using RW2 effects to allow for potentially non-linear relationships.

An exhaustive comparison of all admissible combinations of the selected covariates was then performed. This resulted in a total of 385 candidate models, corresponding to all possible non-redundant subsets of the candidate predictors under the constraint described above. The complete table containing all the models considered can be found in the appendix.

Model comparison was carried out using WAIC and DIC. The effective number of parameters and the marginal log-likelihood were also recorded to support the assessment of model complexity. In latent Gaussian models including a spatial field and RW2 components, model complexity is

expected to be high and was therefore interpreted comparatively across models rather than in absolute terms.

Table 5.3 reports the ten best-performing models according to WAIC. The results indicate that several competing specifications achieve similar predictive performance, suggesting that model selection cannot be based solely on information criteria. For this reason, the final model was selected by jointly considering predictive performance, stability of estimated effects and ecological interpretability.

1	model_id	n_covariates	variables	WAIC	DIC	non_significant_vars	delta_WAIC	delta_DIC	n_param
2	395	6	elev, latitude, distcoast, temperature, tree_cover, popul	80130,82844	35581,73911	latitude	0	160,1224	18457,2
3	469	7	elev, latitude, distcoast, precipitation, temperature, tree_cover, popul	80134,54107	35554,46371	latitude, precipitation	3,71263115	132,847	18474,8
4	478	7	elev, latitude, distcoast, temperature, tree_cover, solar, popul	80136,10193	35620,87271	latitude	5,273493283	199,256	18483
5	504	8	elev, latitude, distcoast, precipitation, temperature, tree_cover, solar, popul	80231,80625	35584,20139	latitude, precipitation	100,9778118	162,5847	18515,5
6	304	5	elev, distcoast, temperature, tree_cover, popul	80471,21088	35474,09019	none	340,3824429	52,4735	18652,2
7	429	6	elev, distcoast, temperature, tree_cover, solar, popul	80495,59263	35421,61669	none	364,7641973	0	18695,6
8	489	7	elev, distcoast, precipitation, temperature, tree_cover, solar, popul	80499,16366	35436,95998	precipitation	368,3352243	15,34329	18707,6
9	420	6	elev, distcoast, precipitation, temperature, tree_cover, popul	80523,79666	35476,30331	precipitation	392,968226	54,68662	18703,4
10	389	6	elev, latitude, distcoast, precipitation, tree_cover, popul	80718,67488	36314,09426	precipitation	587,8464411	892,4776	18568,3

Table 5.3. Best candidate models

Particular attention was also paid to the correlation structure among the predictors. As highlighted before, the final set of predictors is non-redundant, but some of them present a high correlation. Preference was given to model specifications that avoided the simultaneous inclusion of highly correlated variables whenever they provided only marginal improvements in model fit.

After identifying the most competitive covariate combinations, alternative linear and RW2 formulations were compared one covariate at a time using WAIC, DIC and visual inspection of the estimated functional relationships. RW2 effects were retained only when they provided a clear improvement in model fit together with an ecologically interpretable non-linear response.

5.4 Selected model

Based on the model comparison procedure described above, the final specification includes Temperature, Elevation, Tree Cover Density, Distance from the Coast and Population, together with an intercept and a latent Gaussian spatial field. Temperature was retained as a linear effect, whereas the remaining continuous covariates were modelled through second-order random walk (RW2) effects in order to capture potentially non-linear relationships with wildfire occurrence. The selected model is defined as

$$\begin{aligned}
 \log \lambda(s) = & \underbrace{\beta_0 + \beta_1 \text{Temp}(s)}_{\text{Fixed effects}} \\
 & + \underbrace{f_{\text{elev}}(\text{Elev}(s)) + f_{\text{coast}}(\text{Coast}(s)) + f_{\text{tree}}(\text{Tree}(s)) + f_{\text{popul}}(\text{Popul}(s))}_{\text{Non-linear effects (RW2)}} \\
 & + \underbrace{w(s)}_{\text{Latent Gaussian spatial field}}, \quad s \in D
 \end{aligned} \tag{5.1}$$

where $\lambda(s)$ denotes the expected wildfire intensity, β_0 is the intercept and $w(s)$ is the latent Gaussian spatial field accounting for residual spatial dependence.

The SPDE approximation was constructed on a triangular mesh composed of 2106 vertices, while the LGCP likelihood was approximated using 4032 quadrature (integration) points. The

RW2 components were represented using 25 ordered classes for Elevation and 10 ordered classes for Tree Cover Density, Distance from the Coast and Population.

The main goodness-of-fit measures of the selected model are summarised in Table 5.4.

Table 5.4. Model fit diagnostics and information criteria.

Diagnostic / Criterion	Value
WAIC	78879.07
DIC	34167.45
Effective number of parameters (p_{WAIC})	18367.05
Marginal log-likelihood	-25387.65

As said before, after identifying the most competitive covariate combinations, alternative functional forms were evaluated for the continuous predictors. Among these, Population showed the largest improvement when represented through an RW2 effect rather than a linear term. Table 5.5 compares the two formulations.

Table 5.5. Comparison between linear and RW2 formulations for the Population covariate.

Population specification	WAIC	p_{WAIC}
Linear	80414.05	18652.20
RW2	78879.07	18367.05

Replacing the linear specification with an RW2 effect reduced the WAIC by approximately 1535 units while also decreasing the effective number of parameters from 18652.20 to 18367.05. This substantial improvement indicates that a flexible non-linear specification provides a considerably better description of the association between population and wildfire occurrence. Nevertheless, because the functional form was refined using the same dataset employed for model selection, the magnitude of this improvement should be interpreted with some caution and may be optimistic in the absence of external validation.

The effective number of parameters ($p_{\text{WAIC}} = 18367.05$) appears relatively large. However, in LGCP models this quantity should not be interpreted as the actual number of estimated parameters, nor directly compared with the number of observed wildfire events. Instead, it reflects the overall effective complexity of the model, including the latent Gaussian spatial field, the RW2 components and the numerical approximation of the likelihood through the integration points. Consequently, p_{WAIC} is more appropriately interpreted comparatively across competing model specifications than in absolute terms.

The following section examines the posterior distributions of the model parameters and interprets the estimated contribution of each covariate to the spatial intensity of wildfire occurrence.

5.4.1 Fixed effects

Fixed effects quantify the global relationship between covariates and the intensity of the phenomenon, assuming that this relationship is constant across the entire study area. In this model, they consist of two parameters: the global intercept, which defines the baseline intensity level, and temperature, included as a linear effect to capture the average thermal influence at the national scale.

Table 5.6 reports the posterior summaries of the linear coefficients, while Figure 5.3 shows the corresponding posterior marginal distributions.

Table 5.6. Posterior statistics for the fixed effects coefficients of the LGCP model.

Parameter	Mean	SD	0.025 quant	0.5 quant	0.975 quant	Mode	KLD
Intercept	-7.940	0.387	-8.573	-7.998	-7.032	-8.038	0
Temperature	1.369	0.068	1.234	1.369	1.502	1.369	0

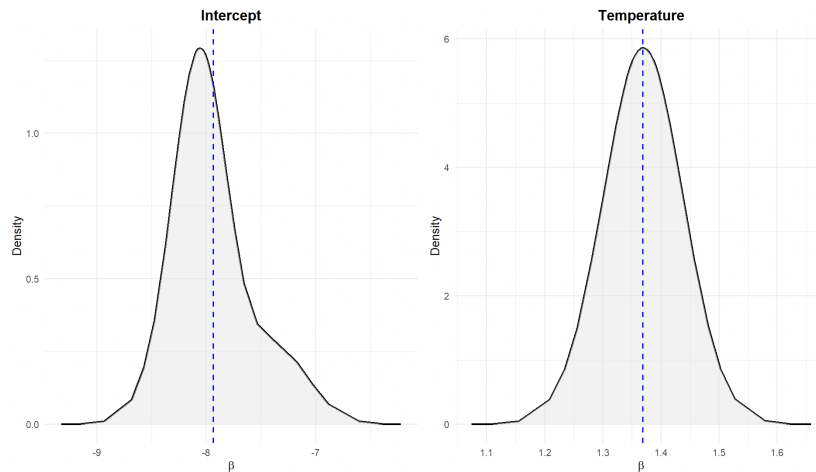


Figure 5.3. Marginal posterior densities for the β coefficients. The dashed lines represent the posterior mean of the densities.

The global intercept is estimated to be strongly negative (posterior mean = -7.940 and 95% posterior credible interval = $[-8.573, -7.032]$). Since the model includes both a latent spatial field and several non-linear covariate effects, the intercept should not be interpreted in isolation. Rather, it represents the baseline log-intensity when the linear predictor is equal to zero after accounting for all model components. Its posterior distribution is slightly right-skewed, with the posterior mean (-7.940) marginally larger than the mode (-8.038), reflecting the uncertainty associated with the decomposition between the global intercept and the structured latent effects. The posterior distribution of the temperature coefficient indicates a strong posterior support for a positive spatial association between temperature and wildfire intensity (posterior mean = 1.369 and 95% posterior credible interval = $[1.234, 1.502]$). Since the temperature covariate was standardised before model fitting, the coefficient refers to a one-standard-deviation increase in long-term temperature conditions. Holding the other model components constant, this corresponds to an estimated intensity approximately 3.9 times larger. The posterior marginal density is unimodal and nearly perfectly symmetric, with the mean, median and mode virtually coinciding. Its relatively narrow spread reflects the high precision of the estimate, while the entire 95% credible interval lies above zero, providing strong evidence for a positive effect of temperature on wildfire occurrence.

It is important to note that the temperature variable considered in this study represents long-term average climatic conditions rather than the temperature at the time of individual wildfire events. Consequently, its estimated effect should be interpreted spatially rather than temporally. The positive coefficient indicates that wildfire occurrences tend to be more frequent in areas

characterized by persistently warmer climates, after accounting for the other environmental covariates and the latent spatial field. This spatial association is consistent with the role of warmer climatic conditions in promoting higher evapotranspiration, lower fuel moisture and increased vegetation flammability, thereby creating environments that are more conducive to wildfire occurrence.

5.4.2 Random effects

Random effects are introduced to capture sources of spatial variability that are not explained by the fixed effects. In particular, they account for residual spatial dependence, as well as potential non-linear relationships between selected covariates and the intensity function.

In this model, the random component is composed of two main elements. First, a continuous latent spatial field $w(s)$, modelled through the SPDE approach, is included to capture residual spatial autocorrelation and unobserved spatial structure. Second, a set of covariates (elevation, distance from the coast, tree cover density and population) is modelled using RW2 priors, allowing for flexible non-linear functional effects while enforcing smoothness through penalisation of second differences.

The corresponding hyperparameters, including the spatial range and marginal standard deviation of the latent field, as well as the precision parameters of the RW2 components, are reported in Table 5.7.

Table 5.7. Posterior statistics for the hyperparameters of the LGCP model.

Hyperparameter	Mean	SD	0.025 quant	0.5 quant	0.975 quant	Mode
Range for space	47.413	4.311	39.600	47.182	56.561	46.650
Stdev for space	2.507	0.114	2.291	2.504	2.740	2.498
Precision for elev	0.277	0.052	0.188	0.272	0.393	0.264
Precision for distcoast	0.006	0.003	0.002	0.006	0.015	0.005
Precision for tree_cover	5.054	1.270	2.998	4.906	7.963	4.628
Precision for popul	0.244	0.120	0.079	0.221	0.539	0.177

Spatial field

The spatial dependence structure is governed by two key hyperparameters: the range parameter ρ and the marginal standard deviation σ of the latent Gaussian field.

The range parameter ρ controls the spatial scale of dependence, determining the distance beyond which the correlation between locations becomes negligible, but never zero.

The marginal standard deviation σ regulates the overall magnitude of spatial variability, quantifying the strength of residual spatial heterogeneity not explained by the covariates.

Figure 5.4 shows the marginal posterior distributions of ρ and σ , both exhibit approximately Gaussian shapes.

The posterior mean of the range parameter ($\rho_{mean} = 47.4$ km) indicates that wildfire occurrence exhibits spatial dependence at a sub-regional scale. Locations within approximately 40–50 km share a substantial degree of latent similarity, while beyond this distance spatial correlation rapidly diminishes, leading to near-independence in practical terms. Compared with the prior specification, which assigned probability 0.5 to ranges below 100 km, the posterior distribution is concentrated around shorter distances, with posterior mean approximately 47 km. This

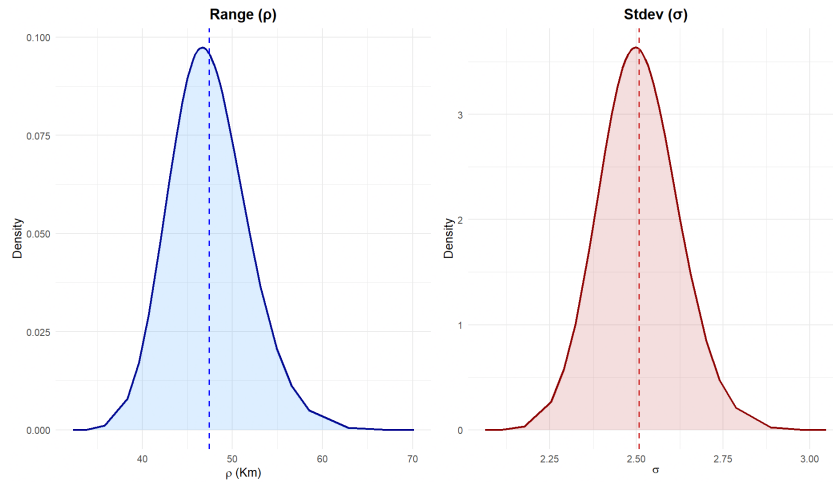


Figure 5.4. Marginal posterior densities of the SPDE parameters: range ρ (left panel) and standard deviation σ (right panel).

indicates that the data support a more local residual dependence structure than the prior median scale.

The marginal standard deviation ($\sigma_{mean} = 2.5$) quantifies the magnitude of residual spatial heterogeneity after accounting for covariate effects. The posterior distribution of σ is clearly shifted above the prior threshold of 1, suggesting substantial residual spatial variability after accounting for the selected covariates. This reflects the inherently complex and partially unobserved nature of wildfire processes, where additional fine-scale drivers and local mechanisms contribute to residual spatial clustering.

Non-linear covariate effects

The following sections examine the estimated non-linear effects of the environmental and anthropogenic covariates included in the final model. Since these variables are modelled using RW2, their estimated relationships with wildfire intensity are represented as smooth functions rather than single regression coefficients. The resulting plots therefore illustrate how the contribution of each covariate to the log-intensity varies across its range of values. The RW2 effects were constrained to sum to zero for identifiability. Consequently, positive and negative values represent deviations from the average covariate effect on the log-intensity scale.

Since RW2 models are defined on an ordered sequence of values rather than on a fully continuous covariate, each continuous predictor is discretised as reported in section 4.1.3.

To facilitate interpretation, raster representations are included selectively for some covariates. In particular, variables characterised by highly skewed distributions or non-intuitive scales, such as population and log-distance from the coast, are accompanied by maps showing their spatial distribution. Conversely, for variables with naturally interpretable scales, such as elevation and tree cover density, only the estimated effect is presented in order to avoid unnecessary redundancy and preserve visual clarity.

The interpretation focuses on the posterior mean function, while intervals overlapping zero indicate regions of the covariate range where the evidence for a positive or negative contribution is weak.

Elevation

The first covariate is elevation, which was discretised using the quantile grouping method. The covariate effect is presented against the original elevation values (metres above sea level) to facilitate interpretation.

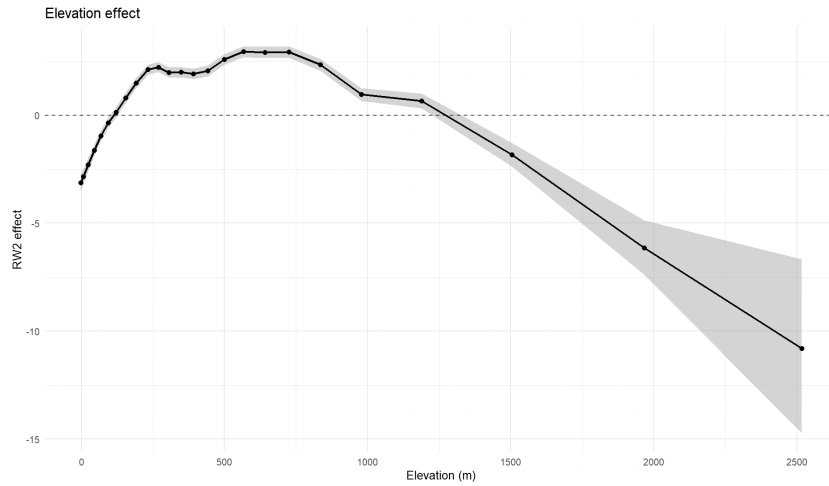


Figure 5.5. Estimated RW2 effect of elevation

Figure 5.5 shows the estimated non-linear effect of elevation on wildfire intensity. The relationship is clearly non-linear and cannot be adequately described by a single regression coefficient.

At low elevations, the estimated effect is negative, indicating a lower wildfire intensity than the baseline. The effect increases progressively with elevation, becoming positive at approximately 125 m above sea level and reaching its maximum around 750 m, where the estimated contribution to the log-intensity is highest. More generally, elevations between approximately 600 and 750 m are associated with the greatest wildfire intensity.

Above this range, the estimated effect gradually decreases and becomes negative again beyond approximately 1250 m. The decline becomes particularly pronounced at the highest elevations, where the posterior credible intervals also widen, reflecting the reduced amount of information available. Overall, the estimated relationship suggests that wildfire occurrence is most strongly associated with hilly and lower-mountain environments, whereas both lowland and high-altitude areas exhibit a lower estimated intensity. The reduced effect observed at the highest elevations is consistent with the environmental characteristics of these areas, where vegetation progressively becomes sparser and extensive rocky surfaces become more prevalent, limiting the availability of combustible fuel. In addition, the highest elevation classes must be interpreted with caution, since they contain relatively few observations within the study domain, which corresponds to wider credible bands and therefore greater uncertainty of the estimated effect.

Tree cover density

To implement the RW2 specification, the continuous tree cover covariate was first discretised into a finite number of ordered classes. In this case, equal-width intervals were used, dividing the original 0–100% range into 10 classes of constant amplitude (0–10%, 10–20%, ..., 90–100%). This choice ensures a uniform coverage of the covariate range, with each class representing the same absolute variation in vegetation density.

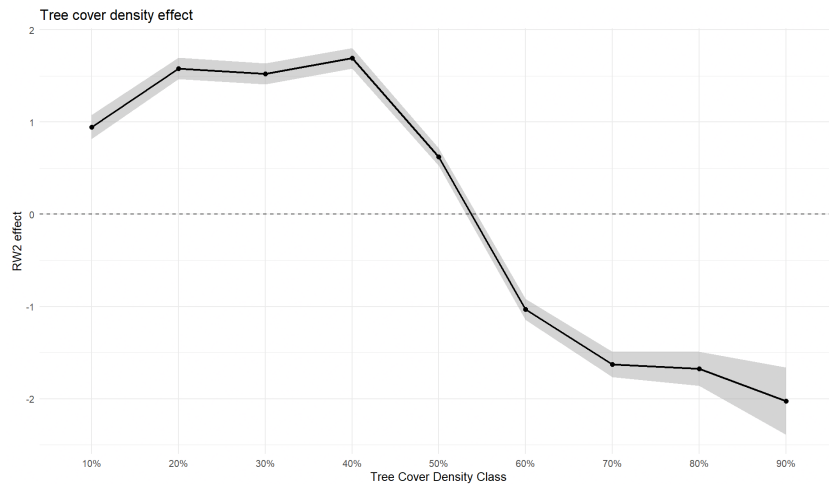


Figure 5.6. Estimated RW2 effect of tree cover density

Figure 5.6 shows the estimated non-linear effect of tree cover density on wildfire intensity. Starting from low tree cover values (0–10%), the estimated effect is positive and increases progressively up to approximately the 30–40% range, where the contribution to the log-intensity reaches its maximum. This suggests that moderately vegetated areas are associated with higher wildfire intensity compared to sparsely vegetated environments.

Beyond this point, the effect begins to decline, with a more pronounced decrease between approximately 40% and 60% tree cover. Around the 50–55% range, the estimated effect crosses zero, transitioning from positive to negative values. For higher levels of tree cover (>60%), the effect remains negative and continues to decrease, although at a slower rate, indicating a progressively lower contribution to wildfire intensity in densely forested areas. It is important to note that zero and low tree cover values should not be interpreted as absence of vegetation, as these areas may include grasslands or other land-cover types that can still be prone to fire, albeit with different fuel characteristics. Based on the posterior estimates, the environments most associated with increased wildfire intensity are those falling within the intermediate 20%–40% tree cover intervals. From a landscape perspective, these classes may correspond to transitional zones, open woodlands, or fragmented forest margins rather than dense, continuous forest stands, although this interpretation should be treated as ecological rather than directly observed from the tree-cover covariate alone.

Distance from coast

The distance from the coast covariate was discretised into 10 ordered classes using the quantile-based grouping procedure implemented in the R-INLA package. This approach assigns observations to classes such that each category contains approximately the same number of spatial units (pixels), rather than equal-width intervals along the covariate range. This ensures a balanced representation of the empirical distribution of the covariate across its full range. For interpretability, the covariate is displayed on its original log-transformed scale, while the spatial representation highlights the corresponding discretised classes.

Figure 5.7 shows the estimated non-linear effect of distance from the coast on wildfire intensity. The relationship is highly non-linear and varies substantially across the spatial gradient from coastal to inland.

At short distances from the coast, the estimated effect is positive and increasing, indicating

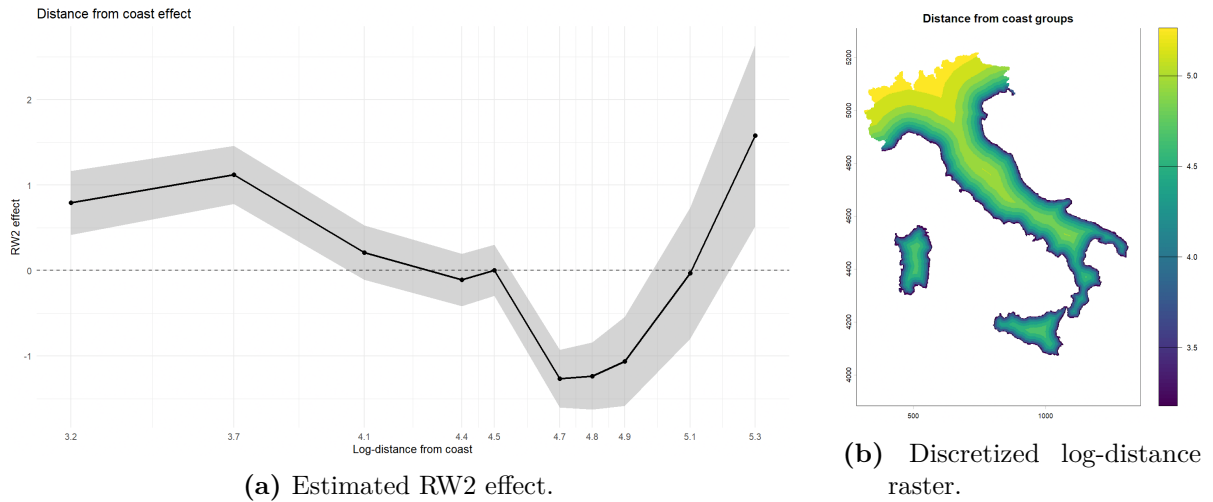


Figure 5.7. Distance from coast covariate: (a) non-linear posterior effect estimated by the RW2 model, and (b) spatial distribution of the corresponding 10 quantile-based classes across the study area.

higher wildfire intensity in coastal and immediately inland areas. Moving further inland, the effect gradually decreases, crossing zero at approximately 4.5 on the log-distance scale, corresponding at 31.6km, and becoming negative around 4.6–4.7 ($\approx 40 - 50\text{km}$). This suggests a local minimum in wildfire intensity in intermediate inland zones.

Beyond this range, the effect increases again, returning to positive values around 5.1 ($\approx 126\text{km}$), corresponding to more internal and high-altitude regions. In this portion of the covariate range, the estimated uncertainty also increases. The estimated effect highlights a complex, non-monotonic relationship between wildfire intensity and distance from the coast. However, given the nature of this covariate, it should be interpreted as a spatial proxy that may capture multiple underlying factors, including but not limited to land use patterns, accessibility and broad-scale environmental gradients. For this reason, the observed pattern should not be interpreted in a strictly causal or single-mechanism sense, but rather as a summary of how wildfire intensity varies along this composite coastal–inland continuum, as learned by the RW2 specification.

Population

Among the covariates included in the model, population is arguably the most challenging to interpret and therefore deserves a more detailed discussion.

Unlike the environmental covariates discussed previously, population is intrinsically a more dynamic concept. A sparsely populated natural area, located close to a densely populated region, is indeed likely to experience a greater degree of human presence than an equally sparsely populated area situated far from major settlements. For example, a natural green area adjacent to a large city is generally expected to be more frequently visited than a similar area located in a remote region.

For this reason, the smoothing procedure is particularly useful. In addition to producing a more homogeneous raster and reducing abrupt local changes, it assigns higher values to sparsely populated natural areas located near densely populated ones by spatially spreading, in a weighted manner, the value recorded in each pixel to its neighbouring cells.

Urban areas themselves are excluded from the modelling domain because they do not belong to

the natural environments, as considered in this study. Nevertheless, this approach allows the model to account for the proximity of natural pixels to urban agglomerations, recognising that wildfire occurrence is influenced not only by the resident population within a pixel but also by the ease with which that location can be accessed by people. After smoothing, the population covariate should be interpreted as a local indicator of surrounding human presence rather than as the resident population strictly contained in each natural pixel.

Following the smoothing procedure, the covariate was discretised into 10 classes using the quantile-based method. The results are then presented graphically. For interpretability, the estimated effects are shown against the log-transformed population class values, and the corresponding raster is displayed alongside to provide spatial context and facilitate interpretation of the class locations.

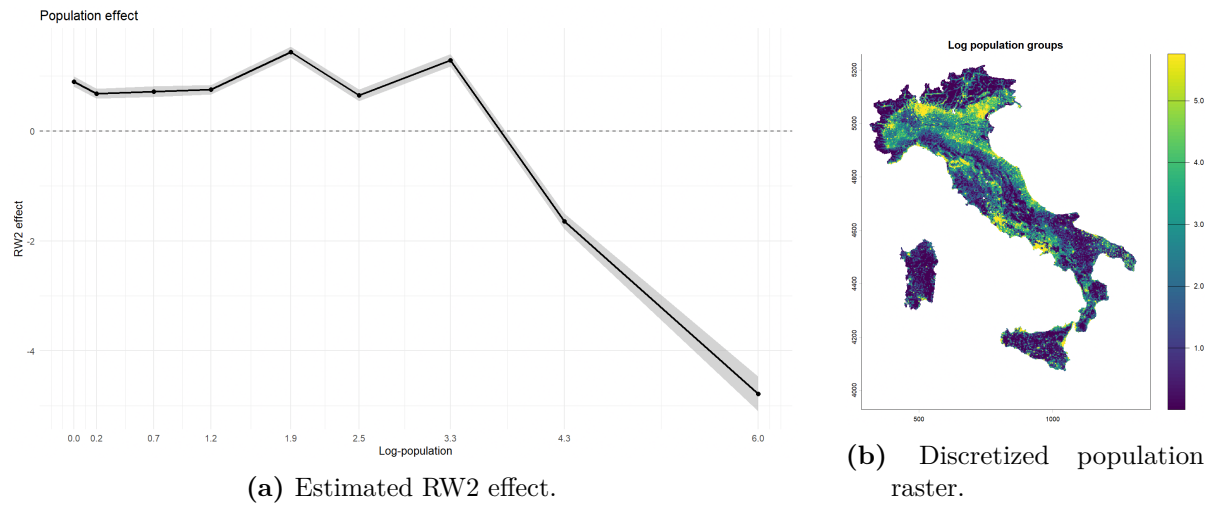


Figure 5.8. Population covariate: (a) non-linear posterior effect estimated by the RW2 model, and (b) spatial distribution of the corresponding 10 quantile-based classes across the study area.

The estimated effect, as shown in Figure 5.8, is positive and oscillatory in the first portion of the covariate range, reaching local maxima around values 1.9 and 3.3. The effect crosses zero at approximately 3.8. Beyond this point, the relationship shows a marked decline, decreasing at a relatively fast rate. This contrasts with the initial segment of the curve, where the effect fluctuates but remains overall more stable.

For the population covariate, two complementary interpretations should be considered. On the one hand, higher human presence may increase the probability of ignition due to anthropogenic activities. On the other hand, more densely populated areas are typically characterised by greater landscape fragmentation, higher accessibility and potentially more effective detection and suppression capabilities.

In addition, in areas where human presence is persistent and intense, rapid fire suppression may prevent fires from spreading and reaching sizes detectable by satellite observations. For this reason, the sign of the effect is not straightforward a priori and should be interpreted with caution.

A further limitation relates to the definition of the spatial unit used in the analysis. The dataset is based on the centroid of each fire polygon, which generally corresponds to the geometrical centre of the burned area rather than the ignition point. As a result, the observed location may be shifted towards more interior portions of the fire extent. Had ignition points been available, the relationship with population might have shown a stronger association with infrastructure

or road networks, consistent with the well-known predominance of anthropogenic causes of wildfires discussed in Chapter 1.

5.5 Estimated spatial intensity

This section presents the posterior mean of the fitted log-intensity function of wildfire occurrence over the study domain. The map represents the combined effect of fixed covariates, non-linear components and the latent spatial field.

Log intensity

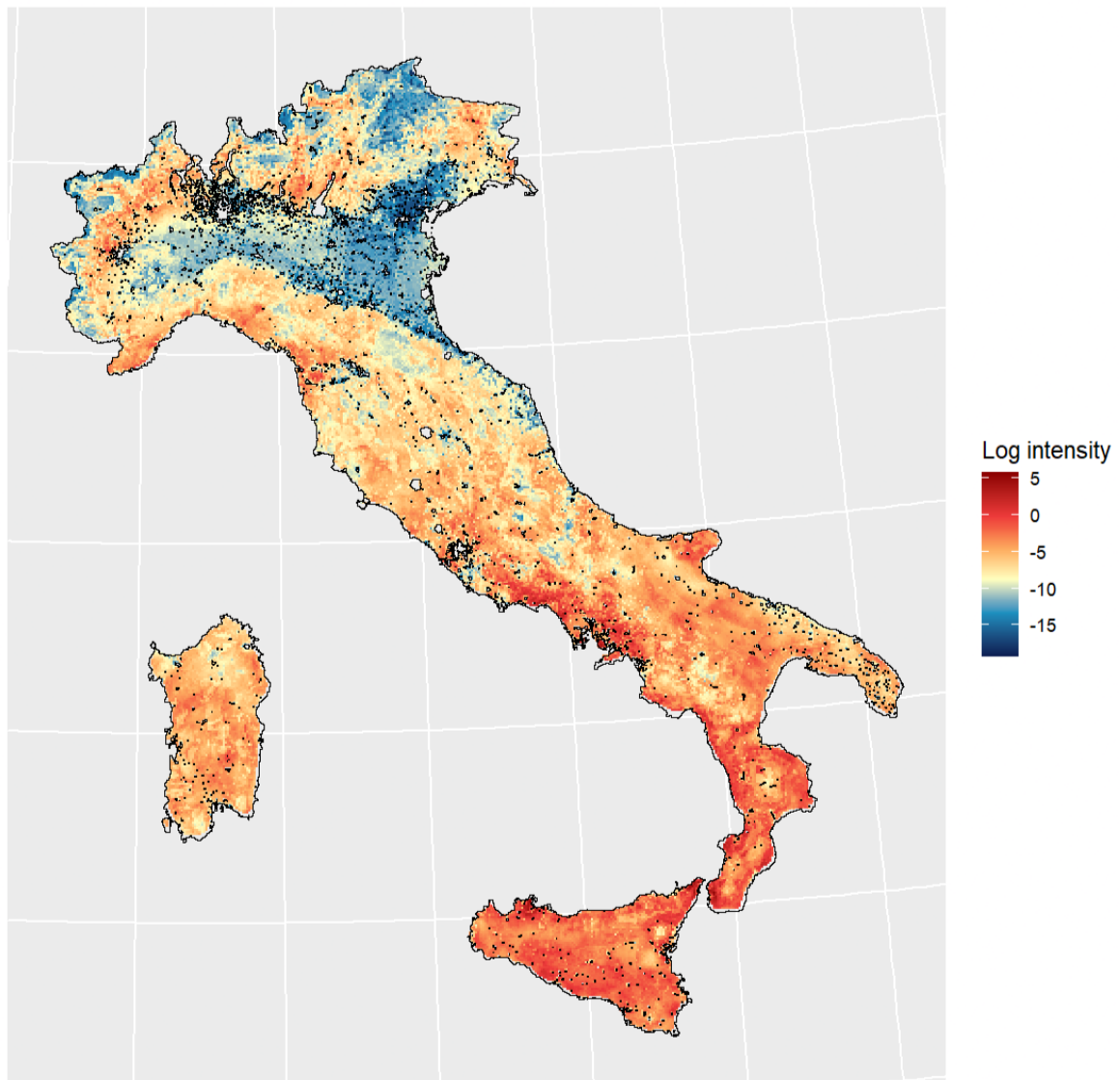


Figure 5.9. Estimated log intensity of wildfires occurrence

The resulting log-intensity can be interpreted as a relative indicator of wildfire occurrence intensity across space, rather than as an absolute probability of fire ignition or burned-area severity. Because the map is shown on the log-intensity scale, differences between areas correspond to multiplicative differences in the expected intensity on the original scale.

The estimated log-intensity map is computed only over the wildfire domain, since assigning wildfire intensity to areas where the phenomenon is not defined would not be meaningful.

The resulting spatial pattern reveals a marked heterogeneity in the estimated intensity of wildfire occurrence across Italy. Higher log-intensity values are concentrated in the southern regions, particularly in Sicily, Calabria, Campania and Lazio, indicating a greater expected occurrence of wildfire events. Sardinia and Apulia also exhibit predominantly medium-to-high estimated values. In contrast, a large portion of northern Italy, especially along the Alpine arc and throughout the Po Valley, is characterised by lower log-intensity values, suggesting a lower expected wildfire intensity.

Although the Alpine region generally displays low estimated values, a belt of medium-to-high log-intensity is visible along the foothill and pre-Alpine areas, extending from eastern Piedmont through Lombardy to Veneto. This indicates that the reduction in wildfire intensity is not spatially uniform across northern Italy but is interrupted by a corridor of comparatively higher expected occurrence.

Central Italy exhibits a more heterogeneous pattern, with alternating areas of medium-to-low intensity and locally higher values.

Overall, the map highlights a clear north–south gradient, with estimated log-intensity generally increasing towards southern and insular regions.

This spatial configuration is broadly consistent with the effects previously estimated for the environmental covariates, particularly temperature and elevation. Areas characterised by warmer climatic conditions and intermediate elevations tend to exhibit higher estimated wildfire intensity, whereas both lowland and high-mountain environments are generally associated with lower values.

5.5.1 Latent spatial field

Figure 5.10 presents the spatial component of the estimated log-intensity, as captured by the latent Gaussian random field. This term represents the residual spatial variation in wildfire occurrence after accounting for the effects of the observed covariates. Its values should therefore be interpreted as spatially structured deviations from the covariate-driven mean intensity, rather than as absolute measures of wildfire risk. Unlike the fitted log-intensity surface, the latent field does not represent wildfire intensity itself but isolates residual spatial structure. Positive values indicate locations where the observed intensity exceeds that predicted by covariates, while negative values indicate the opposite.

The latent field exhibits pronounced spatial heterogeneity, characterised by alternating positive and negative values across the study domain. This pattern indicates the presence of residual spatial dependence at local and regional scales that is not fully explained by the included covariates. These residual patterns should not be attributed to specific omitted drivers without additional evidence. They may reflect unmeasured environmental or anthropogenic factors, local data limitations, centroid approximation, temporal aggregation or remaining model misspecification.

At the regional level, northern Italy shows positive residual clusters along parts of the Alpine arc, interspersed with negative contributions in the adjacent foothill areas. Central Italy displays a mixed structure, with positive residuals along sections of the Tyrrhenian side and negative values along portions of the Apennine ridge. In southern Italy, the field is more fragmented, with alternating positive and negative contributions along coastal and inland areas. Sardinia is predominantly characterised by negative residuals, which indicate that the spatial random component reduces the fitted log-intensity relative to the covariate-only component.

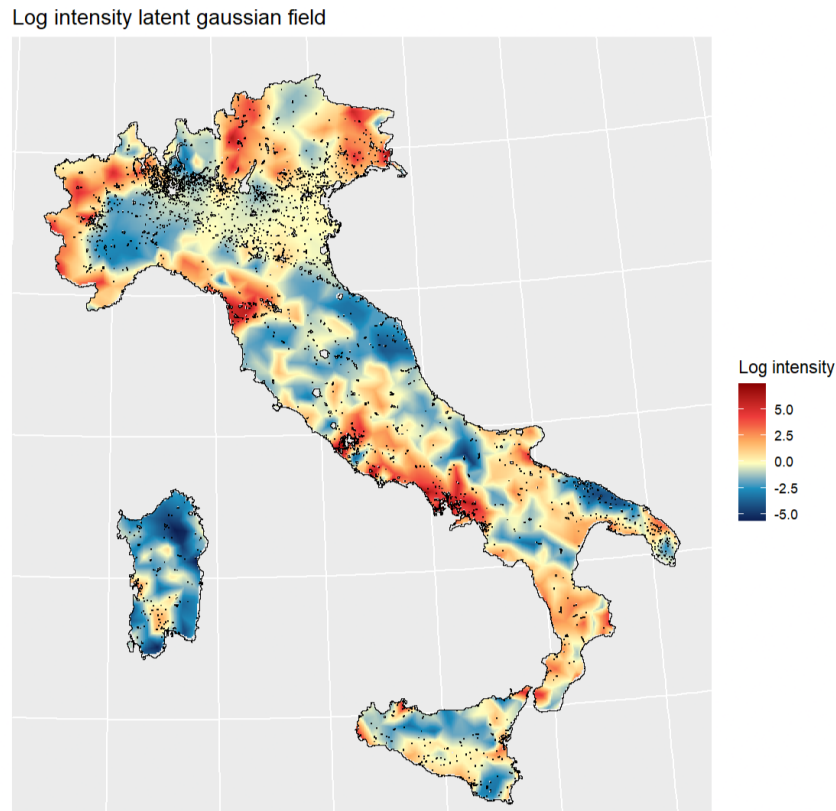


Figure 5.10. Estimated log intensity of latent gaussian random field

Comparison with the fitted log-intensity highlights that the large-scale north–south gradient is substantially reduced in the latent component. The spatial field instead reveals a predominantly local and oscillatory structure, suggesting that broad-scale variation is largely captured by the covariates, whereas the latent process accounts for finer-scale dependence and unobserved spatial heterogeneity.

From a regional perspective, Sicily, despite being among the areas with the highest observed wildfire occurrence, exhibits mostly neutral or negative residuals, with only limited localised positive structures. This indicates that the covariate structure captures a substantial portion of the observed intensity in this region. In contrast, Campania, parts of Lazio and Tuscany, and selected areas in northern Italy display positive residuals, suggesting localised excess intensity not fully explained by the covariates.

Chapter 6

Conclusions

This study proposed a Log-Gaussian Cox Process (LGCP) framework for characterising the long-term spatial intensity of wildfire occurrence across Italy. Rather than providing short-term fire forecasts or a comprehensive wildfire risk assessment, the proposed methodology focuses on modelling the spatial intensity of wildfire occurrence as a point process, allowing the contribution of environmental and anthropogenic drivers to be quantified while accounting for residual spatial dependence through a latent Gaussian field.

The proposed LGCP framework provides a data-driven approach for characterising wildfire occurrence across large spatial domains. By modelling wildfire events as a spatial point pattern, the analysis preserves the information contained in their geographical distribution and allows the contribution of environmental and anthropogenic drivers to be quantified without imposing predefined relationships between covariates and wildfire occurrence. This represents one of the main strengths of the proposed methodology. Unlike approaches in which risk classes or variable weights are specified a priori, the statistical model allows these relationships to emerge directly from the observed data, making it a valuable tool not only for producing spatial intensity maps but also for exploring the ecological processes associated with wildfire occurrence. The use of harmonised Earth Observation products further strengthens this framework by ensuring consistency across the study area and supporting its potential application to other European countries where comparable datasets are available.

The analysis highlighted that the broad spatial gradient of wildfire occurrence across Italy is largely explained by the selected environmental and anthropogenic variables, while the latent Gaussian field mainly captures additional local-scale spatial variability. From an ecological perspective, the residual structures identified in several northern regions are particularly interesting. Although these patterns cannot be directly attributed to specific processes, they suggest the presence of additional drivers that are not fully represented by the available covariates. One possible explanation is that long-term climatic averages are unable to capture more recent or localised environmental changes. These residual patterns therefore represent an interesting direction for future research and motivate the inclusion of higher-resolution climatic information and additional explanatory variables.

Several limitations should nevertheless be acknowledged. The model describes long-term spatial variation in wildfire occurrence intensity and therefore does not explicitly account for the temporal dynamics associated with individual wildfire events. Consequently, some variables, such as precipitation, would likely provide substantially greater explanatory power if considered at the temporal scale of fire occurrence rather than as long-term spatial averages. Similarly, wildfire locations were represented by burned-area centroids instead of ignition points, an

approximation that may influence the estimated relationship with anthropogenic variables related to accessibility and human presence. Finally, model selection and refinement of the functional form were performed on the same dataset, and no external validation was carried out. These aspects should be considered when interpreting the estimated relationships, while also representing natural directions for further methodological development.

Several opportunities exist to further extend the proposed framework. The most natural development consists of moving from a purely spatial to a spatio-temporal LGCP, capable of describing seasonal dynamics, climate-driven variability and changes in human activity over time. The availability of ignition-point data would also allow a more realistic representation of anthropogenic drivers, including proximity to roads, settlements and other infrastructures. Furthermore, extending the response variable to include burned area or fire severity would allow the framework to move beyond wildfire occurrence towards a more comprehensive characterisation of wildfire impacts. Future work should also include an external validation phase to evaluate the predictive performance and transferability of the proposed methodology.

Overall, the spatial intensity maps produced in this study provide a rigorous framework for analysing the spatial heterogeneity of wildfire occurrence in Italy. By identifying environmental conditions associated with increased occurrence intensity, quantifying the effects of the selected covariates and highlighting areas characterised by unexplained residual spatial variability, the proposed methodology contributes to a better understanding of wildfire dynamics and provides a reproducible framework that may support future ecological research, land management, wildfire prevention planning and the allocation of fire suppression resources.

Appendix

Table 6.1. Complete model selection results for all LGCP models ordered by WAIC.

ID	N_{cov}	Included Variables	WAIC	DIC	Non-Significant	ΔWAIC
395	6	elev, latitude, dist_coast, temp, tree_cov, popul	80130.83	35581.74	latitude	0.0
469	7	elev, latitude, dist_coast, precip, temp, tree_cov, popul	80134.54	35554.46	latitude, precip	3.71
478	7	elev, latitude, dist_coast, temp, tree_cov, solar, popul	80136.1	35620.87	latitude	5.27
504	8	elev, latitude, dist_coast, precip, temp, tree_cov, solar, popul	80231.81	35584.2	latitude, precip	100.98
304	5	elev, dist_coast, temp, tree_cov, popul	80471.21	35474.09	—	340.38
429	6	elev, dist_coast, temp, tree_cov, solar, popul	80495.59	35421.62	—	364.76
489	7	elev, dist_coast, precip, temp, tree_cov, solar, popul	80499.16	35436.96	precip	368.34
420	6	elev, dist_coast, precip, temp, tree_cov, popul	80523.8	35476.3	precip	392.97
389	6	elev, latitude, dist_coast, precip, tree_cov, popul	80718.67	36314.09	precip	587.85
268	5	elev, latitude, dist_coast, tree_cov, popul	80734.1	36305.6	—	603.27
400	6	elev, latitude, dist_coast, tree_cov, solar, popul	80736.38	36234.14	solar	605.55
474	7	elev, latitude, dist_coast, precip, tree_cov, solar, popul	80743.56	36287.17	precip, solar	612.73
163	4	elev, dist_coast, tree_cov, popul	80809.39	36955.22	—	678.56
298	5	elev, dist_coast, precip, tree_cov, popul	81006.85	36752.05	—	876.02
309	5	elev, dist_coast, tree_cov, solar, popul	81076.47	36890.11	solar	945.65
425	6	elev, dist_coast, precip, tree_cov, solar, popul	81251.69	36648.21	solar	1120.86
284	5	elev, latitude, temp, tree_cov, popul	81252.89	38165.19	—	1122.06
414	6	elev, latitude, temp, tree_cov, solar, popul	81272.6	37873.61	latitude, solar	1141.77
405	6	elev, latitude, precip, temp, tree_cov, popul	81526.45	38281.07	latitude	1395.62
483	7	elev, latitude, precip, temp, tree_cov, solar, popul	81696.7	38230.13	latitude, solar	1565.87
434	6	elev, precip, temp, tree_cov, solar, popul	82061.47	38012.77	solar	1930.64
29	2	dist_coast, solar	82069.53	49091.46	solar	1938.7
314	5	elev, precip, temp, tree_cov, popul	82251.69	38306.13	—	2120.86
323	5	elev, temp, tree_cov, solar, popul	82267.95	38709.27	solar	2137.13
98	3	dist_coast, precip, solar	82320.54	49065.72	precip, solar	2189.71
179	4	elev, temp, tree_cov, popul	82339.16	38871.74	—	2208.33
78	3	latitude, dist_coast, solar	82426.08	49034.11	—	2295.25
131	4	elev, latitude, dist_coast, precip	82616.31	45635.59	precip	2485.48
503	8	elev, latitude, dist_coast, precip, temp, tree_cov, solar, human_press	82651.53	38334.13	precip	2520.71
154	4	elev, dist_coast, precip, solar	82656.77	45767.57	precip, solar	2525.94
477	7	elev, latitude, dist_coast, temp, tree_cov, solar, human_press	82666.52	38354.97	—	2535.69
4	1	dist_coast	82681.51	49040.47	—	2550.68
26	2	dist_coast, precip	82688.11	49031.26	precip	2557.28
54	3	elev, dist_coast, precip	82705.2	45827.45	precip	2574.37
259	5	elev, latitude, dist_coast, precip, solar	82761.83	45602.68	precip, solar	2631.0
428	6	elev, dist_coast, temp, tree_cov, solar, human_press	82776.92	38339.44	—	2646.09
488	7	elev, dist_coast, precip, temp, tree_cov, solar, human_press	82782.78	38318.55	—	2651.95
19	2	latitude, dist_coast	82788.75	49017.03	—	2657.93
134	4	elev, latitude, dist_coast, solar	82802.44	45629.68	solar	2671.61
57	3	elev, dist_coast, solar	82808.74	45713.41	solar	2677.91
75	3	latitude, dist_coast, precip	82905.97	48999.77	—	2775.14
189	4	latitude, dist_coast, precip, solar	82985.97	48964.49	solar	2855.14
12	2	elev, dist_coast	83048.7	45784.51	—	2917.87
289	5	elev, latitude, tree_cov, solar, popul	83077.17	38173.81	solar	2946.34
47	3	elev, latitude, dist_coast	83134.3	45680.78	—	3003.47
148	4	elev, latitude, tree_cov, popul	83539.95	38213.51	—	3409.12
467	7	elev, latitude, dist_coast, precip, temp, tree_cov, solar	83557.29	41465.55	precip	3426.46
418	6	elev, dist_coast, precip, temp, tree_cov, solar	83557.53	41454.75	—	3426.7
302	5	elev, dist_coast, temp, tree_cov, solar	83629.67	41497.35	—	3498.84
303	5	elev, dist_coast, temp, tree_cov, human_press	83664.78	38458.57	—	3533.96
410	6	elev, latitude, precip, tree_cov, solar, popul	83665.81	38164.66	precip, solar	3534.98
419	6	elev, dist_coast, precip, temp, tree_cov, human_press	83690.28	38448.78	precip	3559.45
394	6	elev, latitude, dist_coast, temp, tree_cov, human_press	83715.97	38493.26	—	3585.15
393	6	elev, latitude, dist_coast, temp, tree_cov, solar	83722.35	41492.72	—	3591.52
468	7	elev, latitude, dist_coast, precip, temp, tree_cov, human_press	83733.69	38485.19	precip	3602.86
278	5	elev, latitude, precip, tree_cov, popul	83809.05	38286.59	precip	3678.22
455	6	latitude, precip, temp, tree_cov, solar, popul	83897.09	41992.09	precip	3766.26

Table 6.1. Complete model selection results for all LGCP models ordered by WAIC.

ID	N_{cov}	Included Variables	WAIC	DIC	Non-Significant	Δ WAIC
257	5	elev, latitude, dist_coast, precip, temp	83906.27	44258.71	precip	3775.45
383	6	elev, latitude, dist_coast, precip, temp, tree_cov	83910.89	41639.98	latitude, precip	3780.06
292	5	elev, dist_coast, precip, temp, tree_cov	83934.03	41631.99	precip	3803.2
446	6	latitude, dist_coast, precip, tree_cov, solar, popul	83960.8	40529.82	solar	3829.97
262	5	elev, latitude, dist_coast, temp, tree_cov	83981.77	41638.29	latitude	3850.95
157	4	elev, dist_coast, temp, tree_cov	84048.12	41619.65	—	3917.29
357	5	latitude, temp, tree_cov, solar, human_press	84059.29	45575.43	—	3928.46
373	5	dist_coast, temp, tree_cov, solar, popul	84062.13	39925.53	—	3931.3
461	6	dist_coast, precip, temp, tree_cov, solar, popul	84063.09	39924.77	precip	3932.26
454	6	latitude, precip, temp, tree_cov, solar, human_press	84072.29	45563.49	precip	3941.46
253	4	temp, tree_cov, solar, popul	84091.35	41950.64	—	3960.52
213	4	latitude, temp, tree_cov, human_press	84149.88	45856.5	—	4019.05
348	5	latitude, precip, temp, tree_cov, human_press	84165.79	45854.2	precip	4034.96
496	7	latitude, dist_coast, precip, temp, tree_cov, solar, popul	84224.34	39981.74	precip	4093.51
252	4	temp, tree_cov, solar, human_press	84264.29	45471.03	—	4133.46
441	6	latitude, dist_coast, precip, temp, tree_cov, popul	84303.18	40112.09	precip	4172.36
293	5	elev, dist_coast, precip, temp, solar	84305.17	44086.81	precip, solar	4174.34
377	5	precip, temp, tree_cov, solar, human_press	84322.68	45458.48	precip	4191.86
152	4	elev, dist_coast, precip, temp	84325.51	44294.31	precip	4194.68
378	5	precip, temp, tree_cov, solar, popul	84334.36	41932.06	precip	4203.53
384	6	elev, latitude, dist_coast, precip, temp, solar	84335.28	44150.98	precip, solar	4204.45
161	4	elev, dist_coast, tree_cov, solar	84356.63	43500.81	—	4225.8
158	4	elev, dist_coast, temp, solar	84372.55	44107.11	solar	4241.72
473	7	elev, latitude, dist_coast, precip, tree_cov, solar, human_press	84395.02	39101.69	precip	4264.2
450	6	latitude, dist_coast, temp, tree_cov, solar, popul	84399.8	39988.84	—	4268.97
358	5	latitude, temp, tree_cov, solar, popul	84418.18	41956.11	—	4287.35
122	3	temp, tree_cov, human_press	84449.23	45849.24	—	4318.4
243	4	precip, temp, tree_cov, human_press	84478.48	45848.2	precip	4347.65
266	5	elev, latitude, dist_coast, tree_cov, solar	84504.04	42664.94	—	4373.21
333	5	latitude, dist_coast, precip, tree_cov, popul	84520.19	40550.99	—	4389.36
353	5	latitude, precip, tree_cov, solar, human_press	84526.95	45907.21	—	4396.12
319	5	elev, precip, tree_cov, solar, popul	84532.61	40662.82	solar	4401.78
387	6	elev, latitude, dist_coast, precip, tree_cov, solar	84533.65	42664.16	precip	4402.82
263	5	elev, latitude, dist_coast, temp, solar	84536.99	44176.76	solar	4406.16
424	6	elev, dist_coast, precip, tree_cov, solar, human_press	84538.77	39477.19	—	4407.95
132	4	elev, latitude, dist_coast, temp	84556.6	44332.29	—	4425.77
296	5	elev, dist_coast, precip, tree_cov, solar	84569.99	43328.64	precip	4439.16
223	4	dist_coast, precip, temp, solar	84570.79	47560.54	precip	4439.96
218	4	latitude, tree_cov, solar, human_press	84601.61	45933.76	—	4470.78
339	5	latitude, dist_coast, temp, tree_cov, popul	84635.02	40114.46	—	4504.19
399	6	elev, latitude, dist_coast, tree_cov, solar, human_press	84665.68	39131.08	—	4534.85
344	5	latitude, dist_coast, tree_cov, solar, popul	84684.79	40609.05	solar	4553.96
55	3	elev, dist_coast, temp	84693.43	44355.1	—	4562.61
332	5	latitude, dist_coast, precip, tree_cov, human_press	84817.59	43495.92	precip	4686.76
267	5	elev, latitude, dist_coast, tree_cov, human_press	84862.8	39110.31	—	4731.98
193	4	latitude, dist_coast, temp, solar	84873.76	47546.37	—	4742.93
248	4	precip, tree_cov, solar, human_press	84890.49	45983.29	precip	4759.66
388	6	elev, latitude, dist_coast, precip, tree_cov, human_press	84903.95	39103.06	precip	4773.12
197	4	latitude, dist_coast, tree_cov, human_press	84913.29	43487.81	—	4782.46
102	3	dist_coast, temp, solar	84915.13	47574.97	—	4784.31
127	3	tree_cov, solar, human_press	84931.71	45992.29	—	4800.88
187	4	latitude, dist_coast, precip, temp	84943.35	47641.26	precip	4812.52
445	6	latitude, dist_coast, precip, tree_cov, solar, human_press	84944.47	43451.7	precip, solar	4813.64
76	3	latitude, dist_coast, temp	84963.16	47639.56	—	4832.33
207	4	latitude, precip, tree_cov, human_press	84963.17	46083.99	precip	4832.34
239	4	dist_coast, tree_cov, solar, popul	84965.95	40696.41	—	4835.12
369	5	dist_coast, precip, tree_cov, solar, popul	84973.59	40694.75	precip	4842.76
328	5	latitude, dist_coast, precip, temp, solar	84973.89	47527.88	precip	4843.06
27	2	dist_coast, temp	84979.75	47653.97	—	4848.92
198	4	latitude, dist_coast, tree_cov, popul	84986.13	40601.06	—	4855.3
96	3	dist_coast, precip, temp	84988.22	47654.2	precip	4857.39
343	5	latitude, dist_coast, tree_cov, solar, human_press	85006.57	43459.04	—	4875.74
234	4	dist_coast, temp, tree_cov, popul	85017.26	40049.57	—	4886.43
364	5	dist_coast, precip, temp, tree_cov, popul	85026.08	40049.87	precip	4895.25
440	6	latitude, dist_coast, precip, temp, tree_cov, human_press	85035.71	43038.66	precip	4904.88
495	7	latitude, dist_coast, precip, temp, tree_cov, solar, human_press	85073.23	43035.28	precip, solar	4942.4
308	5	elev, dist_coast, tree_cov, solar, human_press	85076.45	40002.94	—	4945.63
91	3	latitude, tree_cov, human_press	85082.77	46057.08	—	4951.94
338	5	latitude, dist_coast, temp, tree_cov, human_press	85126.8	43040.2	—	4995.97
206	4	latitude, precip, tree_cov, solar	85145.98	48442.32	—	5015.15
233	4	dist_coast, temp, tree_cov, human_press	85162.85	43054.23	—	5032.02
449	6	latitude, dist_coast, temp, tree_cov, solar, human_press	85163.52	43034.23	solar	5032.69
7	1	tree_cov	85194.07	48598.49	—	5063.24
106	3	dist_coast, tree_cov, human_press	85197.46	43601.2	—	5066.63
227	4	dist_coast, precip, tree_cov, human_press	85209.27	43591.35	precip	5078.44

Table 6.1. Complete model selection results for all LGCP models ordered by WAIC.

ID	N_{cov}	Included Variables	WAIC	DIC	Non-Significant	ΔWAIC
372	5	dist_coast, temp, tree_cov, solar, human_press	85219.04	43037.36	solar	5088.21
80	3	latitude, dist_coast, popul	85259.89	45510.98	—	5129.06
460	6	dist_coast, precip, temp, tree_cov, solar, human_press	85271.53	43047.6	precip, solar	5140.7
363	5	dist_coast, precip, temp, tree_cov, human_press	85289.17	43055.28	precip	5158.34
153	4	elev, dist_coast, precip, tree_cov	85326.03	43612.22	precip	5195.2
42	2	tree_cov, human_press	85335.7	46121.39	—	5204.87
116	3	precip, tree_cov, human_press	85344.39	46120.89	precip	5213.56
368	5	dist_coast, precip, tree_cov, solar, human_press	85359.07	43552.24	precip, solar	5228.24
238	4	dist_coast, tree_cov, solar, human_press	85390.44	43561.24	solar	5259.61
33	2	precip, tree_cov	85435.81	48570.12	precip	5304.98
31	2	dist_coast, popul	85454.83	45546.96	—	5324.0
82	3	latitude, precip, tree_cov	85457.18	48485.52	—	5326.35
173	4	elev, precip, tree_cov, popul	85466.76	40424.48	—	5335.93
22	2	latitude, tree_cov	85473.58	48502.08	—	5342.75
192	4	latitude, dist_coast, temp, tree_cov	85483.97	45101.99	—	5353.14
115	3	precip, tree_cov, solar	85499.75	48517.21	precip	5368.92
41	2	tree_cov, solar	85501.59	48521.01	—	5370.76
327	5	latitude, dist_coast, precip, temp, tree_cov	85504.09	45105.67	precip	5373.26
306	5	elev, dist_coast, temp, solar, popul	85533.1	40235.39	—	5402.27
184	4	elev, tree_cov, solar, popul	85555.41	40696.54	solar	5424.58
107	3	dist_coast, tree_cov, popul	85574.96	40651.17	—	5444.13
228	4	dist_coast, precip, tree_cov, popul	85602.45	40640.64	precip	5471.62
90	3	latitude, tree_cov, solar	85604.91	48452.3	—	5474.08
56	3	elev, dist_coast, tree_cov	85609.31	43767.28	—	5478.48
191	4	latitude, dist_coast, precip, popul	85610.69	45426.45	—	5479.86
162	4	elev, dist_coast, tree_cov, human_press	85629.79	39654.43	—	5498.96
297	5	elev, dist_coast, precip, tree_cov, human_press	85635.03	39807.83	—	5504.2
1	0	—	85637.78	50769.23	—	5506.95
258	5	elev, latitude, dist_coast, precip, tree_cov	85670.59	43001.15	precip	5539.76
133	4	elev, latitude, dist_coast, tree_cov	85710.99	43020.8	—	5580.16
100	3	dist_coast, precip, popul	85745.64	45518.0	—	5614.81
235	4	dist_coast, temp, solar, human_press	85752.76	46356.91	—	5621.93
470	7	elev, latitude, dist_coast, precip, temp, solar, human_press	85776.63	41704.55	precip	5645.8
422	6	elev, dist_coast, precip, temp, solar, popul	85787.5	40184.95	—	5656.67
101	3	dist_coast, temp, tree_cov	85791.08	45057.98	—	5660.25
397	6	elev, latitude, dist_coast, temp, solar, popul	85802.77	40269.49	—	5671.94
362	5	dist_coast, precip, temp, tree_cov, solar	85829.99	45065.51	—	5699.16
222	4	dist_coast, precip, temp, tree_cov	85856.56	45101.48	precip	5725.74
396	6	elev, latitude, dist_coast, temp, solar, human_press	85878.83	41753.25	—	5748.01
312	5	elev, precip, temp, tree_cov, solar	85891.15	43148.54	—	5760.32
403	6	elev, latitude, precip, temp, tree_cov, solar	85896.22	43122.81	latitude, precip	5765.39
232	4	dist_coast, temp, tree_cov, solar	85913.3	45054.42	—	5782.47
282	5	elev, latitude, temp, tree_cov, solar	85928.46	43117.83	—	5797.63
160	4	elev, dist_coast, temp, popul	85934.26	40695.1	—	5803.43
272	5	elev, latitude, precip, temp, tree_cov	85948.23	43127.69	latitude, precip	5817.4
265	5	elev, latitude, dist_coast, temp, popul	85948.52	40727.91	—	5817.7
365	5	dist_coast, precip, temp, solar, human_press	85950.2	46346.38	precip	5819.37
305	5	elev, dist_coast, temp, solar, human_press	85952.66	41684.83	—	5821.83
142	4	elev, latitude, temp, tree_cov	85972.42	43127.65	latitude	5841.59
295	5	elev, dist_coast, precip, temp, popul	85985.69	40697.7	—	5854.86
421	6	elev, dist_coast, precip, temp, solar, human_press	85995.63	41677.39	precip	5864.8
13	2	elev, precip	85996.95	47003.25	precip	5866.12
177	4	elev, temp, tree_cov, solar	86012.15	43135.22	—	5881.32
71	3	elev, tree_cov, popul	86017.47	40559.53	—	5886.64
337	5	latitude, dist_coast, temp, tree_cov, solar	86032.59	45056.48	—	5901.76
171	4	elev, precip, tree_cov, solar	86033.7	45307.59	precip	5902.87
167	4	elev, precip, temp, tree_cov	86034.06	43129.26	precip	5903.23
439	6	latitude, dist_coast, precip, temp, tree_cov, solar	86048.95	45053.11	precip	5918.12
366	5	dist_coast, precip, temp, solar, popul	86071.25	44464.16	—	5940.42
2	1	elev	86075.22	46875.43	—	5944.39
121	3	temp, tree_cov, solar	86087.75	47474.19	solar	5956.92
5	1	precip	86137.56	50724.45	—	6006.74
471	7	elev, latitude, dist_coast, precip, temp, solar, popul	86157.12	40231.45	—	6026.29
65	3	elev, temp, tree_cov	86158.68	43122.65	—	6027.85
81	3	latitude, precip, temp	86163.2	49880.49	—	6032.37
386	6	elev, latitude, dist_coast, precip, temp, popul	86204.68	40729.61	—	6073.85
340	5	latitude, dist_coast, temp, solar, human_press	86213.76	46376.45	—	6082.93
23	2	latitude, solar	86253.37	50715.75	solar	6122.54
138	4	elev, latitude, precip, tree_cov	86262.22	43930.31	precip	6131.39
442	6	latitude, dist_coast, precip, temp, solar, human_press	86270.3	46343.79	—	6139.47
242	4	precip, temp, tree_cov, solar	86276.9	47472.58	precip, solar	6146.07
62	3	elev, precip, solar	86278.1	46848.53	precip	6147.27
3	1	latitude	86318.22	50708.77	—	6187.39
16	2	elev, solar	86326.18	46739.24	—	6195.35
8	1	solar	86333.26	50784.85	solar	6202.43

Table 6.1. Complete model selection results for all LGCP models ordered by WAIC.

ID	N_{cov}	Included Variables	WAIC	DIC	Non-Significant	ΔWAIC
48	3	elev, latitude, precip	86333.57	46629.68	precip	6202.74
21	2	latitude, temp	86340.33	49923.8	—	6209.5
20	2	latitude, precip	86347.27	50694.36	—	6216.44
11	2	elev, latitude	86354.57	46644.26	—	6223.74
212	4	latitude, temp, tree_cov, solar	86391.14	47483.99	solar	6260.32
236	4	dist_coast, temp, solar, popul	86394.39	44494.25	—	6263.56
50	3	elev, latitude, tree_cov	86397.26	43935.5	—	6266.43
347	5	latitude, precip, temp, tree_cov, solar	86411.0	47470.21	precip, solar	6280.17
443	6	latitude, dist_coast, precip, temp, solar, popul	86425.05	44462.77	—	6294.22
32	2	precip, temp	86432.75	49886.42	precip	6301.92
6	1	temp	86437.76	49889.56	—	6306.94
77	3	latitude, dist_coast, tree_cov	86453.85	46689.31	—	6323.02
51	3	elev, latitude, solar	86468.71	46575.69	—	6337.88
123	3	temp, tree_cov, popul	86472.83	42381.91	—	6342.0
37	2	temp, tree_cov	86479.87	47485.49	—	6349.04
330	5	latitude, dist_coast, precip, temp, popul	86487.91	44932.37	—	6357.08
28	2	dist_coast, tree_cov	86501.81	46760.01	—	6370.98
139	4	elev, latitude, precip, solar	86503.0	46585.62	precip	6372.17
349	5	latitude, precip, temp, tree_cov, popul	86521.43	42354.73	precip	6390.6
97	3	dist_coast, precip, tree_cov	86531.4	46751.25	precip	6400.57
385	6	elev, latitude, dist_coast, precip, temp, human_press	86539.73	41692.47	precip	6408.91
188	4	latitude, dist_coast, precip, tree_cov	86541.68	46678.21	—	6410.86
276	5	elev, latitude, precip, tree_cov, solar	86548.17	44096.23	precip	6417.34
391	6	elev, latitude, dist_coast, precip, solar, popul	86564.54	40996.26	—	6433.71
165	4	elev, dist_coast, solar, popul	86587.86	41147.72	—	6457.04
34	2	precip, solar	86593.63	50771.51	solar	6462.8
159	4	elev, dist_coast, temp, human_press	86599.2	41672.51	—	6468.37
214	4	latitude, temp, tree_cov, popul	86608.54	42346.63	—	6477.72
294	5	elev, dist_coast, precip, temp, human_press	86609.83	41667.99	precip	6479.0
264	5	elev, latitude, dist_coast, temp, human_press	86618.88	41666.71	—	6488.05
69	3	elev, tree_cov, solar	86643.19	45621.63	—	6512.36
300	5	elev, dist_coast, precip, solar, popul	86643.76	41118.41	—	6512.93
86	3	latitude, temp, tree_cov	86645.63	47521.45	—	6514.8
200	4	latitude, dist_coast, solar, popul	86651.51	45154.56	—	6520.68
195	4	latitude, dist_coast, temp, popul	86661.2	44986.0	—	6530.38
390	6	elev, latitude, dist_coast, precip, solar, human_press	86670.92	42458.48	precip	6540.09
260	5	elev, latitude, dist_coast, precip, human_press	86675.24	42486.39	precip	6544.42
146	4	elev, latitude, tree_cov, solar	86680.49	44103.67	—	6549.66
202	4	latitude, precip, temp, tree_cov	86702.46	47501.4	precip	6571.64
111	3	precip, temp, tree_cov	86703.69	47507.18	precip	6572.86
18	2	elev, popul	86705.84	43114.08	—	6575.01
270	5	elev, latitude, dist_coast, solar, popul	86710.75	41118.76	—	6579.92
269	5	elev, latitude, dist_coast, solar, human_press	86716.44	42441.79	—	6585.62
164	4	elev, dist_coast, solar, human_press	86727.26	42550.77	—	6596.43
112	3	precip, temp, solar	86749.94	49668.9	precip	6619.11
299	5	elev, dist_coast, precip, solar, human_press	86763.52	42587.95	precip	6632.69
341	5	latitude, dist_coast, temp, solar, popul	86763.77	44542.65	—	6632.94
58	3	elev, dist_coast, human_press	86793.69	42699.83	—	6662.86
244	4	precip, temp, tree_cov, popul	86803.39	42366.1	precip	6672.56
64	3	elev, precip, popul	86822.36	43215.49	precip	6691.54
155	4	elev, dist_coast, precip, human_press	86849.54	42774.76	precip	6718.71
53	3	elev, latitude, popul	86870.59	42029.44	—	6739.77
135	4	elev, latitude, dist_coast, human_press	86876.03	42492.6	—	6745.2
156	4	elev, dist_coast, precip, popul	86888.56	41987.46	—	6757.73
354	5	latitude, precip, tree_cov, solar, popul	86917.15	42182.57	—	6786.33
226	4	dist_coast, precip, tree_cov, solar	86979.84	46696.9	precip	6849.01
15	2	elev, tree_cov	86980.86	45427.85	—	6850.03
204	4	latitude, precip, temp, human_press	86989.99	48836.33	—	6859.16
105	3	dist_coast, tree_cov, solar	86990.1	46731.2	—	6859.27
61	3	elev, precip, tree_cov	86991.17	45076.97	precip	6860.34
38	2	temp, solar	86991.97	49658.28	—	6861.14
225	4	dist_coast, precip, temp, popul	86999.8	45009.26	precip	6868.97
203	4	latitude, precip, temp, solar	87034.31	49587.52	—	6903.48
219	4	latitude, tree_cov, solar, popul	87064.64	42213.83	—	6933.82
30	2	dist_coast, human_press	87075.09	47376.79	—	6944.26
83	3	latitude, precip, solar	87123.31	50719.94	solar	6992.48
350	5	latitude, precip, temp, solar, human_press	87126.49	48592.48	—	6995.67
104	3	dist_coast, temp, popul	87127.66	44985.22	—	6996.83
128	3	tree_cov, solar, popul	87148.72	42229.15	—	7017.89
277	5	elev, latitude, precip, tree_cov, human_press	87166.42	41128.5	precip	7035.59
87	3	latitude, temp, solar	87188.43	49669.24	—	7057.6
79	3	latitude, dist_coast, human_press	87192.4	47337.5	—	7061.57
196	4	latitude, dist_coast, tree_cov, solar	87209.96	46628.74	—	7079.13
409	6	elev, latitude, precip, tree_cov, solar, human_press	87219.56	41059.22	precip, solar	7088.73
113	3	precip, temp, human_press	87240.51	48851.89	precip	7109.68

Table 6.1. Complete model selection results for all LGCP models ordered by WAIC.

ID	N_{cov}	Included Variables	WAIC	DIC	Non-Significant	ΔWAIC
249	4	precip, tree_cov, solar, popul	87243.65	42214.91	precip	7112.82
194	4	latitude, dist_coast, temp, human_press	87251.27	46412.44	—	7120.44
331	5	latitude, dist_coast, precip, tree_cov, solar	87254.12	46620.33	precip	7123.29
322	5	elev, temp, tree_cov, solar, human_press	87254.14	40928.9	solar	7123.32
88	3	latitude, temp, human_press	87263.29	48827.35	—	7132.46
141	4	elev, latitude, precip, popul	87287.08	41954.76	—	7156.25
109	3	dist_coast, solar, popul	87322.31	45143.27	—	7191.49
59	3	elev, dist_coast, popul	87325.32	41642.09	—	7194.49
99	3	dist_coast, precip, human_press	87328.32	47394.84	precip	7197.49
175	4	elev, precip, solar, popul	87350.67	42157.57	precip	7219.85
73	3	elev, solar, popul	87351.35	42022.24	—	7220.52
84	3	latitude, precip, human_press	87360.41	49251.04	—	7229.59
224	4	dist_coast, precip, temp, human_press	87361.97	46407.58	precip	7231.14
39	2	temp, human_press	87377.52	48861.48	—	7246.69
335	5	latitude, dist_coast, precip, solar, popul	87396.82	45103.77	—	7265.99
103	3	dist_coast, temp, human_press	87414.23	46399.44	—	7283.41
229	4	dist_coast, precip, solar, human_press	87422.92	47328.25	precip	7292.09
147	4	elev, latitude, tree_cov, human_press	87423.59	41111.61	—	7292.76
482	7	elev, latitude, precip, temp, tree_cov, solar, human_press	87429.97	40530.01	latitude, solar	7299.14
413	6	elev, latitude, temp, tree_cov, solar, human_press	87433.34	40508.65	solar	7302.51
288	5	elev, latitude, tree_cov, solar, human_press	87434.99	41031.74	solar	7304.16
313	5	elev, precip, temp, tree_cov, human_press	87470.99	40781.39	—	7340.16
433	6	elev, precip, temp, tree_cov, solar, human_press	87488.81	40719.23	solar	7357.98
70	3	elev, tree_cov, human_press	87516.31	43637.09	—	7385.48
316	5	elev, precip, temp, solar, popul	87535.69	41018.77	precip	7404.86
178	4	elev, temp, tree_cov, human_press	87543.04	41038.06	—	7412.21
136	4	elev, latitude, dist_coast, popul	87544.01	41450.9	—	7413.18
407	6	elev, latitude, precip, temp, solar, popul	87544.62	40999.93	—	7413.79
283	5	elev, latitude, temp, tree_cov, human_press	87544.67	40562.8	—	7413.84
245	4	precip, temp, solar, human_press	87550.0	48589.31	precip	7419.17
215	4	latitude, temp, solar, human_press	87550.78	48521.89	—	7419.95
329	5	latitude, dist_coast, precip, temp, human_press	87555.31	46380.39	—	7424.48
334	5	latitude, dist_coast, precip, solar, human_press	87573.04	47285.71	—	7442.21
66	3	elev, temp, solar	87575.58	45251.34	—	7444.75
143	4	elev, latitude, temp, solar	87576.42	45263.65	—	7445.59
124	3	temp, solar, human_press	87576.72	48558.02	—	7445.89
108	3	dist_coast, solar, human_press	87580.51	47353.67	—	7449.68
168	4	elev, precip, temp, solar	87623.12	45219.22	precip	7492.29
183	4	elev, tree_cov, solar, human_press	87624.16	43492.19	solar	7493.33
35	2	precip, human_press	87628.06	49361.35	—	7497.23
210	4	latitude, precip, solar, popul	87630.03	46052.75	—	7499.2
150	4	elev, latitude, solar, popul	87630.16	41784.33	—	7499.33
404	6	elev, latitude, precip, temp, tree_cov, human_press	87637.78	40575.03	latitude	7506.95
9	1	human_press	87651.22	49351.31	—	7520.39
10	1	popul	87653.01	46987.32	—	7522.18
25	2	latitude, popul	87665.05	46965.97	—	7534.22
45	2	solar, popul	87670.07	46106.42	—	7539.24
181	4	elev, temp, solar, popul	87702.25	41002.74	—	7571.42
24	2	latitude, human_press	87766.67	49326.79	—	7635.84
85	3	latitude, precip, popul	87779.39	46801.78	—	7648.56
273	5	elev, latitude, precip, temp, solar	87801.03	45226.04	precip	7670.21
119	3	precip, solar, popul	87803.14	46063.98	—	7672.32
199	4	latitude, dist_coast, solar, human_press	87803.7	47344.41	—	7672.87
94	3	latitude, solar, popul	87820.99	46106.45	—	7690.17
261	5	elev, latitude, dist_coast, precip, popul	87827.17	41433.62	—	7696.34
36	2	precip, popul	87860.82	46960.94	—	7729.99
172	4	elev, precip, tree_cov, human_press	87878.75	43006.21	—	7747.92
318	5	elev, precip, tree_cov, solar, human_press	87899.65	42965.87	solar	7768.82
280	5	elev, latitude, precip, solar, popul	87950.29	41692.26	—	7819.46
286	5	elev, latitude, temp, solar, popul	87984.53	41018.0	latitude	7853.7
190	4	latitude, dist_coast, precip, human_press	87987.87	47311.45	—	7857.04
208	4	latitude, precip, tree_cov, popul	87999.27	42393.46	—	7868.44
230	4	dist_coast, precip, solar, popul	88010.1	45116.63	—	7879.28
118	3	precip, solar, human_press	88108.44	49355.07	—	7977.61
44	2	solar, human_press	88116.77	49255.84	—	7985.94
209	4	latitude, precip, solar, human_press	88124.45	49237.76	—	7993.62
49	3	elev, latitude, temp	88252.27	45362.16	—	8121.45
43	2	tree_cov, popul	88302.87	42454.0	—	8172.04
117	3	precip, tree_cov, popul	88375.93	42451.02	precip	8245.1
93	3	latitude, solar, human_press	88381.04	49191.21	—	8250.22
137	4	elev, latitude, precip, temp	88465.79	45339.64	precip	8334.96
92	3	latitude, tree_cov, popul	88591.37	42379.29	—	8460.54
60	3	elev, precip, temp	88595.78	45342.82	precip	8464.95
14	2	elev, temp	88610.37	45366.03	—	8479.54
72	3	elev, solar, human_press	89429.24	43951.77	—	9298.41

Table 6.1. Complete model selection results for all LGCP models ordered by WAIC.

ID	N_{cov}	Included Variables	WAIC	DIC	Non-Significant	ΔWAIC
174	4	elev, precip, solar, human_press	89475.34	43969.8	precip	9344.51
114	3	precip, temp, popul	89522.81	46664.78	precip	9391.98
279	5	elev, latitude, precip, solar, human_press	89536.41	43646.09	precip	9405.58
67	3	elev, temp, human_press	89615.33	43048.86	—	9484.5
169	4	elev, precip, temp, human_press	89631.78	43049.01	precip	9500.95
149	4	elev, latitude, solar, human_press	89678.71	43680.51	—	9547.88
274	5	elev, latitude, precip, temp, human_press	89682.69	43041.89	precip	9551.86
144	4	elev, latitude, temp, human_press	89757.22	43043.89	—	9626.39
89	3	latitude, temp, popul	89765.53	46707.02	—	9634.7
40	2	temp, popul	89775.82	46713.23	—	9644.99
17	2	elev, human_press	89822.34	44391.37	—	9691.51
68	3	elev, temp, popul	89844.1	41488.74	—	9713.27
205	4	latitude, precip, temp, popul	89911.35	46699.04	—	9780.52
180	4	elev, temp, solar, human_press	89943.05	42874.99	—	9812.22
63	3	elev, precip, human_press	89944.65	44445.72	precip	9813.82
315	5	elev, precip, temp, solar, human_press	89949.44	42875.63	precip	9818.61
145	4	elev, latitude, temp, popul	89951.18	41443.31	—	9820.35
275	5	elev, latitude, precip, temp, popul	89957.07	41361.34	—	9826.24
285	5	elev, latitude, temp, solar, human_press	89979.31	42898.87	—	9848.48
406	6	elev, latitude, precip, temp, solar, human_press	90047.51	42897.75	precip	9916.68
170	4	elev, precip, temp, popul	90122.79	41556.79	precip	9991.96
140	4	elev, latitude, precip, human_press	90216.22	43846.38	precip	10085.39
52	3	elev, latitude, human_press	90233.64	43854.29	—	10102.81
216	4	latitude, temp, solar, popul	90912.84	45789.46	—	10782.01
351	5	latitude, precip, temp, solar, popul	90998.92	45788.5	—	10868.1
125	3	temp, solar, popul	91237.55	45774.22	—	11106.72
246	4	precip, temp, solar, popul	91398.82	45756.97	—	11267.99

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